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KKT Atlas: Learning Low-Dimensional Local Scenario Models for Contextual Stochastic Optimization

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Abstract. Contextual stochastic optimization uses observed operating conditions to guide decisions under uncertainty. Decision-focused learning is a prominent approach that trains conditional models using the operational loss induced by their decisions. Although this approach can improve decision quality, it requires repeated solution of stochastic programs, whose computational cost can grow substantially with the scenario support size during both training and deployment. We propose *KKT Atlas*, a method for learning compact local scenario models; it groups similar contexts into local regions called charts. Within each chart b, Atlas uses gradients of scenario cost functions with respect to decisions to identify an uncertainty subspace of dimension d_b, which is generally much smaller than the original scenario dimension. Atlas then constructs scenarios in this subspace and jointly learns these scenarios and their context-dependent probabilities by minimizing operational loss. Our analysis shows that, under the stated conditions, there exists a model with just $K_b = d_b + 1$ shared scenarios that can be solved to obtain good decisions throughout chart b. We bound the resulting decision error and quantify the tradeoff between finer localization and estimation error from smaller chart samples, explaining why more charts are not always beneficial. In case studies of capacity allocation, portfolio optimization, and network reservation, Atlas uses only 0.8–3.6% of the reference support size per decision, achieves the lowest mean risk in the first two problems and nearly matches the strongest baseline in the third, and reduces model-construction and deployment times by approximately 77–96% and 92–98%, respectively, relative to the full-support comparator. Separate mechanism experiments show how local charts and gradient information help retain decision quality with compact scenario models.

Key words: contextual stochastic optimization; scenario generation; KKT

1. Introduction

Operational decisions often involve uncertainty, while observed operating conditions can provide information about that uncertainty before a decision is made. Contextual stochastic optimization incorporates this information into the uncertainty model used to guide the decision. For example, inventory replenishment decisions depend on uncertain demand. An unconditional approach estimates one demand distribution from observations across operating conditions and uses it in the

stochastic program. Promotions and recent sales, however, provide information about the demand distribution before orders are placed. Because this information distinguishes the current operating condition from others, a model conditioned on these features can represent the demand distribution more accurately than one that only pools observations from all conditions. Empirical studies show that incorporating such information can yield better decisions than ignoring it (Bertsimas and Kallus 2020, Kallus and Mao 2023, Kannan et al. 2025). The practical question is how to turn the available observations into an effective decision rule.

A common approach to contextual stochastic optimization first estimates conditional uncertainty from data and then solves a stochastic program based on that estimate. Statistical learning provides standard methods for the estimation step. For example, multinomial logistic regression fits conditional probabilities by maximum likelihood or, equivalently, by minimizing cross-entropy loss (Hastie et al. 2009); with a fixed scenario support, these probabilities can represent the likelihood of each scenario at a new context. Kannan et al. (2025) construct scenarios from regression predictions and fitted residuals, with the prediction model fitted independently of the stochastic program. In these estimate-then-optimize procedures, the model is first fitted to improve predictive or distributional accuracy and then used in the downstream stochastic program. Because the downstream cost does not enter the fitting objective, training need not give priority to estimation errors that lead to more costly decisions.

Integrated learning instead uses the operational loss of the induced decisions as a training criterion for the statistical model, following the broader principle of decision-focused learning. Elmachtoub and Grigas (2022) develop Smart "Predict, then Optimize" (SPO) for predicting objective coefficients in optimization problems with linear objectives. They measure prediction quality by the excess cost of the induced decision and derive the convex SPO+ surrogate loss for training. Kallus and Mao (2023) apply this principle to forests by choosing splits that improve downstream optimization performance. Qi et al. (2026) develop integrated conditional estimation–optimization (ICEO), which fits a conditional distribution model to minimize the operational loss of the resulting decisions. Their numerical comparisons show that including downstream operational loss can improve on separate distribution fitting based on cross-entropy loss. These gains motivate accounting for operational loss when learning a conditional distribution.

However, training with operational loss can be computationally intensive because each loss evaluation requires solving the stochastic program induced by the current conditional distribution model. This burden becomes substantial when the scenario support is large or recourse is costly. Existing approaches reduce the training cost by approximating quantities that would otherwise require repeated optimization. For example, Kallus and Mao (2023) use perturbation-based criteria to avoid reoptimization at every candidate tree split, while Qi et al. (2026) learn a neural approximation of the solution mapping from solutions computed at sampled probability vectors and use this approximation to reduce repeated optimization during training. These approximations introduce additional error during training, with its magnitude depending on the accuracy of the approximate splitting

criterion or solution mapping. Moreover, these methods reduce computation during training but do not necessarily reduce the cost of downstream optimization at deployment. Instead, we reduce scenario support size to keep downstream optimization problems small during both training and deployment.

We propose *KKT Atlas*, a method for learning compact local scenario models. Atlas uses a dataset of paired observations $\{(x_i, \xi_i)\}$, where x_i is an observed context and ξ_i is the corresponding realization of the uncertain parameter. It clusters the observed contexts to partition the context space into local regions, called charts, that group similar operating conditions. Each chart uses a shared set of generated scenarios with context-dependent probabilities. This localization allows each support to focus on uncertainty within its region, while the collection of local models accommodates differences across regions. For a new context, Atlas solves one small stochastic program using only the assigned chart's scenarios and context-dependent probabilities.

Within each chart, Atlas uses gradients of scenario cost functions to guide the representation of uncertainty. The scenario-dependent cost function $Q(z, \xi; x)$ measures the cost of taking decision z under realization ξ at context x. Its gradient $\nabla_z Q(z, \xi; x)$ determines the realization's contribution to the first-order optimality condition. Our analysis considers a local affine-gradient model in which this gradient is approximately a linear combination of a small number of features of ξ, plus a term independent of ξ. This structure motivates constructing a low-dimensional basis from the cross-covariance between transformed realizations and scenario-cost gradients. Each generated scenario is represented by coordinates in this basis and reconstructed in the original uncertainty space.

Contributions.

(i) We develop KKT Atlas to learn compact local scenario models for contextual stochastic optimization. Contexts within each chart share a small set of generated scenarios with context-dependent probabilities. Gradient-informed low-dimensional coordinates reduce the number of scenario parameters, and integrated fitting jointly learns these coordinates and probability parameters by minimizing observed operational loss. The compact supports reduce the number of scenario terms in downstream stochastic programs, thereby accelerating both training and deployment.

(ii) We establish theoretical guarantees linking local representation dimension, shared-support size, and decision quality. Under a local affine-gradient model, we identify conditions under which the empirical spectral rule recovers the gradient-relevant dimension and the estimated subspace can represent every relevant projection. Under suitable geometric and feasibility conditions, $K_b = d_b + 1$ shared scenarios suffice to cover all relevant conditional means in chart b. Under strong convexity, we translate expected-gradient approximation error into bounds on decision error and the regularized objective gap, and derive an operational-risk bound that accounts for regularization bias. Finally, our finite-sample analysis quantifies the tradeoff between within-chart heterogeneity and sampling error in estimating the gradient signal, explaining why finer localization does not necessarily improve estimation accuracy.

(iii) We compare Atlas with integrated full-support learning and methods based on scenario weights or local observations in capacity allocation, portfolio optimization, and network reservation. The results show that compact atlases achieve the lowest or nearly lowest mean risk while using only 0.8–3.6% of the reference support per decision and reducing model-construction and deployment times by approximately 77–96% and 92–98%, respectively, relative to the full-support comparator. Atlas also outperforms methods based on scenario weights or local observations, reducing deployment time by approximately 57–95% relative to the faster of these methods in each reported setting and achieving mean relative excess 0.5–2.6 percentage points lower than that of the lower-risk method in each application. Targeted mechanism studies examine how support size and chart localization affect decision quality and show that the cross-covariance basis retains decision-relevant directions in the presence of irrelevant variation.

The rest of the paper is organized as follows. Section 2 reviews contextual stochastic optimization, scenario reduction, and scenario generation. Section 3 defines the contextual stochastic program and the scenario-gradient quantities used throughout the paper. In Section 4, we present KKT Atlas, from chart construction and gradient-informed scenario coordinates to structure selection and integrated fitting. Section 5 establishes the corresponding guarantees for localization, representation, support size, and decision quality. In Section 6, we describe the computational study and report the main comparisons and mechanism analyses. Section 7 concludes. The e-companion contains proofs and detailed computational protocols.

2. Related Literature

In Section 2.1, we review the use of contextual data to learn uncertainty models and decision rules. We then discuss scenario reduction in Section 2.2 and methods for scenario generation and dimension reduction in Section 2.3.

2.1. Contextual stochastic optimization

A common approach to contextual stochastic optimization is to learn a context-dependent representation of uncertainty and then solve the stochastic program induced by that representation. Methods along this route differ in both the representation they construct and the criterion used to fit it. Bertsimas and Kallus (2020) represent conditional uncertainty by assigning context-dependent weights to historical observations and solve the corresponding weighted empirical problem. These weights are constructed using predictive methods such as k-nearest neighbors, CART, and random forests, with the goal of capturing the conditional distribution of uncertainty rather than minimizing downstream operational loss. Subsequent tree-based methods incorporate the downstream optimization problem into the construction of these weights. Stratigakos et al. (2022) train ensembles of decision trees by directly minimizing application-specific operating costs and assign larger weights to historical observations that fall in the same terminal leaves as the current context

when constructing a weighted SAA. In a more general contextual stochastic optimization setting, stochastic optimization forests (Kallus and Mao 2023) choose tree splits according to the quality of the downstream decisions they induce. Ribeiro and Fanzeres (2026) consider contextual two-stage linear programs and choose tree splits using an objective that accounts for both how well the conditional uncertainty distribution fits the observations and the quality of the resulting decisions. Other methods construct context-dependent scenarios or probabilities. Kannan et al. (2025) construct context-dependent SAA scenarios by combining a fitted prediction with residuals from the prediction model. The prediction model is fitted independently of the stochastic program, so operational loss does not enter its training objective. ICEO (Qi et al. 2026), in contrast, uses a prespecified finite support and learns context-dependent probabilities by minimizing downstream operational loss.

A distinct route is to learn a direct mapping from context to decisions. Zhang et al. (2026) fit piecewise affine rules by minimizing empirical operational loss, with the number of pieces controlling their flexibility. These rules provide decisions without solving a new stochastic program. In their constrained formulation with general nonlinear constraints, however, feasibility at new contexts is guaranteed with high probability. By contrast, learning the uncertainty distribution and determining the decision through a downstream optimization problem allows the decision constraints to be enforced explicitly at each new context.

2.2. Scenario reduction

Distribution-based scenario reduction seeks a distribution that is supported on fewer scenarios and remains close to the original distribution. Dupačová et al. (2003) retain a subset of the original scenarios and transfer the probability mass of each removed scenario to its nearest retained scenario. Following this approach, Heitsch and Römisch (2003) select the smaller set iteratively, adding or removing original scenarios based on the resulting probability distance. In contrast, Rujeerapaiboon et al. (2022) study continuous scenario reduction, in which new scenarios may be chosen freely rather than selected from the original scenario set.

Optimization-based scenario reduction uses the objective and structure of the stochastic program to guide the choice of a smaller scenario set. Bertsimas and Mundru (2023) define a problem-dependent divergence between distributions that accounts for the quality of the decisions they induce and use it to construct new scenarios. Zhang et al. (2023) select and reweight original scenarios to approximate the recourse function over a pool of candidate decisions, with bounds on solution quality. These two optimization-based methods therefore differ in their approximation criteria and in whether they construct new scenarios or select existing ones. However, both methods account for downstream decision quality, as does Atlas.

2.3. Contextual scenario generation and dimension reduction

Constructing new scenarios adds flexibility even when the number of scenarios is fixed. For linear two-stage stochastic programs, Henrion and Römisch (2022) choose scenario values and probabilities to approximate expected recourse, the expected second-stage cost for a given first-stage decision. Their criterion minimizes the largest discrepancy over feasible first-stage decisions between the original expected recourse and its weighted scenario approximation. Moreover, Homem-de-Mello et al. (2026) show that, for a class of contextual two-stage problems with linear second stages, fixed recourse, and fixed costs, a single suitably constructed scenario, possibly outside the original support, can recover an optimal decision for each context. In contrast, Atlas uses a shared set of generated scenario values within each chart, with probabilities that vary by context.

Contextual scenario generators learn scenario sets from observed data. Islip et al. (2025) map each context to a prescribed number of scenarios, while Hornewall et al. (2025) train a neural scenario generator through a deterministic equivalent regularized by a log barrier, using implicit KKT differentiation. Both formulations specify generated scenarios in the original uncertainty space. In the latter method, KKT differentiation supplies gradients for updating the generator parameters. Atlas combines KKT differentiation during fitting with a preceding representation step that uses scenario-cost gradients to select a low-dimensional basis for each chart. The generated scenarios are fitted through coordinates in this basis, reducing the number of scenario parameters to learn.

Even with a small support, learning scenario values in the full uncertainty space can be costly because each scenario has many components. One way to address this dimensional burden is to identify a few input directions along which a function varies most and use the resulting reduced representation. For example, Zahm et al. (2020) use gradients to identify these directions and approximate a vector-valued function through the corresponding linear combinations of its inputs. Chen et al. (2025) also reduce the output space, jointly selecting input and output subspaces to account for how the choice in each space affects the relevant directions in the other. Both studies select reduced spaces using approximation bounds based on derivatives of the function with respect to its inputs. Atlas uses the cross-covariance between transformed realizations and gradients of the scenario cost Q with respect to decisions to select a representation suited to downstream first-order optimality conditions.

3. Problem Setting and Preliminaries

We first define the contextual stochastic program and its finite-scenario counterpart in Section 3.1. We then examine how the conditional distribution enters the optimality condition in Section 3.2.

Throughout the paper, a hat ($\widehat{a}$) denotes a data-derived quantity, and a star ($a^\star$) denotes an ideal population target or structural reference. Chart-specific structural references are defined relative to the chart construction, preprocessing, SAA reference decision, and other objects held fixed in the surrounding conditional analysis. A tilde ($\widetilde{a}$) is reserved for a numerically computed decision. A short bar ($\bar{a}$) denotes a mean or centroid, whereas a long overline ($\overline{a}$) denotes an explicitly defined upper bound or upper endpoint. Table EC.1 in the e-companion summarizes the notation.

3.1. Contextual stochastic program

Let $X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$ denote the observable context and let $\Xi \in \mathcal{U} \subseteq \mathbb{R}^{d_\xi}$ denote the uncertain parameter. For example, in inventory replenishment, the context X includes promotions and recent sales observed before orders are placed, and Ξ is future demand, which is unknown at that time but affects the operational loss. We use $x \in \mathcal{X}$ and $\xi \in \mathcal{U}$ for their realizations; observed realizations and generated scenarios share the symbol ξ. A decision $z \in \mathcal{Z} \subseteq \mathbb{R}^{d_z}$ incurs the operational loss

$$\ell(z, \xi; x) := f(z; x) + Q(z, \xi; x), \tag{1}$$

where f collects the deterministic terms treated separately in the model and Q is the remaining cost evaluated for a scenario.

We compute a local reference decision and evaluate scenario-cost gradients at the centroid of each chart. The context support need not be convex, so a centroid may lie outside $\mathcal{X}$. To include such centroids, we define the enlarged analysis domain $\mathcal{X}_{\text{an}}$ as the union of $\mathcal{X}$ and these chart centroids. Let P_x denote the conditional probability law of Ξ given $X = x$. We take P_x, f, and Q to be defined throughout $\mathcal{X}_{\text{an}}$.

Let $z_0 : \mathcal{X}_{\text{an}} \to \mathcal{Z}$ be a prespecified measurable mapping from contexts to feasible reference decisions. We use $z_0(x)$ as the regularization center, so the quadratic term below penalizes deviations from this decision. For a prespecified regularization parameter $\eta_{\text{reg}} \geq 0$, define the conditional operational risk R_0 and the regularized objective R by

$$R_0(z; x) := \mathbb{E}_{P_x}[\ell(z, \Xi; x)], \qquad R(z; x) := R_0(z; x) + \frac{\eta_{\text{reg}}}{2} \left\| z - z_0(x) \right\|_2^2, \qquad \eta_{\text{reg}} \geq 0. \tag{2}$$

The target decision for regularization parameter η_{reg} is

$$z^\star(x) := \operatorname*{arg\,min}_{z \in \mathcal{Z}} R(z; x). \tag{3}$$

KKT Atlas replaces P_x with a finite distribution supported on a small number of scenarios. For a positive integer s, let Id_s and $\mathbf{1}_s$ denote the $s \times s$ identity matrix and s-dimensional all-ones vector, respectively; we omit the subscript on $\mathbf{1}$ when its dimension is clear. For a support size K, generated scenarios $\boldsymbol{\xi} = (\xi_1, \ldots, \xi_K) \in \mathcal{U}^K$ and probabilities $p \in \Delta_K := \{p \in \mathbb{R}_+^K : \mathbf{1}^\top p = 1\}$, define the regularized solution mapping

$$z(p, \boldsymbol{\xi}; x) := \operatorname*{arg\,min}_{z \in \mathcal{Z}} \left\{ f(z; x) + \sum_{k=1}^{K} p_k Q(z, \xi_k; x) + \frac{\eta_{\text{reg}}}{2} \left\| z - z_0(x) \right\|_2^2 \right\}. \tag{4}$$

Eq. (4) defines the exact solution mapping used in the analysis. We write $\widetilde{z}$ for its numerically computed counterpart. The added regularization coefficient η_{reg} is distinct from the common curvature bound η specified below.

ASSUMPTION 1 **(Strong convexity)**. *The set $\mathcal{Z}$ is nonempty, closed, and convex. For every $x \in \mathcal{X}_{\mathrm{an}}$ and $\xi \in \mathcal{U}$, $f(\cdot;x)$ and $Q(\cdot,\xi;x)$ are proper, closed, and convex. For every $x \in \mathcal{X}_{\mathrm{an}}$, the relevant expectations under P_x are finite, and the objectives in Eqs. (3) and (4) attain their minima. There exists a common $\eta > 0$ such that the complete population objective and every complete finite-distribution objective in Eqs. (3) and (4) are η-strongly convex on $\mathcal{Z}$ for every $x \in \mathcal{X}_{\mathrm{an}}$. We use the sharpest common lower bound established for these objectives. In particular, for every $x \in \mathcal{X}_{\mathrm{an}}$, $z, z' \in \mathcal{Z} \cap \operatorname{dom} R(\cdot;x)$, and $t \in [0,1]$,*

$$R(tz+(1-t)z';x) \le tR(z;x)+(1-t)R(z';x)-\frac{\eta}{2}t(1-t)\left\|z-z'\right\|_2^2. \tag{5}$$

Each objective in Eq. (4) satisfies the same inequality on its feasible domain.

Assumption 1 ensures that the objectives in Eqs. (3) and (4) are η-strongly convex. Since their minima are attained, both the target decision and the finite-scenario decisions are unique. In the later analysis, we combine strong convexity with first-order optimality to bound decision error in terms of the error in approximating the conditional mean of scenario cost gradients.

3.2. Scenario gradients and optimality

To use fewer scenarios without losing the information needed for a decision, we identify how the uncertain parameter affects the KKT conditions. The derivative of Q with respect to the decision determines the contribution of each realization to the first-order condition. The main analysis relies on the following differentiability assumption.

ASSUMPTION 2 **(Differentiability)**. *For every $x \in \mathcal{X}_{\mathrm{an}}$ and $\xi \in \mathcal{U}$, $f(\cdot;x)$ is differentiable on $\mathcal{Z}$, and $Q(\cdot,\xi;x)$ is differentiable on a neighborhood of $\mathcal{Z}$. For every $z \in \mathcal{Z}$ and $x \in \mathcal{X}_{\mathrm{an}}$, the gradient of Q is integrable under P_x, and differentiation with respect to z can be interchanged with expectation under P_x.*

Under Assumption 2, for a realization ξ, the gradient of $Q(\cdot,\xi;x)$ at z is

$$\phi(z,\xi;x) := \nabla_z Q(z,\xi;x) \in \mathbb{R}^{d_z}. \tag{6}$$

The conditional integrability of the gradients in Eq. (6) allows us to define their conditional mean

$$\bar{\phi}(z;x) := \mathbb{E}_{P_x}[\phi(z,\Xi;x)]. \tag{7}$$

Interchanging differentiation and conditional expectation under the same assumption gives the regularized first-order mapping

$$\nabla_z R(z;x) = \nabla_z f(z;x) + \bar{\phi}(z;x) + \eta_{\mathrm{reg}}\{z - z_0(x)\}. \tag{8}$$

With $N_{\mathcal{Z}}(z) := \{v \in \mathbb{R}^{d_z} : \langle v, u-z\rangle \le 0 \text{ for every } u \in \mathcal{Z}\}$ denoting the normal cone, the target decision satisfies the first-order optimality condition (Boyd and Vandenberghe 2004, Section 4.2.3)

$$0 \in \nabla_z R(z^\star(x);x) + N_{\mathcal{Z}}(z^\star(x)). \tag{9}$$

Eqs. (8) and (9) link scenario construction to decision quality. The deterministic cost, regularizer, and constraints remain unchanged when the conditional law is replaced. The distribution-dependent term is $\bar{\phi}(z;x)$, defined in Eq. (7) as the conditional average of the scenario cost gradients; under the finite scenario distribution, it is replaced by $\sum_{k=1}^{K} p_k \phi(z, \xi_k; x)$.

In Section 4, we use sample gradients $\phi(z, \xi_i; x)$ to construct a cross-covariance basis; in Section 5, we show how error in approximating $\bar{\phi}$ affects the exact optimal decision of the finite-scenario problem.

4. Construction of KKT Atlas

KKT Atlas combines two ideas. First, we partition the context space into charts so that each region can use a scenario support tailored to its local variation. A global support may require many scenarios to represent all contexts, whereas a local support needs to represent only the contexts assigned to its chart. For a new context, Atlas selects the nearest chart and solves only the corresponding local stochastic program. Chart localization therefore allows each query to use a small chart-specific support without combining supports from different regions. Second, within each chart, we use the cross-covariance between transformed observed realizations and cost gradients to identify a low-dimensional scenario space. We reconstruct generated scenarios from coefficients in the retained basis and fix the support size at one more than the retained dimension. The reduced representation lowers the number of scenario coefficients fitted during training, while the compact support reduces the number of scenario terms in downstream problems solved during both model fitting and deployment.

We present an overview in Section 4.1, then describe the context charts and local scenario directions in Sections 4.2 and 4.3. In Section 4.4, we construct the local scenarios and model their context-dependent probabilities. We detail configuration selection and the subsequent joint fitting of the selected local models in Section 4.5.

4.1. Overview

In Algorithm 1, we organize offline construction into *Configuration selection* and *Model fitting*. The first stage compares candidate numbers of charts B. For each candidate, the chart assignments and local sample gradients determine a basis for each chart. The singular-value retention rule sets its dimension d_b, which can differ across charts. We fix $K_b = d_b + 1$, motivated by the shared-support representation in Theorem 4. The number of charts is therefore the only structural parameter selected through candidate comparison. More charts can reduce within-chart variation, but the resulting smaller samples in each chart can increase estimation error. We analyze this tradeoff between localization and sampling in Section 5.

To compare candidate partitions, we use internal training folds. Using the fitting observations of each fold, we initialize the scenario coefficient vectors with projected K-means and fit the probability model to the resulting cluster assignments using cross-entropy loss. During this initial fit, we

keep the scenario coefficient vectors fixed and neither solve nor differentiate through downstream optimization problems, limiting the cost of screening candidate partitions. We then assess the initialized models on the assessment observations by solving the induced downstream problems and computing the operational loss of the resulting decisions. We select the number of charts with the lowest aggregate assessment loss.

In *Model fitting*, we keep the selected partition, local bases, dimensions, and support sizes fixed. Using all training observations in each selected chart, we jointly fit its scenario coefficient vectors and context-dependent probability parameters. Contexts within a chart share the same scenarios, while the probability model maps each context to nonnegative scenario weights that sum to one. We minimize the regularized empirical operational loss, evaluating each induced decision under the corresponding observed realization.

Algorithm 1: KKT Atlas overview

Input: Training observations $\{(x_i, \xi_i)\}_{i \in I_{\mathrm{tr}}}$; candidate numbers of charts $\mathcal{G}_B$.
Step 1. Configuration selection.
Standardize contexts using their training means and sample scales
for $B \in \mathcal{G}_B$ **do**
  Form B context charts using K-means
  for $b = 1, \ldots, B$ **do**
    Compute the local SAA decision and scenario cost gradients
    Extract the local basis and determine d_b using Eq. (15)
    Set $K_b = d_b + 1$ and initialize the scenarios and probability model
    Assess the initialized model using mean operational loss
$\mathcal{G}_B^{\mathrm{elig}} \leftarrow$ candidate numbers in $\mathcal{G}_B$ for which every chart passes the prespecified structural screening
$\widehat{B} \leftarrow$ smallest $B \in \mathcal{G}_B^{\mathrm{elig}}$ minimizing the aggregate operational loss
Step 2. Model fitting.
for $b = 1, \ldots, \widehat{B}$ **do**
  Retain the selected chart dimension d_b and support size $K_b = d_b + 1$
  Fit the local scenario and probability parameters by minimizing the regularized empirical operational loss in Eq. (20)
Return the selected charts, local bases, and fitted scenario distributions

4.2. Context charts from K-means

Let $\{(x_i, \xi_i)\}_{i \in I_{\mathrm{tr}}}$ be the training observations. Let $\widehat{\mu}_x \in \mathbb{R}^{d_x}$ and $\widehat{D}_x \in \mathbb{R}^{d_x \times d_x}$ be their context mean and diagonal matrix of sample standard deviations. We assume that every diagonal entry is positive. K-means uses $\widehat{D}_x^{-1}(x_i - \widehat{\mu}_x)$; the same transformation is applied to new contexts. For $h \in \mathbb{R}^{d_x}$, write $\|h\|_x := \|\widehat{D}_x^{-1} h\|_2$ for the resulting standardized norm.

For a given number B of charts, we apply K-means to the standardized training contexts, alternating between assigning observations to the nearest center and updating each center to the mean of its assigned observations. Throughout Sections 4.2–4.4, all chart-specific quantities refer to the current candidate partition. We use b as the chart index and reserve B for the assignment rule and quantities defined for the partition, including when comparing candidates in Section 4.5. We

transform the fitted centers back to the original context scale and denote the center of chart b by $\bar{x}_b$. We define the assignment rule and mean squared distance to the assigned centers by

$$\tau_B(x) := \min\left(\arg\min_{1\le b\le B} \|x-\bar{x}_b\|_x^2\right), \tag{10}$$

$$\widehat{J}_B := \frac{1}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}} \|x_i - \bar{x}_{\tau_B(x_i)}\|_x^2. \tag{11}$$

The outer minimum in Eq. (10) resolves assignment ties in favor of the smallest chart index. Let $I_b := \{i \in I_{\mathrm{tr}} : \tau_B(x_i) = b\}$ index the training observations assigned to chart b. A finer partition can reduce within-chart variation and hence the approximation error of the local models. However, it leaves fewer observations per chart, which can increase estimation error. To balance these effects, we compare partitions using the mean operational loss defined in Section 4.5 and select the one with the lowest value.

4.3. Scenario directions from sample gradients

For each chart, we compute the local SAA decision $\widetilde{z}_b^{\mathrm{SAA}}$ by solving

$$\min_{z\in\mathcal{Z}}\left\{ f(z;\bar{x}_b) + \frac{1}{|I_b|}\sum_{i\in I_b} Q(z,\xi_i;\bar{x}_b) + \frac{\eta_{\mathrm{reg}}}{2}\|z - z_0(\bar{x}_b)\|_2^2 \right\}. \tag{12}$$

At the solution of Eq. (12), we evaluate $\phi(\widetilde{z}_b^{\mathrm{SAA}}, \xi_i; \bar{x}_b)$ for each $i \in I_b$. Fix a transformation $T : \mathcal{U} \to \mathbb{R}^{d_\xi}$ and a decoder $g : \mathrm{dom}(g) \to \mathcal{U}$, where $\mathrm{dom}(g) \subseteq \mathbb{R}^{d_\xi}$. We need these transformations to ensure that generated scenarios belong to the uncertainty set $\mathcal{U}$. The decoder g enforces this requirement, while T reverses decoding for observed scenarios, satisfying $g(T(\xi)) = \xi$. For example, in an inventory problem with nonnegative demand, we may choose $g(y) = y^2$ and $T(\xi) = \sqrt{\xi}$, applied componentwise. Decoding then produces nonnegative demands and recovers each observed demand from its encoded coordinates. Without this decoding step, generated coordinates may be negative and hence invalid as demand inputs to the stochastic optimization problem. We center and scale the transformed realizations separately within each chart. Let $\widehat{\mu}_b := |I_b|^{-1}\sum_{i\in I_b} T(\xi_i)$. For a small numerical tolerance $\varepsilon_{\mathrm{num}} > 0$, such as 10^{-8}, define the diagonal scale matrix $\widehat{D}_b \in \mathbb{R}^{d_\xi \times d_\xi}$ componentwise by

$$(\widehat{D}_b)_{rr} := \max\left\{ \left(\frac{1}{|I_b|-1}\sum_{i\in I_b} \big([T(\xi_i)]_r - (\widehat{\mu}_b)_r\big)^2 \right)^{1/2}, \varepsilon_{\mathrm{num}} \right\}, \qquad r = 1,\dots,d_\xi. \tag{13}$$

Using the mean and scale in Eq. (13), define the standardized encoder $T_b(\xi) := \widehat{D}_b^{-1}\{T(\xi) - \widehat{\mu}_b\}$ and decoder $g_b(y) := g(\widehat{\mu}_b + \widehat{D}_b y)$. Here $T_b : \mathcal{U} \to \mathbb{R}^{d_\xi}$ and $\mathrm{dom}(g_b) = \{y \in \mathbb{R}^{d_\xi} : \widehat{\mu}_b + \widehat{D}_b y \in \mathrm{dom}(g)\}$. We introduce the notation T_b and g_b because these maps combine T and g with chart-specific standardization and its reversal, respectively, providing compact notation for the complete encoding and decoding operations.

Since $\sum_{i\in I_b} T_b(\xi_i) = 0$, the empirical cross-covariance is

$$\widehat{C}_b := \frac{1}{|I_b|}\sum_{i\in I_b} T_b(\xi_i)\phi(\widetilde{z}_b^{\mathrm{SAA}}, \xi_i; \bar{x}_b)^\top \in \mathbb{R}^{d_\xi \times d_z}. \tag{14}$$

Its leading left singular vectors identify scenario directions associated with variation in the cost gradients.

Let $\sigma_j(\widehat{C}_b)$ denote the singular values in nonincreasing order. For a fixed retention proportion $\alpha_{\mathrm{ret}} \in (0,1)$, choose

$$d_b := \begin{cases} 1, & \|\widehat{C}_b\|_F = 0, \\ \min\left\{ d \in \{1,\dots,\min(d_\xi, d_z)\} : \sum_{j=1}^{d} \sigma_j(\widehat{C}_b)^2 \geq \alpha_{\mathrm{ret}} \|\widehat{C}_b\|_F^2 \right\}, & \|\widehat{C}_b\|_F > 0. \end{cases} \tag{15}$$

4.4. Local scenarios and probabilities

Let $\widehat{V}_b \in \mathbb{R}^{d_\xi \times d_b}$ contain the corresponding leading left singular vectors of the sample cross-covariance as its columns. These vectors define directions in the standardized scenario space. Intuitively, we retain these directions because they capture the strongest linear associations between scenario variation and scenario-cost gradients in the sample, focusing the reduced representation on uncertainty relevant to decisions. Projecting $T_b(\xi) \in \mathbb{R}^{d_\xi}$ onto these directions gives $\widehat{V}_b^\top T_b(\xi) \in \mathbb{R}^{d_b}$, representing the scenario with d_b coordinates instead of d_ξ, thereby reducing the dimension.

For $u \in \mathbb{R}^{d_b}$, we write

$$\xi = g_b(\widehat{V}_b u), \quad \widehat{V}_b u \in \mathrm{dom}(g_b). \tag{16}$$

The vector u specifies the coordinates used to generate a scenario in the retained basis $\widehat{V}_b$. Multiplication by $\widehat{V}_b$ maps the coefficients to a vector in the standardized scenario space; g_b then maps that vector to a feasible scenario.

Motivated by Theorem 4, we use $K_b = d_b + 1$ trainable coefficient vectors $u_{b,1}, \dots, u_{b,K_b} \in \mathbb{R}^{d_b}$ in chart b and define $\xi_{b,k} := g_b(\widehat{V}_b u_{b,k})$. For a context x in chart b, the probability model for these scenarios is

$$p_b(x; a_b, \Gamma_b) := \mathrm{softmax}\Big(a_b + \Gamma_b \widehat{D}_x^{-1}(x - \bar{x}_b)\Big) \in \Delta_{K_b}, \tag{17}$$

where $\mathrm{softmax}(v)_k := \exp(v_k)/\sum_{j=1}^{K_b}\exp(v_j)$ and Δ_{K_b} is the probability simplex. We write $p_{b,k}(x; a_b, \Gamma_b)$ for component k, the probability assigned to $\xi_{b,k}$, and abbreviate it as $p_{b,k}(x)$ when the parameters are clear.

The vector $a_b \in \mathbb{R}^{K_b}$ and matrix $\Gamma_b \in \mathbb{R}^{K_b \times d_x}$ contain the trainable intercepts and context coefficients, respectively. Contexts within a chart share the same scenarios, whose probabilities vary with context. The model has $K_b d_b$ scenario basis coefficients and K_b scenario terms in its stochastic program. We collect the parameters in $\theta_b := (u_{b,1}, \dots, u_{b,K_b}, a_b, \Gamma_b)$ and let $\widetilde{z}_b(x; \theta_b)$ denote the decision obtained by solving

$$\min_{z\in\mathcal{Z}} \left\{ f(z;x) + \sum_{k=1}^{K_b} p_{b,k}(x) Q(z, \xi_{b,k}; x) + \frac{\eta_{\mathrm{reg}}}{2}\|z - z_0(x)\|_2^2 \right\}. \tag{18}$$

In the next subsection, we jointly fit the scenario coefficients and probability parameters in θ_b, keeping $\widehat{V}_b$ and g_b fixed. We use the induced decision $\widetilde{z}_b(x; \theta_b)$ to evaluate operational loss.

4.5. Structure selection and integrated training

We select the number of charts using internal assessment within the training sample. For each candidate B in the fixed grid $\mathcal{G}_B$, the preceding construction determines a partition and the local dimension d_b of each chart.

Fix a candidate partition, chart b, and assessment fold. Let $I^{\text{fit}} \subset I_b$ contain the observations outside that fold; $I_b \setminus I^{\text{fit}}$ is the assessment set. Here we apply K-means again to $\{\widehat{V}_b^\top T_b(\xi_i) : i \in I^{\text{fit}}\}$ and initialize the scenario coefficient vectors at the resulting K_b cluster centers. For $i \in I^{\text{fit}}$, let $k(i) \in \{1, \dots, K_b\}$ index the cluster containing the projection of observation i. Holding the scenario coefficient vectors fixed, we use these cluster assignments to initialize the affine-softmax probability model by solving

$$(a^{(0)}, \Gamma^{(0)}) \in \underset{\substack{a \in \mathbb{R}^{K_b},\, \Gamma \in \mathbb{R}^{K_b \times d_x} \\ \mathbf{1}^\top a = 0,\, \mathbf{1}^\top \Gamma = 0}}{\arg\min} \; -\frac{1}{|I^{\text{fit}}|} \sum_{i \in I^{\text{fit}}} \log p_{b,k(i)}(x_i; a, \Gamma) + \lambda_\Gamma \|\Gamma\|_F^2. \tag{19}$$

Here $\lambda_\Gamma > 0$ is the regularization coefficient for the probability slopes, and $p_{b,k(i)}(x_i; a, \Gamma)$ is the probability that the model in Eq. (17) assigns to the cluster containing observation i. The cross-entropy term in Eq. (19) fits the cluster probabilities conditional on context, and the constraints remove the indeterminacy of the logits under a common shift. Evaluating the objective and its gradient costs $O(|I^{\text{fit}}| K_b d_x)$ and requires no downstream optimization. We exclude candidates that do not meet the requirements for chart size, effective rank, distinctness, or samples per parameter, and denote the remaining candidate set by $\mathcal{G}_B^{\text{elig}}$. We solve the downstream problems only when evaluating the initialized model on the assessment observations of that fold. We aggregate the resulting operational losses across charts, giving each observation equal weight, and select the partition with the lowest aggregate loss. Let $\widehat{B}$ denote the selected number of charts. For the subsequent model fitting, all chart-specific quantities refer to that selected partition, with $K_b = d_b + 1$.

With the selected structure fixed, we reinitialize each local model using all training observations in its chart, indexed by I_b. We then jointly fit the scenario coefficient vectors and probability parameters to improve the operational decisions they induce. With a fixed nonnegative regularization parameter λ_u for the scenario coefficients and λ_Γ as defined above, we minimize the regularized empirical operational loss over parameters with feasible reconstructions,

$$\min_{\theta_b} \frac{1}{|I_b|} \sum_{i \in I_b} \ell(\widetilde{z}_b(x_i; \theta_b), \xi_i; x_i) + \frac{\lambda_u}{K_b} \sum_{k=1}^{K_b} \|u_{b,k}\|_2^2 + \lambda_\Gamma \|\Gamma_b\|_F^2. \tag{20}$$

In Eq. (20), the first term averages the operational losses of the induced decisions under the corresponding observed realizations. The remaining terms regularize the scenario coefficient vectors and context coefficients, respectively.

Write $\widehat{\theta}_b = (\widehat{u}_{b,1}, \dots, \widehat{u}_{b,K_b}, \widehat{a}_b, \widehat{\Gamma}_b)$ for the fitted parameters, and define $\widehat{p}_b(x) := p_b(x; \widehat{a}_b, \widehat{\Gamma}_b)$ and $\widehat{\xi}_{b,k} := g_b(\widehat{V}_b \widehat{u}_{b,k})$. We specify the training schedule and model selection rule for each experiment in Section EC.3.3 of the e-companion. We assign a new context x to chart $b = \tau_{\widehat{B}}(x)$ and solve Eq. (18) using the local scenario distribution defined by its fitted probabilities and scenarios.

5. Theoretical Guarantees

We analyze the structural choices in Section 4. First, we quantify how the number of charts B affects the tradeoff between localization and estimation in Section 5.1. For each chart, we then show in Section 5.2 when the singular-value retention rule in Eq. (15) selects the number of directions needed to represent the decision-relevant part of local gradient variation. We also establish when the resulting subspace can represent every gradient-relevant projection. In Section 5.3, we use these projections to explain the support size $K_b = d_b + 1$. Finally, in Section 5.4, we relate the approximation achieved by the fitted models to decision quality and operational risk.

For the finite-sample statements, let $\{(X_i, \Xi_i) : i \in I_{\mathrm{tr}}\}$ denote the random training observations; x_i and ξ_i denote their realized values used by the construction.

5.1. Chart localization

Increasing B permits finer localization but reduces the average number of observations available to each local model. We quantify this tradeoff through the error in the empirical cross-covariance matrix defined in Eq. (14). The following local representation identifies the ideal signal against which we measure this error and provides the projection structure used in the subsequent dimension and support results.

Fix a candidate partition with B charts and focus on chart b, whose context region is $\mathcal{X}_b := \{x \in \mathcal{X} : \tau_B(x) = b\}$. The maps T_b and g_b from Section 4 express realizations in standardized coordinates and decode those coordinates, respectively; in Section 4.4, we use $g_b(\widehat{V}_b u)$ to construct scenarios. The following assumptions identify the $d_b^\star$-dimensional part of $T_b(\Xi)$ that affects scenario gradients and bound the residual.

For each chart, let $\mathcal{Y}_b \subseteq \mathrm{dom}(g_b)$ denote the standardized reconstruction set on which we assume the generated affine-gradient representation and its bounds hold. The conditions below on the support envelope and the shared support ensure that the observed and constructed coordinates used in the analysis lie in this set.

Assumption 3 **(Affine representation)**. *There exist an integer* $1 \le d_b^\star \le \min\{d_\xi, d_z\}$, *an orthonormal matrix* $V_b^{\mathrm{rel}} \in \mathbb{R}^{d_\xi \times d_b^\star}$, *functions* $\beta_b : \mathcal{Z} \times (\mathcal{X}_b \cup \{\bar{x}_b\}) \to \mathbb{R}^{d_z}$ *and* $H_b : \mathcal{Z} \times (\mathcal{X}_b \cup \{\bar{x}_b\}) \to \mathbb{R}^{d_z \times d_b^\star}$, *and residual functions* $r_b : \mathcal{Z} \times \mathcal{U} \times (\mathcal{X}_b \cup \{\bar{x}_b\}) \to \mathbb{R}^{d_z}$ *and* $r_b^{\mathrm{gen}} : \mathcal{Z} \times \mathcal{Y}_b \times (\mathcal{X}_b \cup \{\bar{x}_b\}) \to \mathbb{R}^{d_z}$ *such that, for every* $z \in \mathcal{Z}$ *and* $x \in \mathcal{X}_b$,

$$\phi(z, \Xi; x) = \beta_b(z, x) + H_b(z, x)(V_b^{\mathrm{rel}})^\top T_b(\Xi) + r_b(z, \Xi, x), \tag{21}$$

$$\phi(z, g_b(y); x) = \beta_b(z, x) + H_b(z, x)(V_b^{\mathrm{rel}})^\top y + r_b^{\mathrm{gen}}(z, y, x), \qquad y \in \mathcal{Y}_b. \tag{22}$$

The same identities hold at $x = \bar{x}_b$. *The superscript* rel *marks the basis governing gradient-relevant variation;* gen *distinguishes the residual for decoded generated inputs from the residual* r_b *for observed realizations.*

ASSUMPTION 4 **(Uniform bounds)**. *For the representation in Assumption 3, there exist finite constants* $\overline{H}_b, \varepsilon_b^{\mathrm{aff}} \geq 0$. *For a matrix* A, *let* $\|A\|_{\mathrm{op}} := \sup_{\|v\|_2=1} \|Av\|_2$ *denote its Euclidean operator norm, equivalently its largest singular value. We require*

$$\sup_{z\in\mathcal{Z},\, x\in\mathcal{X}_b} \left\|H_b(z,x)\right\|_{\mathrm{op}} \leq \overline{H}_b. \tag{23}$$

The residuals satisfy

$$\begin{aligned} \left\|r_b(z,\xi,x)\right\|_2 &\leq \varepsilon_b^{\mathrm{aff}} \quad \text{for every } z\in\mathcal{Z},\ x\in\mathcal{X}_b,\ \text{and for } P_x\text{-almost every } \xi, \\ \left\|r_b^{\mathrm{gen}}(z,y,x)\right\|_2 &\leq \varepsilon_b^{\mathrm{aff}} \quad \text{for every } z\in\mathcal{Z},\ x\in\mathcal{X}_b,\ y\in\mathcal{Y}_b. \end{aligned} \tag{24}$$

The same bounds hold at $x = \bar{x}_b$.

Assumptions 3 and 4 state that the scenario gradient depends on a realization through $(V_b^{\mathrm{rel}})^\top T_b(\Xi)$, up to a bounded residual. The uniform bounds are stated in Eqs. (23)–(24). When the residual vanishes, matching the conditional mean of these projections matches the expected scenario gradient. In the subsequent results, we examine how to estimate a subspace that can represent these projections and how many scenarios are needed to represent their means.

We define $\mu_T(x) := \mathbb{E}[T(\Xi) \mid X = x]$ and $\Sigma_T(x) := \mathrm{Cov}(T(\Xi) \mid X = x)$, and assume that the required conditional moments are defined for every $x \in \mathcal{X}_b$ and at $\bar{x}_b$. The covariance of the standardized transformed realization at the chart centroid is $\Sigma_b := \widehat{D}_b^{-1}\Sigma_T(\bar{x}_b)\widehat{D}_b^{-1}$. The ideal affine-gradient signal is $C_b^\star := \Sigma_b V_b^{\mathrm{rel}} H_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b)^\top \in \mathbb{R}^{d_\xi \times d_z}$.

Let $\varepsilon_b^C := \|\widehat{C}_b - C_b^\star\|_{\mathrm{op}}$ denote the error in estimating the ideal signal. It includes sampling variation, differences among conditional distributions within a chart, and the affine approximation residual. Let $\mathcal{F}_{X,\mathrm{tr}}$ denote the sigma-algebra generated by the training contexts $\{X_i : i \in I_{\mathrm{tr}}\}$.

For the finite-sample covariance bounds, we use the following assumption.

ASSUMPTION 5 **(Finite-sample covariance conditions)**. *For a candidate partition with* B *charts, suppose the following conditions hold conditional on* $\mathcal{F}_{X,\mathrm{tr}}$.

(i) the training realizations are independent with laws P_{X_i};

(ii) Assumptions 3 and 4 hold almost surely for the fitted means, scales, and feasible local SAA decisions, with fixed dimensions $d_b^\star$, *common bounds* $\overline{H}_b$, *and* $\varepsilon_b^{\mathrm{aff}} = 0$;

(iii) every chart has a support envelope $\mathcal{U}_b$, *determined by the training contexts, with* $P_x(\mathcal{U}_b) = 1$ *at its training contexts and centroid, and every* $\xi \in \mathcal{U}_b$ *satisfies* $\|T(\xi)\|_2 \leq R_T$, $g(T(\xi)) = \xi$, *and* $T_b(\xi) \in \mathcal{Y}_b$;

(iv) finite nonnegative constants A_{sc}, R_T, L_Σ, *and* L_μ, *fixed conditional on* $\mathcal{F}_{X,\mathrm{tr}}$, *satisfy* $\overline{H}_b\|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2 \leq A_{\mathrm{sc}}$ *almost surely for every chart, and* $\|\Sigma_T(x) - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}} \leq L_\Sigma\|x - \bar{x}_b\|_x$ *and* $\|\mu_T(x) - \mu_T(\bar{x}_b)\|_2 \leq L_\mu\|x - \bar{x}_b\|_x$ *for every* $x \in \mathcal{X}_b$.

THEOREM 1 **(Tradeoff between localization and sampling)**. *Fix* $\delta \in (0,1)$. *Under Assumption 5, if* $|I_b| \geq \max\{2, \log\{4(d_\xi+1)B/\delta\}\}$ *for every chart, then, conditional on* $\mathcal{F}_{X,\mathrm{tr}}$, *with probability at least* $1-\delta$,

$$\sum_{b=1}^{B}\frac{|I_b|}{|I_{\mathrm{tr}}|}\varepsilon_b^C \le A_{\mathrm{sc}}\left\{\underbrace{L_\Sigma\sqrt{\widehat{J}_B}}_{\substack{\textit{within-chart}\\ \textit{covariance variation}}} + \underbrace{L_\mu^2\widehat{J}_B}_{\substack{\textit{within-chart}\\ \textit{mean variation}}} + \underbrace{8R_T^2\sqrt{\frac{B\log\{4(d_\xi+1)B/\delta\}}{|I_{\mathrm{tr}}|}}}_{\substack{\textit{fewer observations}\\ \textit{per chart}}}\right\}. \tag{25}$$

Proof. See Section EC.1.3 of the e-companion. □

Theorem 1 explains why we compare candidate numbers of charts in Section 4.5. In Eq. (25), the terms involving $\widehat{J}_B$ measure the effect of pooling different contexts in a chart. Finer partitions can reduce these terms. The sampling term grows at a rate of $\sqrt{B/|I_{\mathrm{tr}}|}$, up to logarithmic factors, as the average number of observations per chart decreases. The bound therefore identifies the competing effects that motivate selecting B using the operational loss on internal assessment observations.

5.2. Gradient-relevant projections

For a fixed partition, we study the choice of local scenario subspace in the next two results. Theorem 2 establishes when the singular-value retention rule selects the structural dimension $d_b^\star$. Theorem 3 then shows when a subspace of this dimension can represent every projection used in the affine gradient representation.

Assumption 6 **(Signal nondegeneracy)**. *The matrix $H_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b)$ has column rank $d_b^\star$. The matrix $(V_b^{\mathrm{rel}})^\top \Sigma_b V_b^{\mathrm{rel}}$ is positive definite.*

Under Assumption 6, $C_b^\star$ has rank $d_b^\star$. To compare the selected dimension with this structural dimension, we define the ideal retained fraction

$$\alpha_b^\star(d) := \frac{\sum_{j=1}^{d}\sigma_j(C_b^\star)^2}{\|C_b^\star\|_F^2}, \qquad d = 0,\dots,d_b^\star, \tag{26}$$

where an empty sum is zero. The fraction in Eq. (26) is the ideal counterpart of the empirical retained fraction in Eq. (15).

Theorem 2 **(Spectral dimension recovery)**. *Fix $\delta \in (0,1)$. Suppose Assumption 6 holds and $\alpha_b^\star(d_b^\star - 1) < \alpha_{\mathrm{ret}} < 1$. Let a nonnegative $\bar{\varepsilon}_b^C(\delta)$ satisfy*

$$\mathbb{P}\left(\varepsilon_b^C \le \bar{\varepsilon}_b^C(\delta) \,\middle|\, \mathcal{F}_{X,\mathrm{tr}}\right) \ge 1-\delta. \tag{27}$$

Define

$$\varepsilon_b^{\mathrm{spec}}(\delta) := \frac{\sqrt{\min(d_\xi, d_z)}\,\bar{\varepsilon}_b^C(\delta)}{\|C_b^\star\|_F}\left\{2 + \frac{\sqrt{\min(d_\xi, d_z)}\,\bar{\varepsilon}_b^C(\delta)}{\|C_b^\star\|_F}\right\}. \tag{28}$$

Let $\mathcal{E}_b^{\mathrm{ret}}(\delta)$ be the event on which $\varepsilon_b^{\mathrm{spec}}(\delta) < 1$ and

$$\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)} < \min\{\alpha_{\mathrm{ret}} - \alpha_b^\star(d_b^\star - 1), 1 - \alpha_{\mathrm{ret}}\}. \tag{29}$$

Then $\mathbb{P}(\mathcal{E}_b^{\mathrm{ret}}(\delta) \cap \{d_b \neq d_b^\star\} \mid \mathcal{F}_{X,\mathrm{tr}}) \le \delta$.

Proof. See Section EC.1.5 of the e-companion. □

Theorem 2 explains when the rule in Eq. (15) selects $d_b^\star$. The event $\mathcal{E}_b^{\mathrm{ret}}(\delta)$ requires the error bound for the retained fraction in Eq. (28) to satisfy the separation condition in Eq. (29). When this event and $\varepsilon_b^C \le \bar{\varepsilon}_b^C(\delta)$ both hold, the empirical fraction remains below α_{ret} through $d_b^\star - 1$ directions and exceeds it at $d_b^\star$. The rule therefore selects $d_b = d_b^\star$. Conditional on the training contexts, the probability that the separation event holds but the selected dimension is incorrect is at most δ. Under Assumption 5, Lemma EC.2 gives an explicit error bound satisfying Eq. (27).

We next examine whether the selected subspace can represent the gradient-relevant projections. For a truncation dimension d, write $\widehat{V}_{b,d} \in \mathbb{R}^{d_\xi \times d}$ for the first d left singular vectors of $\widehat{C}_b$, so $\widehat{V}_b = \widehat{V}_{b,d_b}$. The ideal signal has column space range$(\Sigma_b V_b^{\mathrm{rel}})$. Although this space need not equal range(V_b^{rel}), every orthonormal basis $V_b^\star \in \mathbb{R}^{d_\xi \times d_b^\star}$ for this signal space yields a nonsingular matrix $(V_b^{\mathrm{rel}})^\top V_b^\star$. Thus, V_b^{rel} describes gradient dependence, $V_b^\star$ spans the ideal covariance signal, and $\widehat{V}_{b,d}$ is estimated from data. We fix $V_b^\star$ and define $\kappa_b := \|((V_b^{\mathrm{rel}})^\top V_b^\star)^{-1}\|_{\mathrm{op}}$, which measures the conditioning of this representation. Let $\sigma_b := \sigma_{d_b^\star}(C_b^\star)$ denote the smallest nonzero singular value of the ideal signal. Whenever $\varepsilon_b^C < \sigma_b$, define $\varepsilon_b^V := \sqrt{2}\varepsilon_b^C/(\sigma_b - \varepsilon_b^C)$.

THEOREM 3 **(Representation of relevant projections)**. *Suppose Assumptions 3, 4, and 6 hold. If $\varepsilon_b^C < \sigma_b$ and $\kappa_b \varepsilon_b^V < 1$, then every $u^{\mathrm{rel}} \in \mathbb{R}^{d_b^\star}$ can be represented in the empirical scenario subspace using the coefficient vector*

$$u = \left((V_b^{\mathrm{rel}})^\top \widehat{V}_{b,d_b^\star}\right)^{-1} u^{\mathrm{rel}}, \qquad \|u\|_2 \le \frac{\kappa_b}{1 - \kappa_b \varepsilon_b^V} \|u^{\mathrm{rel}}\|_2. \tag{30}$$

Proof. See Section EC.1.7 of the e-companion. □

Theorem 3 establishes the representation property needed for scenario generation. For every gradient-relevant projection vector u^{rel}, it constructs a coefficient vector u that generates a standardized point with gradient-relevant projection u^{rel}. Thus, under the stated signal and conditioning requirements, a subspace of dimension $d_b^\star$ can represent all projection vectors entering the affine gradient model. Eq. (30) bounds the norm of the required coefficient vector. Together with Theorem 2, this result links the dimension selected from data to the representation needed for the shared-support construction below.

5.3. Shared scenario supports

Once the gradient-relevant projections can be represented in the estimated basis, we use the geometry of their conditional means to construct a small shared support. We define $\mu_b^{\mathrm{rel}}(x) := \mathbb{E}[(V_b^{\mathrm{rel}})^\top T_b(\Xi) \mid X = x]$ and $\mathcal{M}_b := \{\mu_b^{\mathrm{rel}}(x) : x \in \mathcal{X}_b\}$, treating the fitted center and scale as fixed and assuming that these conditional moments are defined. A shared support can represent all these means exactly only if the convex hull of its scenario projections contains $\mathcal{M}_b$. The following assumption provides this containment with $d_b^\star + 1$ vertices.

ASSUMPTION 7 **(Simplex containment)**. *There exist $d_b^\star + 1$ vertices $u_{b,1}^{\mathrm{rel}}, \ldots, u_{b,d_b^\star+1}^{\mathrm{rel}} \in \mathbb{R}^{d_b^\star}$ whose convex hull contains $\mathcal{M}_b$.*

With the shared vertices fixed, approximation accuracy also depends on how the probabilities vary with context. For the affine-softmax probability model in Eq. (17), we define the conditional mean approximation error

$$\varepsilon_b^{\mathrm{mean}} := \inf_{\substack{a_b\in\mathbb{R}^{d_b^\star+1},\\ \Gamma_b\in\mathbb{R}^{(d_b^\star+1)\times d_x}}} \sup_{x\in\mathcal{X}_b} \left\| \mu_b^{\mathrm{rel}}(x) - \sum_{k=1}^{d_b^\star+1} p_{b,k}(x;a_b,\Gamma_b) u_{b,k}^{\mathrm{rel}} \right\|_2 . \tag{31}$$

The error in Eq. (31) measures how closely the affine-softmax probabilities on these vertices can reproduce the conditional mean of the gradient-relevant projections uniformly over the chart. Theorem 4 uses feasible representations of the $d_b^\star+1$ vertices to construct a shared support for the local models in Section 4.4. It bounds the errors in the first-order mapping, decision, and regularized objective due to the affine approximation and the probability model. We mark the scenario model and associated quantities constructed in this existence argument with a superscript $\circ$, distinguishing them from fitted models, numerical decisions, and other algorithmic outputs.

THEOREM 4 **(Shared-support sufficiency)**. *Suppose that the conditions of Theorem 3 and Assumptions 1, 2, and 7 hold. For $k=1,\dots,d_b^\star+1$, let $u_{b,k}^\circ$ be the coefficient vector in Eq. (30) for the vertex $u_{b,k}^{\mathrm{rel}}$. Assume $\widehat{V}_{b,d_b^\star} u_{b,k}^\circ \in \mathcal{Y}_b$ for every k. Set $\xi_{b,k}^\circ := g_b(\widehat{V}_{b,d_b^\star} u_{b,k}^\circ)$ and $\boldsymbol{\xi}_b^\circ := (\xi_{b,1}^\circ,\dots,\xi_{b,d_b^\star+1}^\circ)$. For $\varepsilon>0$, define $\varepsilon_b^\circ(\varepsilon) := \overline{H}_b(\varepsilon_b^{\mathrm{mean}}+\varepsilon)+2\varepsilon_b^{\mathrm{aff}}$.*

Then, for every $\varepsilon>0$, there exists an affine-softmax probability model p_b° on this fixed support, depending on ε, such that the induced regularized objective $R_b^\circ(z;x) := \sum_{k=1}^{d_b^\star+1} p_{b,k}^\circ(x)\ell(z,\xi_{b,k}^\circ;x) + \frac{\eta_{\mathrm{reg}}}{2}\|z-z_0(x)\|_2^2$ satisfies

$$\sup_{x\in\mathcal{X}_b,\, z\in\mathcal{Z}} \left\| \nabla_z R(z;x) - \nabla_z R_b^\circ(z;x) \right\|_2 \le \varepsilon_b^\circ(\varepsilon). \tag{32}$$

The resulting decision $z_b^\circ(x) := z(p_b^\circ(x),\boldsymbol{\xi}_b^\circ;x)$ also satisfies

$$\left\| z_b^\circ(x) - z^\star(x) \right\|_2 \le \frac{\varepsilon_b^\circ(\varepsilon)}{\eta}, \tag{33}$$

$$R(z_b^\circ(x);x) - R(z^\star(x);x) \le \frac{\{\varepsilon_b^\circ(\varepsilon)\}^2}{2\eta}, \qquad x\in\mathcal{X}_b. \tag{34}$$

Thus, when $d_b = d_b^\star$, the fixed shared support $\boldsymbol{\xi}_b^\circ$ of $K_b = d_b+1$ generated scenarios is sufficient: for every $\varepsilon>0$, there exists an affine-softmax probability model on this support whose induced objective and decisions satisfy these bounds uniformly over $\mathcal{X}_b$.

Proof. See Section EC.1.10 of the e-companion. □

When the relevant conditional means lie in a simplex with feasible vertices, one scenario at each vertex gives a shared support of size $d_b^\star+1$. Varying the probabilities then changes the represented mean across contexts. The affine-softmax approximation error and the affine gradient residual determine the first-order error in Eq. (32); Eqs. (33)–(34) convert this error into decision and regularized objective bounds.

The preceding results explain the structural choices in Section 4. The chart bound motivates comparing values of B using internal assessment. For each chart, spectral recovery identifies the relevant dimension, Theorem 3 establishes that the empirical subspace can represent every relevant projection, and shared-support capacity motivates $K_b = d_b + 1$ under the stated feasibility and approximation conditions. We now assess the decisions induced by the scenarios and probabilities obtained through fitting.

5.4. Quality of fitted models

Fix a fitted atlas with B charts; for the model selected in Section 4, $B = \widehat{B}$. We retain its dimensions d_b, support sizes K_b, fitted probabilities $\widehat{p}_b(x)$, scenarios $\widehat{\xi}_{b,1}, \ldots, \widehat{\xi}_{b,K_b}$, chart centers, and scale matrices. Let $\eta > 0$ be the common curvature bound for the complete objectives in Assumption 1; the fitted atlas uses the regularization coefficient η_{reg}. We denote the fitted objective by $\widehat{R}_b$ and its gradient by $\nabla_z \widehat{R}_b$. We distinguish its exact minimizer $\widehat{z}_b$ from the numerical decision $\widetilde{z}_b$ used in computation. Write $\widehat{\boldsymbol{\xi}}_b := (\widehat{\xi}_{b,1}, \ldots, \widehat{\xi}_{b,K_b})$. The fitted local objective is

$$\widehat{R}_b(z; x) := \sum_{k=1}^{K_b} \widehat{p}_{b,k}(x)\ell(z, \widehat{\xi}_{b,k}; x) + \frac{\eta_{\mathrm{reg}}}{2}\big\|z - z_0(x)\big\|_2^2. \tag{35}$$

The gradient and exact minimizer of the fitted objective in Eq. (35) are

$$\nabla_z \widehat{R}_b(z; x) = \nabla_z f(z; x) + \sum_{k=1}^{K_b} \widehat{p}_{b,k}(x)\phi(z, \widehat{\xi}_{b,k}; x) + \eta_{\mathrm{reg}}\{z - z_0(x)\}, \tag{36}$$

$$\begin{aligned} \widehat{z}_b(x) &:= \arg\min_{z \in \mathcal{Z}} \widehat{R}_b(z; x) \\ &= z(\widehat{p}_b(x), \widehat{\boldsymbol{\xi}}_b; x). \end{aligned} \tag{37}$$

We measure how accurately the fitted scenarios and probabilities reproduce the population first-order mapping at the chart centroid by

$$\varepsilon_b^{\mathrm{ctr}} := \sup_{z \in \mathcal{Z}} \big\|\nabla_z R(z; \bar{x}_b) - \nabla_z \widehat{R}_b(z; \bar{x}_b)\big\|_2. \tag{38}$$

The following regularity condition controls how both mappings vary from the centroid to other contexts assigned to the chart.

Assumption 8 **(Context regularity)**. *There exists a common finite nonnegative constant L_{pop} and, for each chart b, a finite nonnegative constant L_b such that, for all $z \in \mathcal{Z}$ and all contexts assigned to the chart, $\big\|\nabla_z R(z; x) - \nabla_z R(z; \bar{x}_b)\big\|_2 \le L_{\mathrm{pop}}\|x - \bar{x}_b\|_x$ and $\big\|\nabla_z \widehat{R}_b(z; x) - \nabla_z \widehat{R}_b(z; \bar{x}_b)\big\|_2 \le L_b\|x - \bar{x}_b\|_x$.*

Under this regularity condition, the first-order approximation error at a context is bounded by the centroid error plus the variation of both mappings from the centroid. Define

$$\varepsilon_b(x) := \varepsilon_b^{\mathrm{ctr}} + (L_{\mathrm{pop}} + L_b)\|x - \bar{x}_b\|_x. \tag{39}$$

THEOREM 5 **(Local decision quality)**. *Suppose that Assumptions 1, 2, and 8 hold. If* $\tau_B(x) = b$, *then, with* $\varepsilon_b(x)$ *defined in Eq. (39),*

$$\left\|\widehat{z}_b(x) - z^\star(x)\right\|_2 \le \frac{\varepsilon_b(x)}{\eta}, \tag{40}$$

$$R(\widehat{z}_b(x); x) - R(z^\star(x); x) \le \frac{\varepsilon_b(x)^2}{2\eta}. \tag{41}$$

Proof. See Section EC.1.11 of the e-companion. □

Theorem 5 explains how approximation error in the fitted scenario model affects the induced decision. The centroid error measures how well the local model matches the expected gradient at the chart center. Context regularity controls the additional error away from that center. Eqs. (40)–(41) show how strong convexity converts the sum of these errors into bounds on the decision and its regularized objective. Corollary EC.1 relates the average of these bounds over the training observations to the K-means distortion.

For the following risk bound, we condition on the data used to fit and select the atlas and treat the fitted partition, centroids, scenarios, and probability models as fixed. All probabilities and expectations below are taken over an independent deployment context X. The decision of the fixed atlas is $\widehat{z}(x) := \widehat{z}_{\tau_B(x)}(x)$.

The population distortion and root mean squared centroid error are $J_B := \mathbb{E}[\|X - \bar{x}_{\tau_B(X)}\|_x^2]$ and $\varepsilon_B^{\mathrm{ctr}} := \{\mathbb{E}[(\varepsilon^{\mathrm{ctr}}_{\tau_B(X)})^2]\}^{1/2}$. Write $L_B := \max_b(L_{\mathrm{pop}} + L_b)$ for the largest combined constant for context regularity. These quantities summarize localization, centroid error, and context variation over the fixed atlas.

For comparison with the unregularized operational optimum, let $R_0^\star(x) := \inf_{z\in\mathcal{Z}} R_0(z; x)$. The feasible radius relative to the regularization center is $\mathrm{rad}_{\mathcal{Z},0} := \sup_{x\in\mathcal{X},\, z\in\mathcal{Z}} \left\|z - z_0(x)\right\|_2$.

THEOREM 6 **(Atlas risk)**. *Suppose that the conditions of Theorem 5 hold and* $\mathrm{rad}_{\mathcal{Z},0} < \infty$, $J_B < \infty$, *and* $\varepsilon_B^{\mathrm{ctr}} < \infty$. *Assume that* $R_0^\star(X)$, $R_0(\widehat{z}(X); X)$, *and* $\varepsilon_{\tau_B(X)}(X)$ *are measurable, that the first two random variables are integrable, and that the third has finite expectation. Then the exact finite-support decision* $\widehat{z}(X)$ *of the fixed atlas satisfies*

$$\mathbb{E}[R_0(\widehat{z}(X); X)] - \mathbb{E}[R_0^\star(X)] \le \frac{\left(\varepsilon_B^{\mathrm{ctr}} + L_B\sqrt{J_B}\right)^2}{2\eta} + \frac{\eta_{\mathrm{reg}}\,\mathrm{rad}_{\mathcal{Z},0}^2}{2}. \tag{42}$$

Proof. See Section EC.1.13 of the e-companion. □

Theorem 6 bounds the fitted atlas's expected excess operational risk for a new context relative to the unregularized optimum. In Eq. (42), the first term combines the approximation error of the local models with their variation across each chart, and the second accounts for regularization bias. The bound relates the quality of the fitted scenarios and probabilities, together with the localization achieved by the partition, to expected operational loss.

6. Computational Study

We examine the decision quality and computational cost of compact local scenario models, beginning with the decision problems in Section 6.1 and the experimental design and comparisons in Section 6.2. We present the application results in Section 6.3. We then use mechanism studies in Section 6.4 to examine when adding context charts ceases to improve decision quality and in Section 6.5 to examine whether a basis constructed from gradient information retains directions relevant to decisions in the presence of irrelevant variation.

6.1. Decision problems

We consider three applications in capacity allocation, portfolio optimization, and network reservation. We introduce the common notation and evaluation conventions before describing each model.

In each application, a fitted method maps an observed context to a finite scenario distribution. Let K_{ref} denote the number of scenarios in the full reference catalog $\{\xi^{(r)} : r = 1, \dots, K_{\mathrm{ref}}\} \subseteq \mathcal{U}$. For each application, let $z_{0,\mathrm{app}}(x)$ denote the prescribed regularization center and $\eta_{\mathrm{reg,app}} \geq 0$ the regularization coefficient. We obtain a feasible decision by solving the corresponding regularized stochastic program. For a finite distribution represented by $\{(p_k, \xi_k)\}_{k=1}^{K}$, the downstream objective is

$$\sum_{k=1}^{K} p_k \ell(z, \xi_k; x) + \frac{\eta_{\mathrm{reg,app}}}{2} \|z - z_{0,\mathrm{app}}(x)\|_2^2. \tag{43}$$

Here, $\ell(z, \xi; x)$ is the operational loss for the application, incurred by decision z under realization ξ and context x. The downstream solve uses the objective in Eq. (43). The coefficient $\eta_{\mathrm{reg,app}}$ supplies curvature missing from the operational loss. We evaluate operational risk using ℓ alone. Because the conditional distribution is known in each experiment, we compute the decision that minimizes unregularized operational risk under that distribution and use its risk as the reference for evaluating the true raw and relative excess risks. All methods use the same evaluator for operational loss and the same feasible decision set, with different finite distributions defining their stochastic programs.

Smooth capacity allocation (CAP). The CAP problem allocates total capacity $C > 0$ across d_z resources that jointly serve d_ξ demand classes. We adapt the model of demand uncertainty across multiple resources from Harrison and Van Mieghem (1999) to smooth capacity allocation. Let $A \in \mathbb{R}_+^{d_\xi \times d_z}$ have row sums equal to one and define $\mathcal{Z}_{\mathrm{CAP}}(C) := \{z \in \mathbb{R}_+^{d_z} : \mathbf{1}^\top z = C\}$. With $h_\tau(a) := \tau \log\{1 + \exp(a/\tau)\}$ and overage and underage cost coefficients $c_j^o, c_j^u > 0$, the operational loss under demand realization $\xi \in (0, \infty)^{d_\xi}$ is

$$\ell_{\mathrm{CAP}}(z, \xi) := \frac{1}{d_\xi} \sum_{j=1}^{d_\xi} \left[c_j^o h_\tau((Az)_j - \xi_j) + c_j^u h_\tau(\xi_j - (Az)_j) \right]. \tag{44}$$

In CAP, $d_z < d_\xi$, so the uncertain demand vector has more components than the capacity allocation decision. CAP therefore tests whether compact local scenario models remain effective when the uncertainty dimension exceeds the decision dimension. We specify the parameter values and contextual demand model with four states in Section EC.2.1 of the e-companion.

Smoothed CVaR portfolio optimization (PORT). We use a smoothed CVaR portfolio formulation following Rockafellar and Uryasev (2000). The PORT problem selects portfolio weights $w \in \mathbb{R}^{d_\xi}$ and a threshold $t \in \mathbb{R}$, subject to $\mathbf{1}^\top w = 1$. For a return realization $\xi \in (-r_{\max}, r_{\max})^{d_\xi}$, tail probability $\alpha \in (0,1)$, coefficient $\lambda_\mu \geq 0$ for expected return, and leverage coefficient $\lambda_w \geq 0$, the operational loss is

$$\ell_{\mathrm{PORT}}((w,t),\xi) := -\lambda_\mu \xi^\top w + t + \frac{1}{\alpha} h_\tau(-\xi^\top w - t) + \frac{\lambda_w}{2}\|w\|_2^2. \tag{45}$$

Although $(w,t) \in \mathbb{R}^{d_\xi+1}$, the equality constraint $\mathbf{1}^\top w = 1$ removes one degree of freedom. Thus, the feasible decision set has dimension d_ξ, matching that of the uncertain parameter. PORT therefore tests KKT Atlas when the decision and uncertainty dimensions match. We specify the bounded factor model, including a market-wide crash state that jointly reduces asset returns, and the numerical parameters in Section EC.2.2 of the e-companion.

Multicommodity network reservation (NET). The NET problem models advance capacity reservation under uncertain demand between origins and destinations (Riis and Andersen 2002). The decision $z \in \mathcal{Z}_{\mathrm{NET}} = [0, 1.5]^{48}$, made before demand is observed, reserves link capacity. For a demand realization $\xi \in (0,\infty)^{d_\xi}$, each of the d_ξ commodities can use two fixed paths. Let $P^{(1)}, P^{(2)} \in \{0,1\}^{d_z \times d_\xi}$ denote the link incidence matrices for these paths, and let $q \in \mathbb{R}^{d_\xi}$ denote the routing vector, where $q_j \in [0, \xi_j]$ is the amount of commodity j routed on its first path. The resulting link load is $L(q,\xi) := P^{(2)}\xi + \{P^{(1)} - P^{(2)}\}q$. For reservation costs $c \in \mathbb{R}^{d_z}_+$, congestion coefficient $\lambda > 0$, and routing coefficient $\rho > 0$, the operational loss is

$$\ell_{\mathrm{NET}}(z,\xi) := c^\top z + \min_{0 \leq q \leq \xi} \left\{ \lambda \left\| \max\{L(q,\xi) - z, \mathbf{0}\} \right\|_2^2 + \rho \left\| q - \xi/2 \right\|_2^2 \right\}. \tag{46}$$

In NET, d_z remains fixed while d_ξ determines the number of commodities and routing recourse variables per scenario. NET therefore tests whether compact local scenario supports retain decision quality while reducing the routing recourse work required in downstream solves. We specify the network, demand model, path construction, and numerical parameters in Section EC.2.3 of the e-companion.

In Section EC.2 of the e-companion, we give additional model and data generation details, application-specific parameter settings, and the CAP strong convexity proof.

We compare KKT Atlas with three fitted benchmarks. ICEO-A adapts integrated conditional estimation and optimization (Qi et al. 2026) to the exact objective and predicts probabilities on all K_{ref} reference scenarios. A stochastic optimization forest (SOF) assigns weights to training observations using tree splits designed around the optimization objective (Kallus and Mao 2023). A k-nearest neighbor method (k-NN) selected by validation forms a local empirical SAA from nearby observations (Hastie et al. 2009). For a context assigned to chart b, KKT Atlas instead predicts probabilities on K_b learned scenarios, each represented by d_b basis coefficients before reconstruction in the original realization space.

6.2. Experimental design and comparisons

We first compare Atlas with the three benchmarks in CAP, PORT, and NET using three paired seeds. These comparisons assess decision quality. We then use a CAP support-size study and two controlled quadratic experiments to examine the mechanisms behind compact support construction, chart localization, and the cross-covariance basis. In addition to decision quality, we measure model construction and deployment time.

KKT Atlas selects $\widehat{B} \in \{1, 2, 4\}$ by assessment loss, determines the chart dimensions d_b using Eq. (15), and then sets $K_b = d_b + 1$. Table 1 summarizes the main application settings. Each application has a known finite conditional distribution, and the training and validation samples contain one realization per context. The main comparison uses three paired development seeds and 12,800 training observations, a sample size fixed during the preceding development study.

We evaluate decision quality on 200 additional contexts. For the main comparisons, we compute the exact conditional expected operational loss at each context by summing the losses over the full reference scenario catalog, using the known conditional probabilities as weights. When a setting requires simulation, we instead use 50 conditional draws per context and evaluate every method on the same draws. For these simulations, we estimate operational risk using the Monte Carlo estimator defined in Section EC.3.1. Relative excess is the percentage increase in operational risk over the optimum under the known conditional law.

We report the selected Atlas structure together with construction and deployment time. Construction time covers structure or hyperparameter selection and model fitting. Deployment time includes distribution prediction and computation of the downstream decision. Section EC.3.4 of the e-companion provides detailed definitions of these quality measures and of the reported construction and deployment times.

Atlas and ICEO-A are implemented in PyTorch 2.12.1 (Paszke et al. 2019). For the main comparisons, we train both with Adam (Kingma and Ba 2015) for 1,600 updates, using a learning rate of 0.001, a minibatch size of 64, and gradient-norm clipping at 10. We select the final model only from the saved checkpoints $\{0, 400, 800, 1200, 1600\}$ by minimizing operational loss on the validation sample; checkpoint 0 denotes the initialized model. CAP uses SLSQP in SciPy 1.18.0 (Virtanen et al. 2020), PORT uses Clarabel 0.11.1 (Goulart and Chen 2026), and NET uses Gurobi 13.0.2 (Gurobi Optimization, LLC 2026).

The experiments are run on a Windows server with AMD EPYC 9654 processors running at 2.4 GHz, 192 physical cores, and approximately 1.5 TiB of memory. Each construction run is allocated 32 distinct physical cores, with the same core group used by all methods for a given seed. At most 32 workers run concurrently, and actual core use can be lower in individual phases. Solvers and numerical libraries use one thread per worker. Atlas and ICEO-A parallelize independent decision solves and implicit derivative calculations across contexts, collecting the results before each sequential Adam update. For Atlas, this parallel work is performed within each chart. For k-NN, we assess candidate neighborhood sizes sequentially and parallelize decision solves across validation

contexts for each candidate. SOF divides its 100 trees as evenly as possible among 32 workers for parallel training and also evaluates validation contexts in parallel. We measure deployment times with 1 and 32 physical cores (Table 5). With 32 cores, all four methods distribute a batch of 128 contexts across 32 workers in groups of four. Section EC.3 of the e-companion gives the remaining training settings.

Table 1 Core application dimensions and settings.

Study	K_{ref}	d_ξ	d_z	η_{reg}	$\lvert I_{\text{tr}}\rvert$
CAP	500	64	8	0	12,800
PORT	500	32	33	10^{-6}	12,800
NET	500	16	48	10^{-6}	12,800

6.3. Results across applications

We first compare decision quality across the four methods and then examine the selected scenario supports and the time required for construction and deployment. We specify the experimental settings and paired development seeds in Section EC.3 of the e-companion.

6.3.1. Decision quality Table 2 compares the four methods at the same training sample size $|I_{\text{tr}}| = 12{,}800$, with results averaged over three development seeds. Atlas has the lowest mean operational risk in both CAP and PORT. Its mean relative excess over the optimum under the known conditional distribution is 0.09% in CAP and 2.88% in PORT, compared with 0.38% and 3.08%, respectively, for ICEO-A. SOF and k-NN have larger relative excesses in both applications.

In NET, the mean relative excess is 0.12% for Atlas and 0.09% for ICEO-A. ICEO-A has slightly lower mean operational risk, but the difference is only 0.03% of its mean risk. Atlas has lower risk than SOF and k-NN in all three seeds. Tables 4 and 5 show that Atlas also requires less construction time than ICEO-A and substantially less time to deploy a batch of decisions.

One possible explanation for why neither method uniformly achieves lower mean operational risk is a tradeoff between approximation error and fitting from finite data with a fixed number of updates. Atlas fits substantially fewer parameters, which may ease optimization over the prescribed Adam iterations and reduce estimation error. ICEO-A avoids the additional subspace and support reduction and may benefit from avoiding this approximation when fitted well, although its larger number of parameters alone does not ensure better fitting.

6.3.2. Selected local structures and model sizes The selected models achieve these risk levels with small local scenario supports (Table 3). CAP uses four scenarios per chart in a three-dimensional representation, with one or two charts. PORT uses one chart with 14–18 scenarios in 13–17 dimensions. NET uses two or four charts, each with 10–11 scenarios. Relative to the 500-scenario reference catalog, the number of scenarios needed for a query is reduced by 99.2% in CAP, 96.4%–97.2% in PORT, and 97.8%–98.0% in NET. Atlas learns the scenario locations in the reconstructed realization space.

Table 2 Decision quality at 12,800 training observations.

	CAP		PORT		NET	
Method	Risk ↓	Excess /% ↓	Risk ↓	Excess /% ↓	Risk ↓	Excess /% ↓
Atlas	**1.5799**	**0.09**	**0.0375**	**2.88**	1.7253	0.12
ICEO-A	1.5845	0.38	0.0376	3.08	**1.7247**	**0.09**
SOF	1.6134	2.22	0.0406	11.19	1.7384	0.87
k-NN	1.6073	1.83	0.0385	5.46	1.7343	0.65

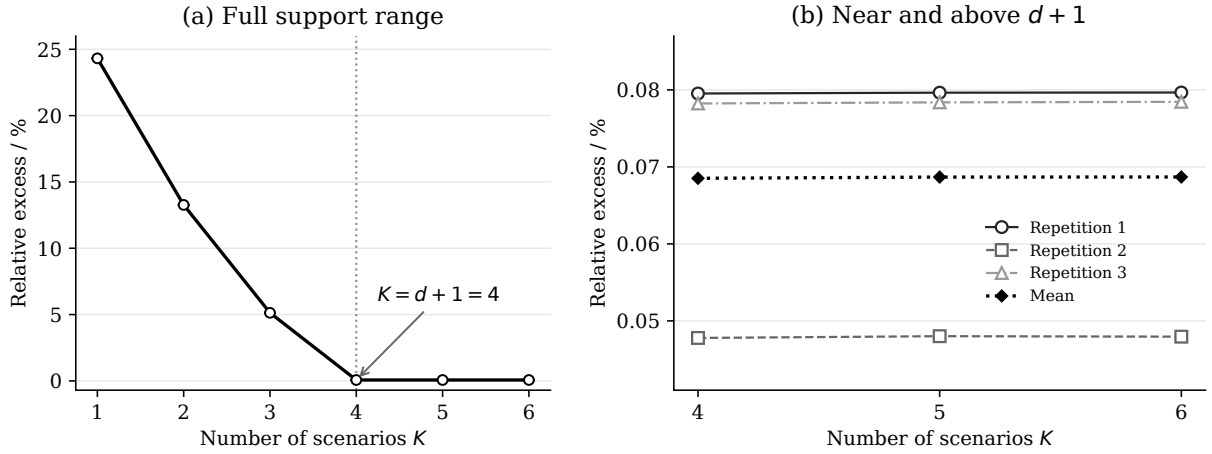
Table 3 Selected Atlas structures and scenario reduction.

Application	Seed 260910+	$\widehat{B}$	Dimensions d_b	Scenarios K_b	Reduction /%
CAP	1	1	3	4	99.2
	2	2	3, 3	4, 4	99.2
	3	1	3	4	99.2
PORT	1	1	13	14	97.2
	2	1	13	14	97.2
	3	1	17	18	96.4
NET	1	4	9, 9, 9, 9	10, 10, 10, 10	98.0
	2	2	9, 10	10, 11	97.8
	3	4	10, 10, 10, 10	11, 11, 11, 11	97.8

6.3.3. Support construction and sensitivity analyses To examine how many local scenarios are needed for good decisions, we vary K in CAP while fixing the number of charts at one and the representation dimension at three. We fit models with $K = 1, \ldots, 6$ on the same data set, using three paired training repetitions for each support size. Figure 1 shows that mean relative excess falls from 24.32% at $K = 1$ to 13.27% at $K = 2$, 5.13% at $K = 3$, and approximately 0.07% at $K = 4, 5, 6$. The plateau begins at $K = d + 1 = 4$, supporting the choice motivated by Theorem 4.

6.3.4. Construction and deployment costs Table 4 reports mean construction wall times, averaged over three seeds, of 0.043 versus 0.190 hours for Atlas and ICEO-A in CAP, 0.029 versus 0.146 hours in PORT, and 0.395 versus 8.941 hours in NET. The corresponding wall-time reductions are 77.1%, 80.0%, and 95.6%, respectively. Both methods repeatedly solve stochastic programs during fitting, but Atlas uses smaller programs defined by its compact supports. Compared with SOF and k-NN, Atlas needs to select charts, estimate bases, and jointly fit scenario locations and probabilities. In contrast, SOF uses approximate-risk scores to compare candidate splits without solving a new stochastic program for each split, and k-NN constructs neighbor weights directly without iterative parameter fitting. Although these weighting methods usually have lower construction costs than Atlas, Atlas achieves much lower mean operational risk than either method in all three applications.

Figure 1 Decision quality versus local support size in CAP.



Notes. The experiment fixes $B = 1$ and $d = 3$. The left plot shows the mean and range over three paired training repetitions; the right plot shows the individual repetitions near $K = d + 1 = 4$.

Table 4 Mean model construction time.

	CAP		PORT		NET	
Method	Wall /h ↓	CPU /h ↓	Wall /h ↓	CPU /h ↓	Wall /h ↓	CPU /h ↓
Atlas	0.043	0.28	0.029	0.11	0.395	2.82
ICEO-A	0.190	3.87	0.146	2.24	8.941	133.39
SOF	0.034	0.75	0.023	0.46	0.806	18.82
k-NN	0.014	0.19	0.013	0.12	0.183	4.27

Table 5 reports deployment wall times, measured from predicting a scenario distribution to returning the decision. Atlas is the fastest method in every application under both core allocations. For a batch of 128 contexts on one core, Atlas takes 0.75, 0.33, and 5.52 seconds in CAP, PORT, and NET, compared with 10.94, 6.17, and 259.88 seconds for ICEO-A. With 32 cores, Atlas is 12.5–61.0 times faster than ICEO-A. The advantage is largest in NET, where reducing the number of scenarios also reduces the size of the recourse problem. For a batch of 128 contexts, using 32 cores reduces the deployment time of Atlas by factors of 25.55, 21.50, and 28.70 in CAP, PORT, and NET, respectively.

6.4. Effects of additional context charts

We use the controlled quadratic decision problem in Eq. (47) because it makes the geometry behind chart localization explicit. The construction holds the contexts and conditional probabilities fixed while rotating the scenario planes associated with the operating modes. The rotation varies the directions that a global scenario model must cover and rotates the optimal decision, while preserving its two-dimensional coordinates $\tanh(t)$ within the mode-specific plane. Let $R \in \{1, 2, 3, 4\}$ denote the operating mode, encoded by $c_R \in \mathbb{R}^6$. The vector $e \in \mathbb{R}^6$ represents a within-mode context

Table 5 Wall time for deployment.

	1 context, 1 core			128 contexts					
	Wall time /10^{-3} s ↓			Wall time /s ↓					
Method	CAP	PORT	NET	CAP		PORT		NET	
				1 core	32 cores	1 core	32 cores	1 core	32 cores
Atlas	**6.07**	**3.01**	**39.30**	**0.75**	**0.03**	**0.33**	**0.02**	**5.52**	**0.19**
ICEO-A	80.29	47.61	2880.62	10.94	0.36	6.17	0.22	259.88	11.74
SOF	32.49	21.36	315.95	4.61	0.17	2.81	0.12	78.13	3.53
k-NN	16.32	32.60	867.60	2.04	0.07	3.83	0.15	131.85	5.65

perturbation, and $t \in \mathbb{R}^2$ indexes the conditional probabilities. Let $V_{\mathrm{c}} \in \mathbb{R}^{10\times 2}$ be a shared basis, and let $V_r^{\mathrm{aux}} \in \mathbb{R}^{10\times 2}$ be mutually orthogonal mode-specific bases. We define

$$\begin{aligned} x &= (c_R + e, t), & V_r(\omega) &= \cos(\omega) V_{\mathrm{c}} + \sin(\omega) V_r^{\mathrm{aux}}, \\ \xi_{r,v} &= V_r(\omega) v, & p(v \mid t) &= \frac{\exp(v^\top t)}{\sum_{w\in\{-1,1\}^2} \exp(w^\top t)}, \qquad v \in \{-1,1\}^2, \\ \ell_{\mathrm{chart}}(z,\xi) &= \frac{1}{2}\|z-\xi\|_2^2, & z^\star(x) &= V_R(\omega)\tanh(t). \end{aligned} \tag{47}$$

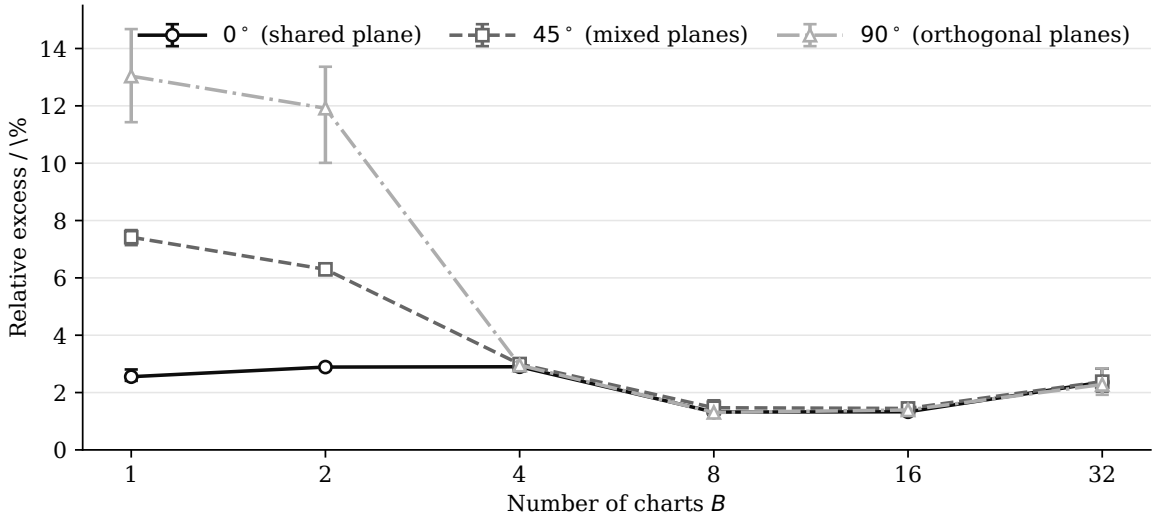
At 0°, all four modes share the plane spanned by V_{c}. At 45°, each plane combines shared and mode-specific directions. At 90°, the four mode-specific planes are mutually orthogonal. Increasing the angle requires a compact global scenario model to cover more mode-specific directions. We test whether chart-specific supports can preserve decision quality with fewer scenarios per query.

For each angle and development seed, we fit a separate Atlas model for each fixed $B \in \{1,2,4,8,16,32\}$, using the same 12,800 training observations and 1,600-update budget. Each fixed-B model selects its local dimensions from the training observations and sets $K_b = d_b + 1$. We compare these models to assess how many charts are needed as the mode-specific scenario planes become more distinct. We give the full data-generating process in Section EC.3.5 of the e-companion.

Figure 2 shows that additional charts become more valuable as the mode-specific planes separate. With $B = 1$, mean relative excess across three seeds rises from 2.55% at 0° to 7.42% at 45° and 13.04% at 90°. With $B = 8$, the corresponding values are 1.32%, 1.47%, and 1.31%. The compact global model loses decision quality when the modes require different scenario directions, whereas a sufficiently localized model remains stable. Across all three angles, decision quality levels off between 8 and 16 charts and worsens at 32 charts. With the global batch size fixed at 64, the 32-chart models process fewer observations per chart in each update, so the deterioration at 32 charts can reflect both estimation and optimization effects.

Table 6 quantifies the tradeoff between decision quality and model size in the heterogeneous 45° and 90° settings. As B increases from 1 to 4, the fitted dimension per chart falls from 8 to 2, and the number of scenarios per decision falls from 9 to 3. Increasing B from 4 to 8 keeps the number of scenarios per decision at 3 and reduces mean relative excess from 3.00% to 1.47% at 45° and from 2.95% to 1.31% at 90°, while the total number of stored scenarios increases from 12 to 24. The

Figure 2 Decision quality as mode-specific scenario planes separate.



Notes. Points show means, and error bars show the range across three development seeds.

Table 6 Decision quality and scenario counts with heterogeneous scenario planes.

B	d_b	K_b	Scenarios per decision	Total stored scenarios	45° /%	90° /%
1	8	9	9	9	7.42	13.04
2	4	5	5	10	6.30	11.92
4	2	3	3	12	3.00	2.95
8	2	3	3	24	1.47	1.31

eight-chart model attains a mean relative excess of 1.39% with only three scenarios per decision. Chart localization therefore keeps the stochastic program small at each query while accommodating mode-specific scenario directions.

6.5. Cross-covariance basis under decision-irrelevant variation

We use a second controlled quadratic decision problem to test whether a basis retains directions that affect decisions when variation that does not affect the decision becomes large. The construction fixes the decision-relevant projection and conditional probabilities while varying only the scale of decision-irrelevant directions. It therefore separates variance from decision relevance. Let $u^{\mathrm{rel}} \in \mathbb{R}^4$ denote the decision-relevant scenario coordinate, and let $e^{\perp} \in \mathbb{R}^8$ denote a decision-irrelevant coordinate. Let $V_{\mathrm{rel}} \in \mathbb{R}^{200\times 4}$ and $V_{\mathrm{noise}} \in \mathbb{R}^{200\times 8}$ have mutually orthonormal columns. We define

$$\ell_{\mathrm{quad}}(z,\xi) = \frac{1}{2}\|z - V_{\mathrm{rel}}^{\top}\xi\|_2^2, \qquad \xi^{(\sigma)} = V_{\mathrm{rel}}u^{\mathrm{rel}} + \sigma V_{\mathrm{noise}}e^{\perp}, \qquad \sigma \in \{1/4, 4\}. \tag{48}$$

Because $V_{\mathrm{rel}}^{\top}V_{\mathrm{noise}} = 0$, we have $V_{\mathrm{rel}}^{\top}\xi^{(\sigma)} = u^{\mathrm{rel}}$, so the optimal decision also remains unchanged as σ varies. Section EC.2.4 of the e-companion gives the full scenario and context construction.

We compare the four-dimensional cross-covariance basis with a four-dimensional principal component analysis (PCA) basis while holding the contexts, probabilities, decision-relevant projections, data, and training procedure fixed. Both methods use $B = 4$, $d_b = 4$, $K_b = 5$, and $\eta_{\mathrm{reg}} = 0$. At evaluation contexts x_i, we use normalized squared decision error $\varepsilon^{\mathrm{norm}}(\widehat{z}) = \sum_i \|\widehat{z}(x_i) - z_i^{\mathrm{ref}}\|_2^2 / \sum_i \|z_i^{\mathrm{ref}} - \bar{z}^{\mathrm{ref}}\|_2^2$, where z_i^{ref} is the decision under the known conditional distribution and $\bar{z}^{\mathrm{ref}}$ is its mean over the evaluation contexts. For each chart, we restore the original component scales of the fitted basis and then orthonormalize its columns to obtain $\widehat{V}_b^{\mathrm{orig}}$. Its alignment with the decision-relevant subspace is $\rho_b := \sigma_{\min}(V_{\mathrm{rel}}^\top \widehat{V}_b^{\mathrm{orig}})$, which equals one for coincident subspaces and approaches zero when the fitted basis misses a decision-relevant direction. We aggregate the alignments across the four charts using weights proportional to their training sample sizes, giving $\bar{\rho} = \sum_{b=1}^4 |I_b|\rho_b / |I_{\mathrm{tr}}|$, and report the mean of $\bar{\rho}$ across the three paired seeds.

Table 7 Cross-covariance basis and PCA under decision-irrelevant variation.

Noise scale σ	Basis	Normalized error ↓	Error reduction /% ↑	Alignment ↑	Risk ↓
1/4	Cross-covariance	0.041	-0.05	0.999	3.4620
	PCA	0.041	0.00	0.999	3.4620
4	Cross-covariance	0.062	93.68	0.884	3.4843
	PCA	0.986	0.00	0.006	4.5003

Table 7 shows that, at the low noise scale, the two bases produce nearly identical decisions. Their mean normalized errors across three paired seeds are both 0.041, and their mean subspace alignments are both 0.999. At the high noise scale, the cross-covariance basis has a mean normalized error of 0.062, compared with 0.986 for PCA, a 93.7% reduction. Its mean subspace alignment is 0.884, compared with 0.006 for PCA, and its mean operational risk is 3.4843, compared with 4.5003. These results show that PCA follows the dominant variation even when that variation does not affect the decision, whereas the cross-covariance basis continues to retain the directions used by the downstream problem.

7. Conclusion

We develop KKT Atlas, a framework for learning compact local scenario models for repeated contextual stochastic optimization. Atlas partitions the context space into charts, identifies local scenario directions from gradients of scenario costs with respect to decisions, and jointly fits scenario locations and probabilities by minimizing operational loss. The theory connects the approximation of local first-order quantities to decision quality and guides two central design choices. The spectral rule selects the local representation dimension, and the shared-support result motivates $K_b = d_b + 1$. For each new context, we then solve only the small stochastic program associated with its chart.

The computational results show that compact supports can preserve decision quality while reducing both model construction time and repeated downstream solution time. Atlas attains the lowest mean risk in CAP and PORT and nearly matches ICEO-A in NET (Table 2), using only 4–18

scenarios per query (Table 3). It requires less wall time for model construction than ICEO-A in all three applications (Table 4) and is 12.5–61.0 times faster than ICEO-A for batches of 128 decisions using 32 cores (Table 5). The CAP support-size study further shows that the $d+1$ support size marks a quality plateau, while the controlled chart and basis studies show, respectively, that charts accommodate mode-specific scenario directions and gradient information protects the scenario basis from decision-irrelevant variation.

Our current analysis relies on structural assumptions that may be restrictive in other applications. Smoothness makes the scenario gradients well defined; the sensitivity calculations used in training also require local regularity of the relevant solution and active set. Strong convexity, either intrinsic or introduced through a small regularizer, converts first-order approximation into decision error. These conditions may fail for nonsmooth losses, discrete decisions, or nonconvex downstream models. We also keep the chart partition fixed after training, so the current construction does not adapt to changes in the contextual distribution over time.

These limitations suggest several extensions. For convex nonsmooth problems, subgradients, proximal residuals, or other generalized first-order objects may replace ordinary gradients. For smooth nonconvex problems, local stationarity conditions and stability around selected solutions may support a chart-specific form of the same construction, although global decision guarantees would require different arguments. Problems that are both nonconvex and nonsmooth may call for surrogate local sensitivity measures or problem-specific stationarity systems. Online updates to the charts and scenario supports are another promising direction when contexts evolve. In each case, the central question remains whether uncertainty can be compressed along the directions that govern the downstream decision.

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E-Companion

Table EC.1: Summary of notation.

Object family	Notation	Role
Data and dimensions	$X \in \mathcal{X} \subseteq \mathbb{R}^{d_x}$	Observable context.
	$x \in \mathcal{X}$	Realized context.
	$\Xi \in \mathcal{U} \subseteq \mathbb{R}^{d_\xi}$	Uncertain parameter.
	$\xi \in \mathcal{U}$	Realization or scenario; observed and generated scenarios use the same symbol.
	$z \in \mathcal{Z} \subseteq \mathbb{R}^{d_z}$	Downstream decision.
	P_x	Conditional law of Ξ given $X = x$.
	$\mathcal{X}_{\mathrm{an}}$	Analysis domain, including the chart centroids.
	d_b	Selected dimension of chart b.
	$d_b^\star$	Structural dimension of chart b.
Charts	B	Candidate number of charts.
	$\widehat{B}$	Selected number of charts.
	$\tau_B(x)$	Assignment of context x to a chart.
	$\bar{x}_b$	Centroid of chart b.
	$I_b \subseteq I_{\mathrm{tr}}$	Training indices assigned to chart b.
	$\widehat{D}_x$	Diagonal matrix of context sample scales.
	$\|h\|_x = \|\widehat{D}_x^{-1} h\|_2$	Standardized norm for context differences.
Representations	$T : \mathcal{U} \to \mathbb{R}^{d_\xi}$	Transformation of realizations.
	$g : \mathrm{dom}(g) \to \mathcal{U}$	Reconstruction of scenarios.
	$\widehat{\mu}_b$	Mean of the transformed observations in chart b.
	$\widehat{D}_b$	Diagonal matrix of transformed sample scales.
	$\varepsilon_{\mathrm{num}}$	Numerical lower bound on the component scales in $\widehat{D}_b$.
	$T_b(\xi) = \widehat{D}_b^{-1}\{T(\xi) - \widehat{\mu}_b\}$	Standardized chart encoder.
	$g_b(y) = g(\widehat{\mu}_b + \widehat{D}_b y)$	Chart decoder.
	$\mathcal{Y}_b \subseteq \mathrm{dom}(g_b)$	Domain used in the affine-gradient analysis.
Gradient information	$\phi(z, \xi; x) = \nabla_z Q(z, \xi; x)$	Scenario gradient.
	$\bar{\phi}(z; x) = \mathbb{E}_{P_x}[\phi(z, \Xi; x)]$	Conditional mean scenario gradient.
	$\widehat{C}_b \in \mathbb{R}^{d_\xi \times d_z}$	Empirical cross-covariance matrix.
	$C_b^\star \in \mathbb{R}^{d_\xi \times d_z}$	Ideal affine-gradient signal.
Bases and coordinates	$\widehat{V}_{b,d}$	First d left singular directions of $\widehat{C}_b$.
	$\widehat{V}_b = \widehat{V}_{b,d_b}$	Selected empirical basis for chart b.
	V_b^{rel}	Basis of the gradient-relevant space.
	$V_b^\star$	Basis of the ideal covariance signal space.
	$u_{b,k}$	Coefficients of generated scenario k.
	u^{rel}	Coordinates in the gradient-relevant space.

Continued on next page

Table EC.1 continued

Object family	Notation	Role
Local scenario model	$\xi_{b,k} = g_b(\widehat{V}_b u_{b,k})$	Generated scenario k in chart b.
	$p_b(x) \in \Delta_{K_b}$	Context-dependent scenario probabilities.
	$K_b = d_b + 1$	Number of scenarios used by the algorithm.
	$a_b \in \mathbb{R}^{K_b}$	Affine-softmax intercepts.
	$\Gamma_b \in \mathbb{R}^{K_b \times d_x}$	Affine-softmax context coefficients.
	$\theta_b = (u_{b,1}, \ldots, u_{b,K_b}, a_b, \Gamma_b)$	Parameters fitted for chart b.
Loss and objectives	$\ell = f + Q$	Operational loss.
	$R_0(z;x) = \mathbb{E}_{P_x}[\ell(z,\Xi;x)]$	Operational risk.
	$R(z;x)$	Regularized population objective.
	$\widehat{R}_b(z;x)$	Regularized fitted objective for chart b.
	$\nabla_z R(z;x)$	Population first-order mapping.
	$\nabla_z \widehat{R}_b(z;x)$	Fitted first-order mapping.
Decisions and bounds	$z^\star(x)$	Exact minimizer of $R(\cdot;x)$.
	$z(p,\boldsymbol{\xi};x)$	Exact finite-scenario solution.
	$\widehat{z}_b(x)$	Exact fitted decision.
	$\widetilde{z}_b(x)$	Numerically computed fitted decision.
	$\eta_{\mathrm{reg}} \geq 0$	Added quadratic regularization coefficient.
	$\eta > 0$	Certified curvature lower bound.
	ε with role superscripts	Error quantities distinguished by role.
	$\overline{\varepsilon}_b^C(\delta)$	Upper bound on ε_b^C.

EC.1. Proofs and Auxiliary Results

EC.1.1. Cross-covariance estimation

THEOREM EC.1 **(Matrix Bernstein inequality, Tropp 2012, Theorem 1.6)**. *Let I be a finite index set and let Z_i, $i \in I$, be independent, self-adjoint random matrices in $\mathbb{R}^{d_{\mathrm{mat}} \times d_{\mathrm{mat}}}$ satisfying $\mathbb{E}[Z_i] = 0$ and $\|Z_i\|_{\mathrm{op}} \leq L$ almost surely, where $L > 0$. Then, for every $t > 0$,*

$$\mathbb{P}\left(\left\|\sum_{i\in I} Z_i\right\|_{\mathrm{op}} \geq t\right) \leq 2d_{\mathrm{mat}} \exp\left\{-\frac{t^2}{2\|\sum_{i\in I}\mathbb{E}[Z_i^2]\|_{\mathrm{op}} + (2L/3)t}\right\}. \tag{EC.1}$$

LEMMA EC.1. *Let I be a finite index set and let $\mathcal{F}$ be a sigma-algebra. Conditional on $\mathcal{F}$, let Z_i, $i \in I$, be independent, symmetric random matrices in $\mathbb{R}^{d_{\mathrm{mat}} \times d_{\mathrm{mat}}}$ satisfying $\mathbb{E}[Z_i \mid \mathcal{F}] = 0$. Suppose the nonnegative $\mathcal{F}$-measurable quantities L and v satisfy $\|Z_i\|_{\mathrm{op}} \leq L$ and $\|\sum_{i\in I}\mathbb{E}[Z_i^2 \mid \mathcal{F}]\|_{\mathrm{op}} \leq v$ almost surely. Then, for every $u > 0$,*

$$\mathbb{P}\left(\left\|\sum_{i\in I} Z_i\right\|_{\mathrm{op}} > \sqrt{2vu} + \frac{2Lu}{3} \,\middle|\, \mathcal{F}\right) \leq 2d_{\mathrm{mat}} \exp(-u). \tag{EC.2}$$

Proof. Under the conditional distribution, if $L = 0$, all Z_i are zero matrices, so Eq. (EC.2) holds. Otherwise, the matrices satisfy the hypotheses of Theorem EC.1, with norm bound L and variance parameter at most v. Set t to $\sqrt{2vu} + 2Lu/3 > 0$ in Theorem EC.1. Expanding the square gives $t^2 - u\{2v + (2L/3)t\} = (2Lu/3)\sqrt{2vu} \geq 0$. Thus, $t^2/\{2v + (2L/3)t\} \geq u$, and we also have $\|\sum_{i\in I}\mathbb{E}[Z_i^2 \mid \mathcal{F}]\|_{\mathrm{op}} \leq v$ almost surely. Together, we can obtain $\frac{t^2}{2\|\sum_{i\in I}\mathbb{E}[Z_i^2|\mathcal{F}]\|_{\mathrm{op}} + (2L/3)t} \geq \frac{t^2}{2v+(2L/3)t} \geq u$. Substitution proves Eq. (EC.2) almost surely. □

For a candidate partition and chart b, write $\bar{\mu}_{T,b} := |I_b|^{-1} \sum_{i \in I_b} \mu_T(x_i)$.

DEFINITION EC.1. For $\delta \in (0,1)$, define

$$\varepsilon_b^{\mathrm{loc}} := \left\| \frac{1}{|I_b|} \sum_{i \in I_b} \left[\Sigma_T(x_i) + \{\mu_T(x_i) - \bar{\mu}_{T,b}\}\{\mu_T(x_i) - \bar{\mu}_{T,b}\}^\top \right] - \Sigma_T(\bar{x}_b) \right\|_{\mathrm{op}}, \tag{EC.3}$$

$$\varepsilon_b^{\mathrm{samp}}(\delta) := R_T^2 \left(3\sqrt{\frac{2\log\{4(d_\xi+1)/\delta\}}{|I_b|}} + \frac{10\log\{4(d_\xi+1)/\delta\}}{3|I_b|} \right), \tag{EC.4}$$

$$\bar{\varepsilon}_b^C(\delta) := \overline{H}_b \|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2 \{\varepsilon_b^{\mathrm{loc}} + \varepsilon_b^{\mathrm{samp}}(\delta)\}. \tag{EC.5}$$

LEMMA EC.2. *Fix a candidate partition satisfying Assumption 5 in Section 5.1 and a chart b with $|I_b| \geq 2$. Then, for every $\delta \in (0,1)$, the following bound holds with probability at least $1-\delta$ conditional on the training contexts,*

$$\varepsilon_b^C = \left\| \widehat{C}_b - C_b^\star \right\|_{\mathrm{op}} \leq \bar{\varepsilon}_b^C(\delta). \tag{EC.6}$$

Proof. All expectations in this proof are conditional on the training contexts.

Step 1. Relating cross-covariance error to covariance error. Let $W_i := T(\Xi_i)$, $\widehat{\Sigma}_W := |I_b|^{-1} \sum_{i \in I_b} (W_i - \widehat{\mu}_b)(W_i - \widehat{\mu}_b)^\top$, $Y_i := T_b(\Xi_i) = \widehat{D}_b^{-1}(W_i - \widehat{\mu}_b)$, $\widehat{\Sigma}_Y := |I_b|^{-1} \sum_{i \in I_b} Y_i Y_i^\top$, and $H := H_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b)$. Since $\widehat{\mu}_b = |I_b|^{-1} \sum_{i \in I_b} W_i$, we have $\sum_{i \in I_b} Y_i = \widehat{D}_b^{-1}(\sum_{i \in I_b} W_i - |I_b|\widehat{\mu}_b) = 0$. Assumption 5(iii) gives $Y_i \in \mathcal{Y}_b$ and $g_b(Y_i) = \Xi_i$, and Assumption 5(ii) gives the exact representation $\phi(\widetilde{z}_b^{\mathrm{SAA}}, \Xi_i; \bar{x}_b) = \beta_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b) + H(V_b^{\mathrm{rel}})^\top Y_i$. Substitution into Eq. (14) cancels the intercept because $\sum_{i \in I_b} Y_i = 0$

$$\widehat{C}_b = \frac{1}{|I_b|} \sum_{i \in I_b} Y_i \phi(\widetilde{z}_b^{\mathrm{SAA}}, \Xi_i; \bar{x}_b)^\top = \left(\frac{1}{|I_b|} \sum_{i \in I_b} Y_i \right) \beta_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b)^\top + \widehat{\Sigma}_Y V_b^{\mathrm{rel}} H^\top = \widehat{\Sigma}_Y V_b^{\mathrm{rel}} H^\top.$$

Since $C_b^\star = \Sigma_b V_b^{\mathrm{rel}} H^\top$,

$$\widehat{C}_b - C_b^\star = (\widehat{\Sigma}_Y - \Sigma_b) V_b^{\mathrm{rel}} H^\top. \tag{EC.7}$$

Thus,

$$\|\widehat{C}_b - C_b^\star\|_{\mathrm{op}} \leq \|\widehat{\Sigma}_Y - \Sigma_b\|_{\mathrm{op}} \|V_b^{\mathrm{rel}}\|_{\mathrm{op}} \|H^\top\|_{\mathrm{op}} \leq \overline{H}_b \|\widehat{\Sigma}_Y - \Sigma_b\|_{\mathrm{op}}, \tag{EC.8}$$

where the first inequality follows from Eq. (EC.7), and the second uses orthonormality of V_b^{rel} and $\|H^\top\|_{\mathrm{op}} = \|H\|_{\mathrm{op}} \leq \overline{H}_b$ from Eq. (23).

Diagonality of $\widehat{D}_b$ gives $\widehat{\Sigma}_Y = \widehat{D}_b^{-1} \widehat{\Sigma}_W \widehat{D}_b^{-1}$. Since $\Sigma_b = \widehat{D}_b^{-1} \Sigma_T(\bar{x}_b) \widehat{D}_b^{-1}$,

$$\begin{aligned} \widehat{\Sigma}_Y - \Sigma_b &= \widehat{D}_b^{-1} \{\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\} \widehat{D}_b^{-1}, \\ \|\widehat{\Sigma}_Y - \Sigma_b\|_{\mathrm{op}} &\leq \|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2 \|\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}}. \end{aligned} \tag{EC.9}$$

It remains to bound $\|\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}}$.

Step 2. Bounding the covariance error. Expanding the definition of $\widehat{\Sigma}_W$ gives $\widehat{\Sigma}_W = |I_b|^{-1} \sum_{i \in I_b} W_i W_i^\top - \widehat{\mu}_b \widehat{\mu}_b^\top$. The identities $\mathbb{E}[W_i W_i^\top] = \Sigma_T(x_i) + \mu_T(x_i)\mu_T(x_i)^\top$ and $\sum_{i \in I_b} \mu_T(x_i) = |I_b| \bar{\mu}_{T,b}$ give

$$\begin{aligned} \frac{1}{|I_b|} \sum_{i \in I_b} \mathbb{E}[W_i W_i^\top] - \bar{\mu}_{T,b} \bar{\mu}_{T,b}^\top &= \frac{1}{|I_b|} \sum_{i \in I_b} \{\Sigma_T(x_i) + \mu_T(x_i)\mu_T(x_i)^\top\} - \bar{\mu}_{T,b} \bar{\mu}_{T,b}^\top \\ &= \frac{1}{|I_b|} \sum_{i \in I_b} \Sigma_T(x_i) + \frac{1}{|I_b|} \sum_{i \in I_b} \{\mu_T(x_i) - \bar{\mu}_{T,b}\}\{\mu_T(x_i) - \bar{\mu}_{T,b}\}^\top. \end{aligned}$$

Let $Z_i := W_i W_i^\top - \mathbb{E}[W_i W_i^\top]$. Adding and subtracting the average second moment and the outer product of $\bar{\mu}_{T,b}$ gives

$$\begin{aligned}\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b) &= \frac{1}{|I_b|}\sum_{i\in I_b}\{W_i W_i^\top - \mathbb{E}[W_i W_i^\top]\} + \frac{1}{|I_b|}\sum_{i\in I_b}\mathbb{E}[W_i W_i^\top] - \widehat{\mu}_b\widehat{\mu}_b^\top - \Sigma_T(\bar{x}_b)\\ &= \frac{1}{|I_b|}\sum_{i\in I_b} Z_i - \{\widehat{\mu}_b\widehat{\mu}_b^\top - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top\} + \left[\frac{1}{|I_b|}\sum_{i\in I_b}\mathbb{E}[W_i W_i^\top] - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top - \Sigma_T(\bar{x}_b)\right].\end{aligned} \tag{EC.10}$$

Taking operator norms in Eq. (EC.10) and applying the triangle inequality yields

$$\begin{aligned}\|\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}} &\le \frac{1}{|I_b|}\left\|\sum_{i\in I_b} Z_i\right\|_{\mathrm{op}} + \|\widehat{\mu}_b\widehat{\mu}_b^\top - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top\|_{\mathrm{op}}\\ &\quad + \left\|\frac{1}{|I_b|}\sum_{i\in I_b}\mathbb{E}[W_i W_i^\top] - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top - \Sigma_T(\bar{x}_b)\right\|_{\mathrm{op}}\\ &= \frac{1}{|I_b|}\left\|\sum_{i\in I_b} Z_i\right\|_{\mathrm{op}} + \|\widehat{\mu}_b\widehat{\mu}_b^\top - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top\|_{\mathrm{op}} + \varepsilon_b^{\mathrm{loc}}.\end{aligned} \tag{EC.11}$$

By Eqs. (EC.8) and (EC.9), it now suffices to bound the sum of the two random terms in Eq. (EC.11) by $\varepsilon_b^{\mathrm{samp}}(\delta)$ with conditional probability at least $1-\delta$.

Step 3. Bounding the second moment error. Let $t := \log\{4(d_\xi+1)/\delta\}$. For symmetric matrices, $A \preceq B$ means that $B - A$ is positive semidefinite. Assumption 5(iii) gives $\|W_i\|_2 \le R_T$. Both $W_i W_i^\top$ and $\mathbb{E}[W_i W_i^\top]$ lie between 0 and $R_T^2 \mathrm{Id}_{d_\xi}$, so $-R_T^2\mathrm{Id}_{d_\xi} \preceq Z_i \preceq R_T^2\mathrm{Id}_{d_\xi}$ and $\|Z_i\|_{\mathrm{op}} \le R_T^2$. Expanding the square and using conditional measurability of $\mathbb{E}[W_i W_i^\top]$ gives

$$\begin{aligned}\mathbb{E}[Z_i^2] &= \mathbb{E}\big[(W_i W_i^\top)^2 - W_i W_i^\top \mathbb{E}[W_i W_i^\top] - \mathbb{E}[W_i W_i^\top] W_i W_i^\top + \{\mathbb{E}[W_i W_i^\top]\}^2\big]\\ &= \mathbb{E}[\|W_i\|_2^2 W_i W_i^\top] - \{\mathbb{E}[W_i W_i^\top]\}^2,\\ 0 \preceq \mathbb{E}[Z_i^2] &\preceq \mathbb{E}[\|W_i\|_2^2 W_i W_i^\top] \preceq R_T^2\mathbb{E}[W_i W_i^\top] \preceq R_T^4 \mathrm{Id}_{d_\xi}.\end{aligned}$$

Hence $\|\sum_{i\in I_b}\mathbb{E}[Z_i^2]\|_{\mathrm{op}} \le |I_b| R_T^4$. Assumption 5(i) gives conditional independence; centering gives $\mathbb{E}[Z_i] = 0$. Lemma EC.1, conditional on $\mathcal{F}_{X,\mathrm{tr}}$, with $L = R_T^2$, $v = |I_b| R_T^4$, and $u = t$, gives

$$\left\|\frac{1}{|I_b|}\sum_{i\in I_b}\{W_i W_i^\top - \mathbb{E}[W_i W_i^\top]\}\right\|_{\mathrm{op}} \le R_T^2\sqrt{\frac{2t}{|I_b|}} + \frac{2R_T^2 t}{3|I_b|}, \tag{EC.12}$$

with conditional failure probability at most $2d_\xi \exp(-t) \le \delta/2$.

Step 4. Bounding the sample mean error. Let $E_i := W_i - \mu_T(x_i)$. Lemma EC.1 bounds sums of symmetric matrices, so we express the sample mean error as

$$\|\widehat{\mu}_b - \bar{\mu}_{T,b}\|_2 = \frac{1}{|I_b|}\left\|\sum_{i\in I_b} E_i\right\|_2 = \frac{1}{|I_b|}\left\|\sum_{i\in I_b}\begin{pmatrix} 0 & E_i^\top \\ E_i & 0\end{pmatrix}\right\|_{\mathrm{op}}. \tag{EC.13}$$

Assumption 5(i) gives conditional independence of the block matrices. Since $\mathbb{E}[W_i] = \mu_T(x_i)$, we have $\mathbb{E}[E_i] = 0$, so the block matrices have conditional mean zero. The bounds $\|\mu_T(x_i)\|_2 \le \mathbb{E}[\|W_i\|_2] \le R_T$ and $\|W_i\|_2 \le R_T$ give $\|E_i\|_2 \le 2R_T$. Expanding the squared norm gives

$$\mathbb{E}[\|E_i\|_2^2] = \mathbb{E}[\|W_i\|_2^2] - 2\mu_T(x_i)^\top \mathbb{E}[W_i] + \|\mu_T(x_i)\|_2^2 = \mathbb{E}[\|W_i\|_2^2] - \|\mu_T(x_i)\|_2^2 \le R_T^2.$$

Since $E_iE_i^\top \preceq \|E_i\|_2^2 \mathrm{Id}_{d_\xi}$, the symmetric block matrices satisfy

$$\begin{pmatrix} 0 & E_i^\top \\ E_i & 0 \end{pmatrix}^2 = \begin{pmatrix} \|E_i\|_2^2 & 0 \\ 0 & E_iE_i^\top \end{pmatrix},$$

$$\sum_{i\in I_b} \mathbb{E}\left[\begin{pmatrix} 0 & E_i^\top \\ E_i & 0 \end{pmatrix}^2\right] = \begin{pmatrix} \sum_{i\in I_b} \mathbb{E}[\|E_i\|_2^2] & 0 \\ 0 & \sum_{i\in I_b} \mathbb{E}[E_iE_i^\top] \end{pmatrix} \preceq |I_b| R_T^2 \mathrm{Id}_{d_\xi+1}.$$

We also have $\left\|\begin{pmatrix} 0 & E_i^\top \\ E_i & 0 \end{pmatrix}\right\|_{\mathrm{op}} = \|E_i\|_2 \le 2R_T$. Applying Lemma EC.1 to the block matrices with $L = 2R_T$, $v = |I_b|R_T^2$, and $u = t$, and using Eq. (EC.13) gives

$$\|\widehat{\mu}_b - \bar{\mu}_{T,b}\|_2 \le R_T\sqrt{\frac{2t}{|I_b|}} + \frac{4R_T t}{3|I_b|}, \tag{EC.14}$$

with conditional failure probability at most $2(d_\xi+1)\exp(-t) = \delta/2$.

Both $\|\widehat{\mu}_b\|_2$ and $\|\bar{\mu}_{T,b}\|_2$ are bounded by R_T, so on the event in Eq. (EC.14),

$$\widehat{\mu}_b\widehat{\mu}_b^\top - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top = (\widehat{\mu}_b - \bar{\mu}_{T,b})\widehat{\mu}_b^\top + \bar{\mu}_{T,b}(\widehat{\mu}_b - \bar{\mu}_{T,b})^\top,$$

$$\|\widehat{\mu}_b\widehat{\mu}_b^\top - \bar{\mu}_{T,b}\bar{\mu}_{T,b}^\top\|_{\mathrm{op}} \le 2R_T\|\widehat{\mu}_b - \bar{\mu}_{T,b}\|_2 \le 2R_T^2\sqrt{\frac{2t}{|I_b|}} + \frac{8R_T^2 t}{3|I_b|}.$$

Step 5. Combining the error bounds. By a union bound, Eqs. (EC.12) and (EC.14) hold together with conditional probability at least $1-\delta$. Substituting Eq. (EC.12) and the preceding outer-product bound into Eq. (EC.11), and using Eq. (EC.4), gives

$$\|\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}} \le \varepsilon_b^{\mathrm{loc}} + 3\sqrt{2}R_T^2\sqrt{\frac{t}{|I_b|}} + \frac{10R_T^2 t}{3|I_b|} = \varepsilon_b^{\mathrm{loc}} + \varepsilon_b^{\mathrm{samp}}(\delta). \tag{EC.15}$$

Eqs. (EC.8), (EC.9), and (EC.15) now yield

$$\varepsilon_b^C = \|\widehat{C}_b - C_b^\star\|_{\mathrm{op}} \le \overline{H}_b\|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2\|\widehat{\Sigma}_W - \Sigma_T(\bar{x}_b)\|_{\mathrm{op}} \le \overline{H}_b\|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2\{\varepsilon_b^{\mathrm{loc}} + \varepsilon_b^{\mathrm{samp}}(\delta)\} = \bar{\varepsilon}_b^C(\delta).$$

□

EC.1.2. Covariance localization

For each chart, define $\widehat{J}_b := |I_b|^{-1}\sum_{i\in I_b}\|x_i - \bar{x}_b\|_x^2$.

LEMMA EC.3. *Suppose that there exist finite nonnegative constants $L_{\mu,b}$ and $L_{\Sigma,b}$ such that, for every context assigned to chart b, $\left\|\mu_T(x) - \mu_T(\bar{x}_b)\right\|_2 \le L_{\mu,b}\|x - \bar{x}_b\|_x$ and $\left\|\Sigma_T(x) - \Sigma_T(\bar{x}_b)\right\|_{\mathrm{op}} \le L_{\Sigma,b}\|x - \bar{x}_b\|_x$. Then*

$$\varepsilon_b^{\mathrm{loc}} \le L_{\Sigma,b}\sqrt{\widehat{J}_b} + L_{\mu,b}^2\widehat{J}_b. \tag{EC.16}$$

Proof. Let $M_b := |I_b|^{-1}\sum_{i\in I_b}\{\mu_T(x_i)-\bar{\mu}_{T,b}\}\{\mu_T(x_i)-\bar{\mu}_{T,b}\}^\top$. Eq. (EC.3) gives

$$\begin{aligned}
\varepsilon_b^{\mathrm{loc}} &= \left\| \frac{1}{|I_b|}\sum_{i\in I_b}\{\Sigma_T(x_i)-\Sigma_T(\bar{x}_b)\} + M_b \right\|_{\mathrm{op}} \\
&\le \left\| \frac{1}{|I_b|}\sum_{i\in I_b}\{\Sigma_T(x_i)-\Sigma_T(\bar{x}_b)\} \right\|_{\mathrm{op}} + \|M_b\|_{\mathrm{op}} \\
&\le \frac{L_{\Sigma,b}}{|I_b|}\sum_{i\in I_b}\|x_i-\bar{x}_b\|_x + \|M_b\|_{\mathrm{op}} \\
&\le L_{\Sigma,b}\left(\frac{1}{|I_b|}\sum_{i\in I_b}\|x_i-\bar{x}_b\|_x^2\right)^{1/2} + \|M_b\|_{\mathrm{op}} \\
&= L_{\Sigma,b}\sqrt{\widehat{J}_b} + \|M_b\|_{\mathrm{op}}.
\end{aligned}$$

Since $\sum_{i\in I_b}\{\mu_T(x_i)-\bar{\mu}_{T,b}\}=0$, expanding the squared norms yields

$$\begin{aligned}
\sum_{i\in I_b}\|\mu_T(x_i)-\mu_T(\bar{x}_b)\|_2^2 &= \sum_{i\in I_b}\|\mu_T(x_i)-\bar{\mu}_{T,b}\|_2^2 + |I_b|\|\bar{\mu}_{T,b}-\mu_T(\bar{x}_b)\|_2^2 \\
&\quad + 2\left\langle \sum_{i\in I_b}\{\mu_T(x_i)-\bar{\mu}_{T,b}\}, \bar{\mu}_{T,b}-\mu_T(\bar{x}_b)\right\rangle \\
&= \sum_{i\in I_b}\|\mu_T(x_i)-\bar{\mu}_{T,b}\|_2^2 + |I_b|\|\bar{\mu}_{T,b}-\mu_T(\bar{x}_b)\|_2^2.
\end{aligned}$$

As an average of outer products, M_b is positive semidefinite. Its operator norm is therefore at most its trace, and we then have

$$\begin{aligned}
\|M_b\|_{\mathrm{op}} \le \mathrm{tr}(M_b) &= \frac{1}{|I_b|}\sum_{i\in I_b}\|\mu_T(x_i)-\bar{\mu}_{T,b}\|_2^2 \\
&\le \frac{1}{|I_b|}\sum_{i\in I_b}\|\mu_T(x_i)-\mu_T(\bar{x}_b)\|_2^2 \\
&\le \frac{L_{\mu,b}^2}{|I_b|}\sum_{i\in I_b}\|x_i-\bar{x}_b\|_x^2 = L_{\mu,b}^2\widehat{J}_b.
\end{aligned}$$

Combining the two bounds proves Eq. (EC.16). □

EC.1.3. Proof of Theorem 1

Proof. Let $t := \log\{4(d_\xi+1)B/\delta\}$. The sample size requirement in Theorem 1 gives $|I_b| \ge \max\{2,t\}$ for every chart. Under Assumption 5, Lemma EC.2, with conditional failure probability at most δ/B for each chart, gives

$$\varepsilon_b^C \le \overline{\varepsilon}_b^C(\delta/B) = \overline{H}_b\|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2\{\varepsilon_b^{\mathrm{loc}} + \varepsilon_b^{\mathrm{samp}}(\delta/B)\}. \tag{EC.17}$$

To bound the weighted average over all charts, we need these bounds to hold simultaneously. A union bound gives

$$\mathbb{P}\left(\bigcup_{b=1}^{B}\{\varepsilon_b^C > \overline{\varepsilon}_b^C(\delta/B)\}\,\middle|\,\mathcal{F}_{X,\mathrm{tr}}\right) \le \sum_{b=1}^{B}\mathbb{P}\left(\varepsilon_b^C > \overline{\varepsilon}_b^C(\delta/B)\,\middle|\,\mathcal{F}_{X,\mathrm{tr}}\right) \le \delta.$$

Thus, Eq. (EC.17) holds for every chart with conditional probability at least $1-\delta$. We work on this simultaneous event for the remainder of the proof.

Since $|I_b| \geq t$, we have $0 < t/|I_b| \leq 1$ and $t/|I_b| \leq \sqrt{t/|I_b|}$. Using the definition in Eq. (EC.4) and $3\sqrt{2} + 10/3 < 8$, we obtain

$$\varepsilon_b^{\mathrm{samp}}(\delta/B) = R_T^2 \left(3\sqrt{2}\sqrt{\frac{t}{|I_b|}} + \frac{10t}{3|I_b|} \right) \leq \left(3\sqrt{2} + \frac{10}{3} \right) R_T^2 \sqrt{\frac{t}{|I_b|}} \leq 8R_T^2 \sqrt{\frac{t}{|I_b|}}. \tag{EC.18}$$

Lemma EC.3, with the common constants L_Σ and L_μ, gives $\varepsilon_b^{\mathrm{loc}} \leq L_\Sigma \sqrt{\widehat{J}_b} + L_\mu^2 \widehat{J}_b$. Assumption 5(iv) gives $\overline{H}_b \|\widehat{D}_b^{-1}\|_{\mathrm{op}}^2 \leq A_{\mathrm{sc}}$. Substituting these two bounds and Eq. (EC.18) into Eq. (EC.17) yields, for every chart,

$$\varepsilon_b^C \leq A_{\mathrm{sc}} \left\{ L_\Sigma \sqrt{\widehat{J}_b} + L_\mu^2 \widehat{J}_b + 8R_T^2 \sqrt{\frac{t}{|I_b|}} \right\}. \tag{EC.19}$$

We now average Eq. (EC.19) with weights $|I_b|/|I_{\mathrm{tr}}|$. The partition and distortion definitions give $\sum_{b=1}^B |I_b|/|I_{\mathrm{tr}}| = 1$ and $\sum_{b=1}^B |I_b| \widehat{J}_b / |I_{\mathrm{tr}}| = \widehat{J}_B$. The two remaining weighted sums satisfy, by Cauchy–Schwarz,

$$\sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \sqrt{\widehat{J}_b} = \sum_{b=1}^B \sqrt{\frac{|I_b|}{|I_{\mathrm{tr}}|}} \sqrt{\frac{|I_b| \widehat{J}_b}{|I_{\mathrm{tr}}|}} \leq \left(\sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \right)^{1/2} \left(\sum_{b=1}^B \frac{|I_b| \widehat{J}_b}{|I_{\mathrm{tr}}|} \right)^{1/2} = \sqrt{\widehat{J}_B},$$

$$\sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \sqrt{\frac{t}{|I_b|}} = \sum_{b=1}^B \sqrt{\frac{|I_b|}{|I_{\mathrm{tr}}|}} \sqrt{\frac{t}{|I_{\mathrm{tr}}|}} \leq \left(\sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \right)^{1/2} \left(\sum_{b=1}^B \frac{t}{|I_{\mathrm{tr}}|} \right)^{1/2} = \sqrt{\frac{Bt}{|I_{\mathrm{tr}}|}}.$$

Multiplying Eq. (EC.19) by $|I_b|/|I_{\mathrm{tr}}|$, summing over b, and using these identities and inequalities gives

$$\begin{aligned} \sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \varepsilon_b^C &\leq A_{\mathrm{sc}} \sum_{b=1}^B \frac{|I_b|}{|I_{\mathrm{tr}}|} \left\{ L_\Sigma \sqrt{\widehat{J}_b} + L_\mu^2 \widehat{J}_b + 8R_T^2 \sqrt{\frac{t}{|I_b|}} \right\} \\ &\leq A_{\mathrm{sc}} \left\{ L_\Sigma \sqrt{\widehat{J}_B} + L_\mu^2 \widehat{J}_B + 8R_T^2 \sqrt{\frac{Bt}{|I_{\mathrm{tr}}|}} \right\}. \end{aligned}$$

Substituting $t = \log\{4(d_\xi + 1)B/\delta\}$ proves Eq. (25) with conditional probability at least $1-\delta$. □

EC.1.4. Ideal cross-covariance subspace

Lemma EC.4. *Suppose Assumptions 3 and 6 hold. Then*

$$\mathrm{range}(C_b^\star) = \mathrm{range}(\Sigma_b V_b^{\mathrm{rel}}), \qquad \mathrm{rank}(C_b^\star) = d_b^\star. \tag{EC.20}$$

Moreover, for every orthonormal matrix $V_b^\star \in \mathbb{R}^{d_\xi \times d_b^\star}$ spanning $\mathrm{range}(C_b^\star)$, $(V_b^{\mathrm{rel}})^\top V_b^\star$ is nonsingular.

Proof. Let $H := H_b(\widetilde{z}_b^{\mathrm{SAA}}, \bar{x}_b)$. Full column rank of H gives $a^\top H^\top H a = \|Ha\|_2^2 > 0$ for every nonzero $a \in \mathbb{R}^{d_b^\star}$. Thus, $H^\top H$ is nonsingular, and $H(H^\top H)^{-1}$ is a right inverse of $H^\top$, since $H^\top H (H^\top H)^{-1} = \mathrm{Id}_{d_b^\star}$. The identities $C_b^\star = \Sigma_b V_b^{\mathrm{rel}} H^\top$ and $C_b^\star H (H^\top H)^{-1} = \Sigma_b V_b^{\mathrm{rel}}$ give both range inclusions

$$\mathrm{range}(C_b^\star) = \mathrm{range}(\Sigma_b V_b^{\mathrm{rel}} H^\top) \subseteq \mathrm{range}(\Sigma_b V_b^{\mathrm{rel}}),$$
$$\mathrm{range}(\Sigma_b V_b^{\mathrm{rel}}) = \mathrm{range}\big(C_b^\star H (H^\top H)^{-1}\big) \subseteq \mathrm{range}(C_b^\star).$$

If $\Sigma_b V_b^{\mathrm{rel}} a = 0$, then $a^\top (V_b^{\mathrm{rel}})^\top \Sigma_b V_b^{\mathrm{rel}} a = (V_b^{\mathrm{rel}} a)^\top (\Sigma_b V_b^{\mathrm{rel}} a) = 0$. Positive definiteness of $(V_b^{\mathrm{rel}})^\top \Sigma_b V_b^{\mathrm{rel}}$ forces $a = 0$. Hence $\Sigma_b V_b^{\mathrm{rel}}$ has full column rank $d_b^\star$, and both matrices have the same range of dimension $d_b^\star$, proving Eq. (EC.20).

For any orthonormal basis $V_b^\star$ of range$(\Sigma_b V_b^{\mathrm{rel}})$, each column of $V_b^\star$ is a linear combination of the columns of $\Sigma_b V_b^{\mathrm{rel}}$. Hence we can write $V_b^\star = \Sigma_b V_b^{\mathrm{rel}} A_0$, where $A_0 \in \mathbb{R}^{d_b^\star \times d_b^\star}$. If $A_0 a = 0$, then $V_b^\star a = 0$, and orthonormality gives $\|a\|_2^2 = \|V_b^\star a\|_2^2 = 0$. Hence a must be the zero vector. Thus, A_0 is nonsingular. The factorization $(V_b^{\mathrm{rel}})^\top V_b^\star = \{(V_b^{\mathrm{rel}})^\top \Sigma_b V_b^{\mathrm{rel}}\} A_0$ is therefore a product of two nonsingular matrices, proving the remaining claim. □

EC.1.5. Proof of Theorem 2

Proof. By Eq. (15), $d_b = d_b^\star$ follows if the empirical retained fraction is below α_{ret} for every $1 \le d < d_b^\star$ and above α_{ret} at $d_b^\star$.

Fix the training contexts and suppose that $\varepsilon_b^C \le \overline{\varepsilon}_b^C(\delta)$ and $\mathcal{E}_b^{\mathrm{ret}}(\delta)$ both hold. The Frobenius error satisfies $\|\widehat{C}_b - C_b^\star\|_F \le \sqrt{\min(d_\xi, d_z)}\|\widehat{C}_b - C_b^\star\|_{\mathrm{op}} \le \sqrt{\min(d_\xi, d_z)}\,\overline{\varepsilon}_b^C(\delta)$. Write $S := \|C_b^\star\|_F^2 > 0$, $\widehat{S} := \|\widehat{C}_b\|_F^2$, $S_b(d) := \sum_{j=1}^d \sigma_j(C_b^\star)^2$, and $\widehat{S}_b(d) := \sum_{j=1}^d \sigma_j(\widehat{C}_b)^2$ for $0 \le d \le \min(d_\xi, d_z)$, with empty sums equal to zero. By Lemma EC.4, $C_b^\star$ has rank $d_b^\star$, so $\sigma_j(C_b^\star) = 0$ for $j > d_b^\star$. Consequently,

$$S_b(d_b^\star) = \sum_{j=1}^{d_b^\star} \sigma_j(C_b^\star)^2 = \sum_{j=1}^{\min(d_\xi, d_z)} \sigma_j(C_b^\star)^2 = \|C_b^\star\|_F^2 = S. \tag{EC.21}$$

The equality with $\|C_b^\star\|_F^2$ uses the identity that the squared Frobenius norm equals the sum of all squared singular values.

The difference between the sums of the first d squared singular values satisfies

$$\begin{aligned}
|\widehat{S}_b(d) - S_b(d)| &= \left| \sum_{j=1}^d \{\sigma_j(\widehat{C}_b) - \sigma_j(C_b^\star)\}\{\sigma_j(\widehat{C}_b) + \sigma_j(C_b^\star)\} \right| \\
&\le \left(\sum_{j=1}^d \{\sigma_j(\widehat{C}_b) - \sigma_j(C_b^\star)\}^2 \right)^{1/2} \left(\sum_{j=1}^d \{\sigma_j(\widehat{C}_b) + \sigma_j(C_b^\star)\}^2 \right)^{1/2} \\
&\le \|\widehat{C}_b - C_b^\star\|_F \left(\sum_{j=1}^d \{\sigma_j(\widehat{C}_b) + \sigma_j(C_b^\star)\}^2 \right)^{1/2} \qquad \text{(Mirsky Inequality)} \\
&\le \|\widehat{C}_b - C_b^\star\|_F \{\sqrt{\widehat{S}_b(d)} + \sqrt{S_b(d)}\} \\
&\le \|\widehat{C}_b - C_b^\star\|_F \{\sqrt{\widehat{S}} + \sqrt{S}\} \\
&= \|\widehat{C}_b - C_b^\star\|_F \{\|\widehat{C}_b\|_F + \|C_b^\star\|_F\} \\
&= \|\widehat{C}_b - C_b^\star\|_F \{\|C_b^\star + (\widehat{C}_b - C_b^\star)\|_F + \|C_b^\star\|_F\} \\
&\le \|\widehat{C}_b - C_b^\star\|_F \{2\|C_b^\star\|_F + \|\widehat{C}_b - C_b^\star\|_F\} \\
&= \|\widehat{C}_b - C_b^\star\|_F \{2\sqrt{S} + \|\widehat{C}_b - C_b^\star\|_F\} \\
&\le \sqrt{\min(d_\xi, d_z)}\,\overline{\varepsilon}_b^C(\delta) \left\{ 2\sqrt{S} + \sqrt{\min(d_\xi, d_z)}\,\overline{\varepsilon}_b^C(\delta) \right\} \\
&= S\varepsilon_b^{\mathrm{spec}}(\delta).
\end{aligned} \tag{EC.22}$$

At $d = \min(d_\xi, d_z)$, the definitions give $S_b(d) = S$ and $\widehat{S}_b(d) = \widehat{S}$. Substituting $d = \min(d_\xi, d_z)$ into Eq. (EC.22) therefore yields $|\widehat{S} - S| \le S\varepsilon_b^{\mathrm{spec}}(\delta)$. Since $\varepsilon_b^{\mathrm{spec}}(\delta) < 1$ on $\mathcal{E}_b^{\mathrm{ret}}(\delta)$, $\widehat{S} \ge S - |\widehat{S} - S| \ge S\{1 - \varepsilon_b^{\mathrm{spec}}(\delta)\} > 0$. For every integer $0 \le d \le \min(d_\xi, d_z)$, $0 \le S_b(d) \le S$, so the retained fractions

satisfy

$$\begin{aligned}\left|\frac{\widehat{S}_b(d)}{\widehat{S}}-\frac{S_b(d)}{S}\right| &= \left|\frac{\widehat{S}_b(d)-S_b(d)}{\widehat{S}}+\frac{S_b(d)(S-\widehat{S})}{\widehat{S}S}\right| \\ &\le \frac{|\widehat{S}_b(d)-S_b(d)|}{\widehat{S}}+\frac{S_b(d)|\widehat{S}-S|}{\widehat{S}S} \\ &\le \frac{2S\varepsilon_b^{\mathrm{spec}}(\delta)}{\widehat{S}} \le \frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)}.\end{aligned} \tag{EC.23}$$

Eq. (EC.23) bounds the difference from above and below, giving

$$\frac{S_b(d)}{S}-\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)} \le \frac{\widehat{S}_b(d)}{\widehat{S}} \le \frac{S_b(d)}{S}+\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)} \qquad \text{for every } 0\le d\le \min(d_\xi,d_z).$$

For $1\le d<d_b^\star$, monotonicity gives $S_b(d)/S=\alpha_b^\star(d)\le\alpha_b^\star(d_b^\star-1)$. At $d=d_b^\star$, Eq. (EC.21) gives $S_b(d_b^\star)/S=1$. On $\mathcal{E}_b^{\mathrm{ret}}(\delta)$, Eq. (29) makes the error bound smaller than both $\alpha_{\mathrm{ret}}-\alpha_b^\star(d_b^\star-1)$ and $1-\alpha_{\mathrm{ret}}$. Hence,

$$\begin{aligned}\frac{\widehat{S}_b(d)}{\widehat{S}} &\le \frac{S_b(d)}{S}+\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)} \\ &\le \alpha_b^\star(d_b^\star-1)+\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)}<\alpha_{\mathrm{ret}}, \qquad 1\le d<d_b^\star, \\ \frac{\widehat{S}_b(d_b^\star)}{\widehat{S}} &\ge \frac{S_b(d_b^\star)}{S}-\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)}=1-\frac{2\varepsilon_b^{\mathrm{spec}}(\delta)}{1-\varepsilon_b^{\mathrm{spec}}(\delta)}>\alpha_{\mathrm{ret}}.\end{aligned}$$

Thus, the rule in Eq. (15) selects $d_b=d_b^\star$ on the intersection of the two events. Consequently, $\mathcal{E}_b^{\mathrm{ret}}(\delta)\cap\{d_b\ne d_b^\star\}\subseteq\{\varepsilon_b^C>\overline{\varepsilon}_b^C(\delta)\}$. Eq. (27) therefore gives

$$\mathbb{P}\left(\mathcal{E}_b^{\mathrm{ret}}(\delta)\cap\{d_b\ne d_b^\star\}\,|\,\mathcal{F}_{X,\mathrm{tr}}\right)\le\mathbb{P}\left(\varepsilon_b^C>\overline{\varepsilon}_b^C(\delta)\,\middle|\,\mathcal{F}_{X,\mathrm{tr}}\right)\le\delta.$$

We have therefore established the probability bound stated, completing the proof. □

EC.1.6. Perturbation of the empirical subspace

Fix chart b. With a slight abuse of notation, we omit the subscript b from $d_b^\star$, V_b^{rel}, $C_b^\star$, $\widehat{C}_b$, $V_b^\star$, $\widehat{V}_{b,d^\star}$, σ_b, κ_b, ε_b^C, and ε_b^V in the following lemma and the proof of Theorem 3.

LEMMA EC.5. *Suppose Assumptions 3 and 6 hold. Let $\sigma=\sigma_{d^\star}(C^\star)$ denote the smallest nonzero singular value of $C^\star$, and let $\widehat{V}_{d^\star}$ contain the leading $d^\star$ left singular vectors of $\widehat{C}$. If $\varepsilon^C<\sigma$ and $\kappa\varepsilon^V<1$, then there exists an orthogonal matrix $O\in\mathbb{R}^{d^\star\times d^\star}$ such that $\|\widehat{V}_{d^\star}O-V^\star\|_{\mathrm{op}}\le\varepsilon^V$. Moreover, $(V^{\mathrm{rel}})^\top\widehat{V}_{d^\star}$ is nonsingular, and*

$$\left\|\left((V^{\mathrm{rel}})^\top\widehat{V}_{d^\star}\right)^{-1}\right\|_{\mathrm{op}}\le\frac{\kappa}{1-\kappa\varepsilon^V}. \tag{EC.24}$$

Proof. Let $A^\star:=(V^{\mathrm{rel}})^\top V^\star$. Lemma EC.4 gives that $A^\star$ is nonsingular and $\|(A^\star)^{-1}\|_{\mathrm{op}}=\kappa$. For any orthogonal matrix $O\in\mathbb{R}^{d^\star\times d^\star}$, let $E_V:=(V^{\mathrm{rel}})^\top(\widehat{V}_{d^\star}O-V^\star)$. The projected matrix after right multiplication by O factors as

$$(V^{\mathrm{rel}})^\top\widehat{V}_{d^\star}O=A^\star+E_V=A^\star\{\mathrm{Id}_{d^\star}+(A^\star)^{-1}E_V\}.$$

If $\|E_V\|_{\mathrm{op}}\le\varepsilon^V$, then $\|(A^\star)^{-1}E_V\|_{\mathrm{op}}\le\kappa\|E_V\|_{\mathrm{op}}\le\kappa\varepsilon^V<1$. For any $v\in\mathbb{R}^{d^\star}$ satisfying $\{\mathrm{Id}_{d^\star}+(A^\star)^{-1}E_V\}v=0$, we have $v=-(A^\star)^{-1}E_Vv$, so $\|v\|_2\le\kappa\varepsilon^V\|v\|_2$. Since $\kappa\varepsilon^V<1$, this forces $v=0$.

Thus, $\mathrm{Id}_{d^\star} + (A^\star)^{-1}E_V$ is nonsingular. Since $A^\star$ is nonsingular, the above factorization also makes $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O$ nonsingular. The Neumann series converges in operator norm and gives

$$\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O\}^{-1} = \{\mathrm{Id}_{d^\star} + (A^\star)^{-1}E_V\}^{-1}(A^\star)^{-1} = \left(\sum_{k=0}^{\infty} \{-(A^\star)^{-1}E_V\}^k\right)(A^\star)^{-1},$$

$$\left\|\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O\}^{-1}\right\|_{\mathrm{op}} \le \kappa \sum_{k=0}^{\infty} \|(A^\star)^{-1}E_V\|_{\mathrm{op}}^k = \frac{\kappa}{1 - \|(A^\star)^{-1}E_V\|_{\mathrm{op}}} \le \frac{\kappa}{1 - \kappa\varepsilon^V},$$

Under $\|E_V\|_{\mathrm{op}} \le \varepsilon^V$, $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} = \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O\}O^\top$ is a product of nonsingular matrices and is therefore nonsingular. Hence,

$$\left\|\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1}\right\|_{\mathrm{op}} = \left\|O\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O\}^{-1}\right\|_{\mathrm{op}} = \left\|\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} O\}^{-1}\right\|_{\mathrm{op}} \le \frac{\kappa}{1 - \kappa\varepsilon^V}.$$

The inverse bound in Eq. (EC.24) therefore holds whenever $\|E_V\|_{\mathrm{op}} \le \varepsilon^V$.

Orthonormality of V^{rel} gives

$$\|E_V\|_{\mathrm{op}} \le \|(V^{\mathrm{rel}})^\top\|_{\mathrm{op}} \|\widehat{V}_{d^\star} O - V^\star\|_{\mathrm{op}} = \|\widehat{V}_{d^\star} O - V^\star\|_{\mathrm{op}}.$$

The preceding argument establishes nonsingularity and Eq. (EC.24) whenever an orthogonal matrix O satisfies $\|\widehat{V}_{d^\star} O - V^\star\|_{\mathrm{op}} \le \varepsilon^V$. It remains to construct such a matrix.

Let $E_C := \widehat{C} - C^\star$ and $P_\perp := \mathrm{Id}_{d_\xi} - V^\star(V^\star)^\top$. Lemma EC.4 gives $\mathrm{rank}(C^\star) = d^\star$ and $C^\star = V^\star(V^\star)^\top C^\star$. Thus $P_\perp C^\star = 0$, since $P_\perp V^\star = 0$; moreover, $P_\perp^\top = P_\perp^2 = P_\perp$ and $\|P_\perp\|_{\mathrm{op}} \le 1$. Let $\widehat{W}_{d^\star}$ contain the corresponding $d^\star$ right singular vectors of $\widehat{C}$, and write $\widehat{\Lambda}_{d^\star} := \mathrm{Diag}(\sigma_1(\widehat{C}), \ldots, \sigma_{d^\star}(\widehat{C}))$. Then $\widehat{C}\widehat{W}_{d^\star} = \widehat{V}_{d^\star}\widehat{\Lambda}_{d^\star}$, and Weyl's inequality for singular values gives

$$\sigma_{d^\star}(\widehat{C}) \ge \sigma_{d^\star}(C^\star) - \|E_C\|_{\mathrm{op}} = \sigma - \varepsilon^C > 0,$$
$$\|\widehat{\Lambda}_{d^\star}^{-1}\|_{\mathrm{op}} = \max_{1\le j\le d^\star} \frac{1}{\sigma_j(\widehat{C})} = \frac{1}{\sigma_{d^\star}(\widehat{C})} \le \frac{1}{\sigma - \varepsilon^C}.$$

Projecting the singular-vector equation and using $\|\widehat{W}_{d^\star}\|_{\mathrm{op}} = 1$ yields

$$P_\perp \widehat{V}_{d^\star}\widehat{\Lambda}_{d^\star} = P_\perp(C^\star + E_C)\widehat{W}_{d^\star} = P_\perp E_C \widehat{W}_{d^\star},$$
$$\|P_\perp \widehat{V}_{d^\star}\|_{\mathrm{op}} = \|P_\perp E_C \widehat{W}_{d^\star}\widehat{\Lambda}_{d^\star}^{-1}\|_{\mathrm{op}}$$
$$\le \|P_\perp\|_{\mathrm{op}}\|E_C\|_{\mathrm{op}}\|\widehat{W}_{d^\star}\|_{\mathrm{op}}\|\widehat{\Lambda}_{d^\star}^{-1}\|_{\mathrm{op}} \le \frac{\varepsilon^C}{\sigma - \varepsilon^C}.$$

Let $G := (V^\star)^\top \widehat{V}_{d^\star}$. For $j = 1, \ldots, d^\star$, orthonormality gives $0 \le \sigma_j(G) \le \|G\|_{\mathrm{op}} \le 1$. Write $\theta_j := \arccos(\sigma_j(G))$ for the principal angles, ordered as $0 \le \theta_1 \le \cdots \le \theta_{d^\star} \le \pi/2$, and let $\theta_{\max} := \theta_{d^\star}$. Take the SVD $G = O_1 \,\mathrm{Diag}(\cos\theta_1, \ldots, \cos\theta_{d^\star}) O_2^\top$, where $O_1, O_2 \in \mathbb{R}^{d^\star \times d^\star}$ are orthogonal. Using $P_\perp^\top P_\perp = P_\perp$, the Gram matrix of the projected basis is

$$(P_\perp \widehat{V}_{d^\star})^\top (P_\perp \widehat{V}_{d^\star}) = \mathrm{Id}_{d^\star} - G^\top G = O_2 \,\mathrm{Diag}(\sin^2\theta_1, \ldots, \sin^2\theta_{d^\star}) O_2^\top.$$

Its largest eigenvalue is $\sin^2\theta_{\max}$, so $\|P_\perp \widehat{V}_{d^\star}\|_{\mathrm{op}} = \sin\theta_{\max}$.

It remains to convert the principal-angle bound into the required bound on $\|\widehat{V}_{d^\star} O - V^\star\|_{\mathrm{op}}$. Choose $O := O_2 O_1^\top$, which is orthogonal since O_1 and O_2 are orthogonal. Then $GO = O^\top G^\top = O_1 \,\mathrm{Diag}(\cos\theta_1, \ldots, \cos\theta_{d^\star}) O_1^\top$, and

$$(\widehat{V}_{d^\star} O - V^\star)^\top (\widehat{V}_{d^\star} O - V^\star) = 2\mathrm{Id}_{d^\star} - O^\top G^\top - GO$$
$$= 4 O_1 \,\mathrm{Diag}\big(\sin^2(\theta_1/2), \ldots, \sin^2(\theta_{d^\star}/2)\big) O_1^\top.$$

Taking the square root of its largest eigenvalue and using $\cos(\theta_{\max}/2) \ge 1/\sqrt{2}$ gives

$$\|\widehat{V}_{d^\star} O - V^\star\|_{\mathrm{op}} = 2\sin(\theta_{\max}/2) = \frac{\sin\theta_{\max}}{\cos(\theta_{\max}/2)} \le \sqrt{2}\sin\theta_{\max} \le \frac{\sqrt{2}\varepsilon^C}{\sigma - \varepsilon^C} = \varepsilon^V.$$

Thus, $\|E_V\|_{\mathrm{op}} \le \varepsilon^V$, so $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}$ is nonsingular and its inverse satisfies Eq. (EC.24), completing the proof. □

EC.1.7. Proof of Theorem 3

Proof. Let $u^{\mathrm{rel}} \in \mathbb{R}^{d^\star}$ be arbitrary. To represent u^{rel}, it suffices to find $u \in \mathbb{R}^{d^\star}$ satisfying $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u = u^{\mathrm{rel}}$ and the coefficient bound in Eq. (30). Under the stated perturbation conditions, Lemma EC.5 gives that $A := (V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}$ is nonsingular and satisfies Eq. (EC.24). Set

$$u := A^{-1} u^{\mathrm{rel}} = \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u^{\mathrm{rel}}. \tag{EC.25}$$

Then $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u = A A^{-1} u^{\mathrm{rel}} = u^{\mathrm{rel}}$, and

$$\|u\|_2 \le \|A^{-1}\|_{\mathrm{op}} \|u^{\mathrm{rel}}\|_2 \le \frac{\kappa}{1 - \kappa \varepsilon^V} \|u^{\mathrm{rel}}\|_2.$$

Since u^{rel} was arbitrary, the proof is complete. □

EC.1.8. Gradient approximation error

Fix chart b. With a slight abuse of notation, we omit the chart index b from d_b, $d_b^\star$, V_b^{rel}, $\widehat{V}_{b,d_b^\star}$, $\mathcal{X}_b$, $\mathcal{Y}_b$, g_b, β_b, H_b, $\overline{H}_b$, r_b, r_b^{gen}, μ_b^{rel}, $\varepsilon_b^{\mathrm{mean}}$, $\varepsilon_b^{\mathrm{aff}}$, ε_b^V, ε_b°, κ_b, a_b, a_b°, Γ_b, Γ_b°, p_b, p_b°, $p_{b,k}^\circ$, $u_{b,k}^{\mathrm{rel}}$, $u_{b,k}^\circ$, $\xi_{b,k}^\circ$, R_b°, z_b°, and $R_{y,b}$ in Lemmas EC.6 and EC.7 and the proof of Theorem 4.

Lemma EC.6. *Suppose Assumptions 1, 2, 3, and 4 hold. Let $u_1^{\mathrm{rel}}, \dots, u_K^{\mathrm{rel}} \in \mathbb{R}^{d^\star}$, with coefficient vectors u_k given by Eq. (EC.25). Assume $\widehat{V}_{d^\star} u_k \in \mathcal{Y}$ for every $k = 1, \dots, K$. Set $\xi_k := g(\widehat{V}_{d^\star} u_k)$. For any probability model $p : \mathcal{X} \to \Delta_K$, define $\varepsilon^{\mathrm{mean}}(p) := \sup_{x \in \mathcal{X}} \big\| \mu^{\mathrm{rel}}(x) - \sum_{k=1}^K p_k(x) u_k^{\mathrm{rel}} \big\|_2$ and let $R_p(z;x) := \sum_{k=1}^K p_k(x) \ell(z, \xi_k; x) + \eta_{\mathrm{reg}} \|z - z_0(x)\|_2^2/2$ be the induced regularized objective. Then*

$$\sup_{x \in \mathcal{X},\, z \in \mathcal{Z}} \big\| \nabla_z R(z;x) - \nabla_z R_p(z;x) \big\|_2 \le \overline{H} \varepsilon^{\mathrm{mean}}(p) + 2\varepsilon^{\mathrm{aff}}. \tag{EC.26}$$

Proof. Fix $z \in \mathcal{Z}$ and $x \in \mathcal{X}$. Assumption 2 permits differentiation under the conditional expectation in Eq. (2). Using $\ell = f + Q$ and $\phi = \nabla_z Q$, we obtain

$$\begin{aligned} \nabla_z R(z;x) &= \mathbb{E}[\nabla_z \ell(z, \Xi; x) \mid X = x] + \eta_{\mathrm{reg}}(z - z_0(x)) \\ &= \nabla_z f(z;x) + \mathbb{E}[\phi(z, \Xi; x) \mid X = x] + \eta_{\mathrm{reg}}(z - z_0(x)). \end{aligned}$$

The weights $p_k(x)$ do not depend on z, so the definition of R_p gives

$$\begin{aligned} \nabla_z R_p(z;x) &= \sum_{k=1}^K p_k(x) \nabla_z \ell(z, \xi_k; x) + \eta_{\mathrm{reg}}(z - z_0(x)) \\ &= \left(\sum_{k=1}^K p_k(x) \right) \nabla_z f(z;x) + \sum_{k=1}^K p_k(x) \phi(z, \xi_k; x) + \eta_{\mathrm{reg}}(z - z_0(x)) \\ &= \nabla_z f(z;x) + \sum_{k=1}^K p_k(x) \phi(z, \xi_k; x) + \eta_{\mathrm{reg}}(z - z_0(x)). \end{aligned}$$

Subtracting the two expressions cancels $\nabla_z f(z;x)$ and $\eta_{\mathrm{reg}}(z - z_0(x))$, yielding

$$\nabla_z R(z;x) - \nabla_z R_p(z;x) = \mathbb{E}[\phi(z, \Xi; x) \mid X = x] - \sum_{k=1}^K p_k(x) \phi(z, \xi_k; x).$$

Taking conditional expectations in Eq. (21) and using the definition of $\mu^{\mathrm{rel}}(x)$ gives

$$\mathbb{E}[\phi(z, \Xi; x) \mid X = x] = \beta(z,x) + H(z,x) \mu^{\mathrm{rel}}(x) + \mathbb{E}[r(z, \Xi, x) \mid X = x]. \tag{EC.27}$$

The feasible coefficients satisfy $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u_k = u_k^{\mathrm{rel}}$ by Eq. (EC.25). Thus, Eq. (22) and $\sum_{k=1}^K p_k(x) = 1$ give

$$\sum_{k=1}^K p_k(x)\phi(z,\xi_k;x) = \beta(z,x) + H(z,x)\sum_{k=1}^K p_k(x)u_k^{\mathrm{rel}} + \sum_{k=1}^K p_k(x) r^{\mathrm{gen}}(z, \widehat{V}_{d^\star}u_k, x). \tag{EC.28}$$

Substituting the two affine mean expressions therefore yields

$$\begin{aligned}
&\nabla_z R(z;x) - \nabla_z R_p(z;x) \\
&\quad = \mathbb{E}[\phi(z,\Xi;x) \mid X = x] - \sum_{k=1}^K p_k(x)\phi(z,\xi_k;x) \\
&\quad = H(z,x)\left\{\mu^{\mathrm{rel}}(x) - \sum_{k=1}^K p_k(x)u_k^{\mathrm{rel}}\right\} + \mathbb{E}[r(z,\Xi,x) \mid X = x] - \sum_{k=1}^K p_k(x) r^{\mathrm{gen}}(z, \widehat{V}_{d^\star}u_k, x).
\end{aligned}$$

Eq. (23) and the definition of $\varepsilon^{\mathrm{mean}}(p)$ bound the first term on the right by $\overline{H}\varepsilon^{\mathrm{mean}}(p)$ in norm. To obtain Eq. (EC.26), it remains to bound each residual average by $\varepsilon^{\mathrm{aff}}$. Using Eq. (24) and $p(x) \in \Delta_K$, we obtain

$$\begin{aligned}
&\left\|\mathbb{E}[r(z,\Xi,x) \mid X = x]\right\|_2 \le \mathbb{E}[\|r(z,\Xi,x)\|_2 \mid X = x] \le \varepsilon^{\mathrm{aff}}, && \text{(Jensen's inequality)} \\
&\left\|\sum_{k=1}^K p_k(x) r^{\mathrm{gen}}(z, \widehat{V}_{d^\star}u_k, x)\right\|_2 \le \sum_{k=1}^K p_k(x)\|r^{\mathrm{gen}}(z, \widehat{V}_{d^\star}u_k, x)\|_2 \le \varepsilon^{\mathrm{aff}}\sum_{k=1}^K p_k(x) = \varepsilon^{\mathrm{aff}}.
\end{aligned}$$

Combining the bounds above gives

$$\|\nabla_z R(z;x) - \nabla_z R_p(z;x)\|_2 \le \overline{H}\varepsilon^{\mathrm{mean}}(p) + 2\varepsilon^{\mathrm{aff}}.$$

Taking the supremum over $x \in \mathcal{X}$ and $z \in \mathcal{Z}$ proves Eq. (EC.26). □

EC.1.9. Decision and objective error bounds

LEMMA EC.7. *Suppose $\mathcal{Z}$ is nonempty, closed, and convex. Let $\eta > 0$, and let R_1 and R_2 be differentiable convex objectives on $\mathcal{Z}$, where R_1 is η-strongly convex. For $i = 1, 2$, let $z_i \in \mathcal{Z}$ satisfy $0 \in \nabla R_i(z_i) + N_{\mathcal{Z}}(z_i)$. If $\sup_{z\in\mathcal{Z}}\left\|\nabla R_1(z) - \nabla R_2(z)\right\|_2 \le \varepsilon$ for some $\varepsilon \ge 0$, then $\left\|z_2 - z_1\right\|_2 \le \varepsilon/\eta$ and $R_1(z_2) - R_1(z_1) \le \varepsilon^2/(2\eta)$.*

Proof. Let $\Delta z := z_2 - z_1$ and $e := \nabla R_1(z_2) - \nabla R_2(z_2)$, so $\|e\|_2 \le \varepsilon$. If $\Delta z = 0$, both bounds hold. Otherwise, the distance bound is equivalent to $\eta\|\Delta z\|_2^2 \le \varepsilon\|\Delta z\|_2$. It therefore suffices to establish this inequality. Choose $v_1 \in N_{\mathcal{Z}}(z_1)$ and $v_2 \in N_{\mathcal{Z}}(z_2)$ satisfying $\nabla R_1(z_1) + v_1 = 0$ and $\nabla R_2(z_2) + v_2 = 0$. The definition of the normal cone gives $\langle v_1, \Delta z\rangle \le 0$ and $\langle v_2, -\Delta z\rangle \le 0$. Hence $\langle v_2, \Delta z\rangle \ge 0$ and $\langle v_2 - v_1, \Delta z\rangle \ge 0$.

Strong convexity at the two decisions gives

$$\begin{aligned}
R_1(z_2) &\ge R_1(z_1) + \langle \nabla R_1(z_1), \Delta z\rangle + \frac{\eta}{2}\|\Delta z\|_2^2, \\
R_1(z_1) &\ge R_1(z_2) - \langle \nabla R_1(z_2), \Delta z\rangle + \frac{\eta}{2}\|\Delta z\|_2^2.
\end{aligned}$$

Adding the two inequalities gives

$$\begin{aligned}
\eta\|\Delta z\|_2^2 &\le \langle \nabla R_1(z_2) - \nabla R_1(z_1), \Delta z\rangle \\
&\le \langle \nabla R_1(z_2) - \nabla R_1(z_1) + v_2 - v_1, \Delta z\rangle && (\langle v_2 - v_1, \Delta z\rangle \ge 0) \\
&= \langle \nabla R_1(z_2) - \nabla R_2(z_2), \Delta z\rangle && (\nabla R_i(z_i) = -v_i,\ i = 1, 2) \\
&= \langle e, \Delta z\rangle \\
&\le \|e\|_2\|\Delta z\|_2 \\
&\le \varepsilon\|\Delta z\|_2.
\end{aligned}$$

Division by $\eta\|\Delta z\|_2 > 0$ gives $\|z_2 - z_1\|_2 \le \varepsilon/\eta$.

It remains to bound $R_1(z_2) - R_1(z_1)$. Strong convexity at z_2 gives

$$\begin{aligned}
R_1(z_2) - R_1(z_1) &\le \langle \nabla R_1(z_2), \Delta z\rangle - \frac{\eta}{2}\|\Delta z\|_2^2 \\
&= \langle e, \Delta z\rangle - \langle v_2, \Delta z\rangle - \frac{\eta}{2}\|\Delta z\|_2^2 \quad (\nabla R_1(z_2) = e - v_2) \\
&\le \langle e, \Delta z\rangle - \frac{\eta}{2}\|\Delta z\|_2^2 \qquad (\langle v_2, \Delta z\rangle \ge 0) \\
&\le \|e\|_2\|\Delta z\|_2 - \frac{\eta}{2}\|\Delta z\|_2^2 \\
&\le \varepsilon\|\Delta z\|_2 - \frac{\eta}{2}\|\Delta z\|_2^2 \\
&= \frac{\varepsilon^2}{2\eta} - \frac{\eta}{2}\left(\|\Delta z\|_2 - \frac{\varepsilon}{\eta}\right)^2 \\
&\le \frac{\varepsilon^2}{2\eta}.
\end{aligned}$$

We have therefore established the decision and objective error bounds, completing the proof. □

EC.1.10. Proof of Theorem 4

Proof. Let $K := d^\star + 1$ and fix $\varepsilon > 0$. For any feasible support representing the given vertices and any affine-softmax probability model p, Lemma EC.6 gives

$$\sup_{x\in\mathcal{X},\, z\in\mathcal{Z}} \|\nabla_z R(z;x) - \nabla_z R_p(z;x)\|_2 \le \overline{H}\varepsilon^{\mathrm{mean}}(p) + 2\varepsilon^{\mathrm{aff}}.$$

For any candidate feasible scenario model with induced objective R°, fix $x \in \mathcal{X}$ and consider Lemma EC.7 with $R_1 = R(\cdot;x)$ and $R_2 = R^\circ(\cdot;x)$. Assumption 1 makes $\mathcal{Z}$ nonempty, closed, and convex, ensures that both objectives attain their minima, and gives the required convexity, with R_1 being η-strongly convex. Assumption 2 makes both objectives differentiable. Their exact minimizers $z_1 = z^\star(x)$ and $z_2 = z^\circ(x)$ therefore satisfy the optimality conditions in Lemma EC.7. If Eq. (32) holds with $\varepsilon^\circ(\varepsilon) < \infty$, its uniformity over x gives the remaining hypothesis,

$$\sup_{z\in\mathcal{Z}} \|\nabla_z R(z;x) - \nabla_z R^\circ(z;x)\|_2 \le \varepsilon^\circ(\varepsilon).$$

Applying Lemma EC.7 with gradient error bound $\varepsilon^\circ(\varepsilon)$ then gives Eqs. (33) and (34) at every $x \in \mathcal{X}$.

It remains to construct a fixed feasible support of K scenarios and a probability model p° satisfying Eq. (32). For a fixed feasible support representing the given vertices, suppose that p° satisfies $\varepsilon^{\mathrm{mean}}(p^\circ) \le \varepsilon^{\mathrm{mean}} + \varepsilon$. Applying Lemma EC.6 with $p = p^\circ$ gives

$$\begin{aligned}
\sup_{x\in\mathcal{X},\, z\in\mathcal{Z}} \|\nabla_z R(z;x) - \nabla_z R^\circ(z;x)\|_2 &\le \overline{H}\varepsilon^{\mathrm{mean}}(p^\circ) + 2\varepsilon^{\mathrm{aff}} \\
&\le \overline{H}(\varepsilon^{\mathrm{mean}} + \varepsilon) + 2\varepsilon^{\mathrm{aff}} \\
&= \varepsilon^\circ(\varepsilon).
\end{aligned}$$

Thus, after constructing the support, it suffices to choose p° with mean error at most $\varepsilon^{\mathrm{mean}} + \varepsilon$.

Step 1. Construct the support. To construct the support required by Lemma EC.6, we seek, for each vertex u_k^{rel}, a coefficient vector u_k° satisfying

$$(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u_k^\circ = u_k^{\mathrm{rel}}, \qquad \widehat{V}_{d^\star} u_k^\circ \in \mathcal{Y}.$$

The hypotheses include Assumptions 3 and 6, together with $\varepsilon^C < \sigma$ and $\kappa\varepsilon^V < 1$. Lemma EC.5 therefore ensures that $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}$ is nonsingular. For each $k = 1, \dots, K$, set $u_k^\circ := \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}}$. Then

$$(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u_k^\circ = \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}} = u_k^{\mathrm{rel}}.$$

For these coefficient vectors, the feasibility assumption in Theorem 4 gives $\widehat{V}_{d^\star} u_k^\circ \in \mathcal{Y} \subseteq \mathrm{dom}(g)$. Since $g : \mathrm{dom}(g) \to \mathcal{U}$, we can therefore define

$$\xi_k^\circ := g(\widehat{V}_{d^\star} u_k^\circ) = g\Big(\widehat{V}_{d^\star}\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}}\Big) \in \mathcal{U}, \qquad k = 1, \dots, K.$$

This defines one scenario tuple $\boldsymbol{\xi}^\circ := (\xi_1^\circ, \dots, \xi_K^\circ) \in \mathcal{U}^K$, without reference to x or ε. For any probability model $p : \mathcal{X} \to \Delta_K$, the projection identities give

$$\sum_{k=1}^K p_k(x)(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u_k^\circ = \sum_{k=1}^K p_k(x) u_k^{\mathrm{rel}}, \qquad x \in \mathcal{X}.$$

Thus, the same $K = d^\star + 1$ scenarios satisfy the support requirements of Lemma EC.6 for every $x \in \mathcal{X}$; the probability model determines their weights and the resulting mean error.

Step 2. Choose the probability model. We first show that the infimum in Eq. (31) is finite by evaluating one model in this class. Setting $a = 0$ and $\Gamma = 0$ in Eq. (17) gives

$$p_k(x; 0, 0) = \frac{\exp(0)}{\sum_{j=1}^K \exp(0)} = \frac{1}{K}, \qquad k = 1, \dots, K, \quad x \in \mathcal{X}.$$

To bound the error of this uniform model, fix $x \in \mathcal{X}$. Assumption 7 supplies $\lambda_k(x) \geq 0$ with $\sum_{k=1}^K \lambda_k(x) = 1$ and $\mu^{\mathrm{rel}}(x) = \sum_{k=1}^K \lambda_k(x) u_k^{\mathrm{rel}}$. Thus,

$$\begin{aligned} \|\mu^{\mathrm{rel}}(x)\|_2 &= \left\| \sum_{k=1}^K \lambda_k(x) u_k^{\mathrm{rel}} \right\|_2 \leq \sum_{k=1}^K \lambda_k(x) \|u_k^{\mathrm{rel}}\|_2 \\ &\leq \left(\max_{1 \leq k \leq K} \|u_k^{\mathrm{rel}}\|_2 \right) \sum_{k=1}^K \lambda_k(x) = \max_{1 \leq k \leq K} \|u_k^{\mathrm{rel}}\|_2, \end{aligned}$$

$$\left\| \mu^{\mathrm{rel}}(x) - \frac{1}{K} \sum_{k=1}^K u_k^{\mathrm{rel}} \right\|_2 \leq \|\mu^{\mathrm{rel}}(x)\|_2 + \frac{1}{K} \sum_{k=1}^K \|u_k^{\mathrm{rel}}\|_2 \leq 2 \max_{1 \leq k \leq K} \|u_k^{\mathrm{rel}}\|_2.$$

The last bound holds for every $x \in \mathcal{X}$. Since the uniform model is included in the infimum in Eq. (31),

$$0 \leq \varepsilon^{\mathrm{mean}} \leq \varepsilon^{\mathrm{mean}}\big(p(\cdot; 0, 0)\big) = \sup_{x \in \mathcal{X}} \left\| \mu^{\mathrm{rel}}(x) - \frac{1}{K} \sum_{k=1}^K u_k^{\mathrm{rel}} \right\|_2 \leq 2 \max_{1 \leq k \leq K} \|u_k^{\mathrm{rel}}\|_2 < \infty.$$

Suppose, for contradiction, that every finite pair $a \in \mathbb{R}^K$, $\Gamma \in \mathbb{R}^{K \times d_x}$ has mean error greater than $\varepsilon^{\mathrm{mean}} + \varepsilon$. Then $\varepsilon^{\mathrm{mean}} + \varepsilon$ is a lower bound strictly larger than $\varepsilon^{\mathrm{mean}}$, contradicting the definition of the infimum. Thus, there is a finite pair (a°, Γ°) such that, with $p^\circ(x) := p(x; a^\circ, \Gamma^\circ)$,

$$\varepsilon^{\mathrm{mean}}(p^\circ) = \sup_{x \in \mathcal{X}} \left\| \mu^{\mathrm{rel}}(x) - \sum_{k=1}^K p_k^\circ(x) u_k^{\mathrm{rel}} \right\|_2 \leq \varepsilon^{\mathrm{mean}} + \varepsilon.$$

For this probability model, it remains to verify the bounds in Eqs. (32)–(34). Using $\ell = f + Q$ and $\sum_k p_k^\circ(x) = 1$, the induced objective is

$$\begin{aligned} R^\circ(z;x) &:= \sum_{k=1}^{K} p_k^\circ(x)\ell(z,\xi_k^\circ;x) + \frac{\eta_{\mathrm{reg}}}{2}\|z - z_0(x)\|_2^2 \\ &= f(z;x) + \sum_{k=1}^{K} p_k^\circ(x) Q(z,\xi_k^\circ;x) + \frac{\eta_{\mathrm{reg}}}{2}\|z - z_0(x)\|_2^2. \end{aligned}$$

We now verify the conditions of Lemma EC.6 for this model. Step 1 chose $u_k^\circ = \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}}$, which is the coefficient formula in Eq. (EC.25). Multiplying by $(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}$ gives

$$(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star} u_k^\circ = \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}\{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}} = u_k^{\mathrm{rel}}.$$

Thus, the generated point $\widehat{V}_{d^\star} u_k^\circ$ has the prescribed projection coordinates u_k^{rel}. The separate requirement $\widehat{V}_{d^\star} u_k^\circ \in \mathcal{Y}$ is a feasibility assumption of Theorem 4. With $\xi_k^\circ = g(\widehat{V}_{d^\star} u_k^\circ)$, the coefficients and scenarios therefore have the form required by Lemma EC.6. The affine-softmax model chosen in Step 2 satisfies $p^\circ : \mathcal{X} \to \Delta_K$ by Eq. (17). Assumptions 1, 2, 3, and 4, also required by Lemma EC.6, are included in the hypotheses of Theorem 4. We may therefore apply Lemma EC.6 with $u_k = u_k^\circ$, $\xi_k = \xi_k^\circ$, and $p = p^\circ$. The induced objective R_p is then R°, so

$$\begin{aligned} &\sup_{\substack{x\in\mathcal{X}\\ z\in\mathcal{Z}}} \|\nabla_z R(z;x) - \nabla_z R^\circ(z;x)\|_2 \\ &\quad \le \overline{H}\,\varepsilon^{\mathrm{mean}}(p^\circ) + 2\varepsilon^{\mathrm{aff}} \\ &\quad \le \overline{H}(\varepsilon^{\mathrm{mean}} + \varepsilon) + 2\varepsilon^{\mathrm{aff}} = \varepsilon^\circ(\varepsilon). \end{aligned}$$

This establishes Eq. (32). For each fixed $x \in \mathcal{X}$, taking the supremum over z alone cannot increase the supremum just bounded, so

$$\sup_{z\in\mathcal{Z}} \|\nabla_z R(z;x) - \nabla_z R^\circ(z;x)\|_2 \le \varepsilon^\circ(\varepsilon).$$

The bound $\varepsilon^\circ(\varepsilon)$ is finite and nonnegative by Assumption 4 and the finiteness of $\varepsilon^{\mathrm{mean}}$ established above.

Step 3. Establish the decision and objective error bounds. It remains to apply Lemma EC.7 to the exact decisions. The second expression for R° has the form of the objective in Eq. (4). Since the scenarios lie in $\mathcal{U}$ and the probabilities lie in Δ_K, Assumption 1 ensures that $R^\circ(\cdot;x)$ and $R(\cdot;x)$ attain their minima on $\mathcal{Z}$. Both objectives are convex, and $R(\cdot;x)$ is η-strongly convex. Assumption 2 makes both objectives differentiable. Their exact minimizers $z^\star(x)$ and $z^\circ(x) := z(p^\circ(x), (\xi_1^\circ, \ldots, \xi_K^\circ); x)$ therefore satisfy

$$0 \in \nabla_z R(z^\star(x);x) + N_{\mathcal{Z}}(z^\star(x)), \qquad 0 \in \nabla_z R^\circ(z^\circ(x);x) + N_{\mathcal{Z}}(z^\circ(x)).$$

All hypotheses of Lemma EC.7 hold with $R_1 = R(\cdot;x)$, $R_2 = R^\circ(\cdot;x)$, $z_1 = z^\star(x)$, $z_2 = z^\circ(x)$, and gradient error bound $\varepsilon^\circ(\varepsilon)$. Hence,

$$\|z^\circ(x) - z^\star(x)\|_2 \le \frac{\varepsilon^\circ(\varepsilon)}{\eta}, \qquad R(z^\circ(x);x) - R(z^\star(x);x) \le \frac{\{\varepsilon^\circ(\varepsilon)\}^2}{2\eta}.$$

Because x was arbitrary, these are the uniform bounds in Eqs. (33) and (34). The construction uses the same $K = d^\star + 1$ scenarios for every $\varepsilon > 0$. When $d = d^\star$, $K = d + 1$, proving the claimed shared-support sufficiency.

The assumed feasibility can also be verified using a radius bound. If $R_y > 0$ satisfies $\{y \in \mathbb{R}^{d_\xi} : \|y\|_2 \le R_y\} \subseteq \mathcal{Y}$, a sufficient feasibility condition is

$$\frac{\kappa}{1 - \kappa\varepsilon^V} \max_{1\le k\le K} \|u_k^{\mathrm{rel}}\|_2 \le R_y. \tag{EC.29}$$

Suppose Eq. (EC.29) holds. Since $\widehat{V}_{d^\star}^\top \widehat{V}_{d^\star} = \mathrm{Id}_{d^\star}$, $\|\widehat{V}_{d^\star} u_k^\circ\|_2^2 = (u_k^\circ)^\top \widehat{V}_{d^\star}^\top \widehat{V}_{d^\star} u_k^\circ = \|u_k^\circ\|_2^2$. Taking square roots and using the coefficient formula gives

$$\begin{aligned}
\|\widehat{V}_{d^\star} u_k^\circ\|_2 = \|u_k^\circ\|_2 &= \left\| \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} u_k^{\mathrm{rel}} \right\|_2 \\
&\le \left\| \{(V^{\mathrm{rel}})^\top \widehat{V}_{d^\star}\}^{-1} \right\|_{\mathrm{op}} \|u_k^{\mathrm{rel}}\|_2 \\
&\le \frac{\kappa}{1-\kappa\varepsilon^V} \|u_k^{\mathrm{rel}}\|_2 && \text{(Eq. (EC.24))} \\
&\le \frac{\kappa}{1-\kappa\varepsilon^V} \max_{1\le j\le K} \|u_j^{\mathrm{rel}}\|_2 \\
&\le R_y. && \text{(Eq. (EC.29))}
\end{aligned}$$

Hence, $\widehat{V}_{d^\star} u_k^\circ \in \{y \in \mathbb{R}^{d_\xi} : \|y\|_2 \le R_y\} \subseteq \mathcal{Y}$ for every k. □

EC.1.11. Proof of Theorem 5

Proof. Fix $x \in \mathcal{X}_b$. The assertions hold when $\varepsilon_b(x) = +\infty$. Suppose henceforth that $\varepsilon_b(x) < \infty$. We apply Lemma EC.7 with $R_1 = R(\cdot; x)$, $R_2 = \widehat{R}_b(\cdot; x)$, $z_1 = z^\star(x)$, and $z_2 = \widehat{z}_b(x)$. Assumption 1 gives a nonempty, closed, convex feasible set, convex objectives that attain their minima, and η-strong convexity of $R(\cdot; x)$. Assumption 2 gives differentiability. The exact minimizers therefore satisfy

$$0 \in \nabla_z R(z^\star(x); x) + N_{\mathcal{Z}}(z^\star(x)), \qquad 0 \in \nabla_z \widehat{R}_b(\widehat{z}_b(x); x) + N_{\mathcal{Z}}(\widehat{z}_b(x)).$$

It remains to establish the gradient error bound required by Lemma EC.7. For every $z \in \mathcal{Z}$, adding and subtracting the two gradients at $\bar{x}_b$ gives

$$\begin{aligned}
\nabla_z R(z; x) - \nabla_z \widehat{R}_b(z; x) &= \{\nabla_z R(z; x) - \nabla_z R(z; \bar{x}_b)\} \\
&\quad + \{\nabla_z R(z; \bar{x}_b) - \nabla_z \widehat{R}_b(z; \bar{x}_b)\} + \{\nabla_z \widehat{R}_b(z; \bar{x}_b) - \nabla_z \widehat{R}_b(z; x)\}.
\end{aligned}$$

The triangle inequality and context bounds give

$$\begin{aligned}
\|\nabla_z R(z; x) - \nabla_z \widehat{R}_b(z; x)\|_2 &\le \|\nabla_z R(z; x) - \nabla_z R(z; \bar{x}_b)\|_2 \\
&\quad + \|\nabla_z R(z; \bar{x}_b) - \nabla_z \widehat{R}_b(z; \bar{x}_b)\|_2 + \|\nabla_z \widehat{R}_b(z; \bar{x}_b) - \nabla_z \widehat{R}_b(z; x)\|_2 \\
&\le L_{\mathrm{pop}} \|x - \bar{x}_b\|_x + \varepsilon_b^{\mathrm{ctr}} + L_b \|x - \bar{x}_b\|_x = \varepsilon_b(x),
\end{aligned}$$

uniformly over $z \in \mathcal{Z}$. Taking the supremum over $z \in \mathcal{Z}$ gives

$$\sup_{z \in \mathcal{Z}} \|\nabla_z R(z; x) - \nabla_z \widehat{R}_b(z; x)\|_2 \le \varepsilon_b(x).$$

Lemma EC.7, with error bound $\varepsilon_b(x)$, now yields

$$\|\widehat{z}_b(x) - z^\star(x)\|_2 \le \frac{\varepsilon_b(x)}{\eta}, \qquad R(\widehat{z}_b(x); x) - R(z^\star(x); x) \le \frac{\varepsilon_b(x)^2}{2\eta}.$$

These are Eqs. (40) and (41), completing the proof. □

EC.1.12. Average localization refinement

Corollary EC.1 **(Average localization bound)**. *Under the conditions of Theorem 5, define* $L_B := \max_b (L_{\mathrm{pop}} + L_b)$ *and* $\widehat{\varepsilon}_B^{\mathrm{ctr}} := \{|I_{\mathrm{tr}}|^{-1} \sum_{i \in I_{\mathrm{tr}}} (\varepsilon^{\mathrm{ctr}}_{\tau_B(x_i)})^2\}^{1/2}$. *The average squared decision error then satisfies*

$$\frac{1}{|I_{\mathrm{tr}}|} \sum_{i \in I_{\mathrm{tr}}} \left\| \widehat{z}_{\tau_B(x_i)}(x_i) - z^\star(x_i) \right\|_2^2 \le \frac{1}{\eta^2} \left(\widehat{\varepsilon}_B^{\mathrm{ctr}} + L_B \sqrt{\widehat{J}_B} \right)^2. \tag{EC.30}$$

Proof. If the right-hand side of Eq. (EC.30) is infinite, the inequality holds. Otherwise, write $b_i := \tau_B(x_i)$. Note that

$$\begin{aligned}
&\frac{1}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}} \varepsilon^{\mathrm{ctr}}_{b_i}\|x_i-\bar{x}_{b_i}\|_x \\
&\quad\le \left\{\frac{1}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}}(\varepsilon^{\mathrm{ctr}}_{b_i})^2\right\}^{1/2}\left\{\frac{1}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}}\|x_i-\bar{x}_{b_i}\|_x^2\right\}^{1/2} \\
&\quad= \widehat{\varepsilon}^{\mathrm{ctr}}_B\sqrt{\widehat{J}_B}.
\end{aligned}$$

Eq. (40) and the definition of L_B give $\|\widehat{z}_{b_i}(x_i)-z^\star(x_i)\|_2 \le \{\varepsilon^{\mathrm{ctr}}_{b_i}+L_B\|x_i-\bar{x}_{b_i}\|_x\}/\eta$. Squaring this bound and averaging over the training observations yields

$$\begin{aligned}
&\frac{1}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}}\|\widehat{z}_{b_i}(x_i)-z^\star(x_i)\|_2^2 \\
&\quad\le \frac{1}{\eta^2|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}}\{\varepsilon^{\mathrm{ctr}}_{b_i}+L_B\|x_i-\bar{x}_{b_i}\|_x\}^2 \\
&\quad= \frac{1}{\eta^2}\left\{(\widehat{\varepsilon}^{\mathrm{ctr}}_B)^2+\frac{2L_B}{|I_{\mathrm{tr}}|}\sum_{i\in I_{\mathrm{tr}}}\varepsilon^{\mathrm{ctr}}_{b_i}\|x_i-\bar{x}_{b_i}\|_x+L_B^2\widehat{J}_B\right\} \\
&\quad\le \frac{1}{\eta^2}\left(\widehat{\varepsilon}^{\mathrm{ctr}}_B+L_B\sqrt{\widehat{J}_B}\right)^2.
\end{aligned}$$

This proves Eq. (EC.30). □

EC.1.13. Proof of Theorem 6

Proof. Write $\varepsilon(x) := \varepsilon_{\tau_B(x)}(x)$. To bound the expected excess operational risk, we first compare $R_0(\widehat{z}(x);x)$ with $R_0(z;x)$ for an arbitrary $z\in\mathcal{Z}$. Fix x with $\varepsilon(x)<\infty$. The regularization penalty is nonnegative, so $R_0(\widehat{z}(x);x)\le R(\widehat{z}(x);x)$. Theorem 5 then gives

$$R_0(\widehat{z}(x);x)\le R(z^\star(x);x)+\frac{\varepsilon(x)^2}{2\eta}.$$

Since $z^\star(x)$ minimizes $R(\cdot;x)$ over $\mathcal{Z}$, $R(z^\star(x);x)\le R(z;x)$ for every $z\in\mathcal{Z}$. Hence,

$$\begin{aligned}
R_0(\widehat{z}(x);x) &\le R(z;x)+\frac{\varepsilon(x)^2}{2\eta} \\
&= R_0(z;x)+\frac{\eta_{\mathrm{reg}}}{2}\|z-z_0(x)\|_2^2+\frac{\varepsilon(x)^2}{2\eta}.
\end{aligned}$$

By the definition of $\mathrm{rad}_{\mathcal{Z},0}$, $\|z-z_0(x)\|_2\le \mathrm{rad}_{\mathcal{Z},0}$. Substituting this bound yields

$$R_0(\widehat{z}(x);x)\le R_0(z;x)+\frac{\eta_{\mathrm{reg}}}{2}\,\mathrm{rad}_{\mathcal{Z},0}^2+\frac{\varepsilon(x)^2}{2\eta}.$$

The two error terms do not depend on z. Taking the infimum over $z\in\mathcal{Z}$ therefore gives

$$R_0(\widehat{z}(x);x)-R_0^\star(x)\le \frac{\varepsilon(x)^2}{2\eta}+\frac{\eta_{\mathrm{reg}}}{2}\,\mathrm{rad}_{\mathcal{Z},0}^2. \tag{EC.31}$$

The inequality also holds when $\varepsilon(x)=+\infty$. Because both operational risks are integrable, taking expectations gives

$$\mathbb{E}[R_0(\widehat{z}(X);X)]-\mathbb{E}[R_0^\star(X)]\le \frac{\mathbb{E}[\varepsilon(X)^2]}{2\eta}+\frac{\eta_{\mathrm{reg}}}{2}\,\mathrm{rad}_{\mathcal{Z},0}^2.$$

It remains to show that $\mathbb{E}[\varepsilon(X)^2]\le(\varepsilon^{\mathrm{ctr}}_B+L_B\sqrt{J_B})^2$.

The definition of $\varepsilon_b(x)$ and $L_{\mathrm{pop}} + L_b \le L_B$ give

$$0 \le \varepsilon(X) \le \varepsilon^{\mathrm{ctr}}_{\tau_B(X)} + L_B \|X - \bar{x}_{\tau_B(X)}\|_x.$$

By definition, $\mathbb{E}[(\varepsilon^{\mathrm{ctr}}_{\tau_B(X)})^2] = (\varepsilon^{\mathrm{ctr}}_B)^2$ and $\mathbb{E}[\|X - \bar{x}_{\tau_B(X)}\|_x^2] = J_B$. Both quantities are finite by hypothesis. The cross term satisfies

$$\begin{aligned}
&\mathbb{E}\left[\varepsilon^{\mathrm{ctr}}_{\tau_B(X)} \|X - \bar{x}_{\tau_B(X)}\|_x\right] \\
&\quad \le \left\{\mathbb{E}[(\varepsilon^{\mathrm{ctr}}_{\tau_B(X)})^2]\right\}^{1/2} \left\{\mathbb{E}[\|X - \bar{x}_{\tau_B(X)}\|_x^2]\right\}^{1/2} \\
&\quad = \varepsilon^{\mathrm{ctr}}_B \sqrt{J_B}.
\end{aligned}$$

Squaring the nonnegative bound on $\varepsilon(X)$ and taking expectations yields

$$\begin{aligned}
\mathbb{E}[\varepsilon(X)^2] &\le \mathbb{E}\left[\{\varepsilon^{\mathrm{ctr}}_{\tau_B(X)} + L_B \|X - \bar{x}_{\tau_B(X)}\|_x\}^2\right] \\
&= (\varepsilon^{\mathrm{ctr}}_B)^2 + 2L_B \mathbb{E}\left[\varepsilon^{\mathrm{ctr}}_{\tau_B(X)} \|X - \bar{x}_{\tau_B(X)}\|_x\right] + L_B^2 J_B \\
&\le \left(\varepsilon^{\mathrm{ctr}}_B + L_B \sqrt{J_B}\right)^2.
\end{aligned}$$

The last inequality uses the cross-term bound above. Substituting this second-moment bound into the expected risk bound gives

$$\mathbb{E}[R_0(\widehat{z}(X); X)] - \mathbb{E}[R_0^\star(X)] \le \frac{(\varepsilon^{\mathrm{ctr}}_B + L_B\sqrt{J_B})^2}{2\eta} + \frac{\eta_{\mathrm{reg}}}{2} \operatorname{rad}^2_{\mathcal{Z},0},$$

which proves Eq. (42). □

EC.2. Models and Experimental Settings

In this section, we give data generation details and application-specific parameter choices for the computational study in Section 6.2.

EC.2.1. Smooth capacity allocation

We construct A by drawing its entries independently from a gamma distribution with shape 0.65 and unit scale, then normalizing each row. We set $C = 5d_z$ and $\tau = 0.35$. For $j = 1, \ldots, d_\xi$, we set $c^u_j = 3 + 0.4\{1 + \sin(0.31(j-1))\}$ and $c^o_j = 0.7 + 0.15\{1 + \cos(0.23(j-1))\}$.

We generate four demand states indexed by $v_r \in \{(-1,-1), (-1,1), (1,-1), (1,1)\}$. We draw a $d_z \times 2$ matrix with independent standard normal entries, subtract each column mean, and let V_{res} contain the orthonormal columns of its QR factorization. We fix this exogenous resource basis before fitting any method. Define $z^{(r)}_{\mathrm{req}} = 5d_z \operatorname{softmax}(V_{\mathrm{res}} v_r)$. The reference scenario set contains $K_{\mathrm{ref}}/4$ scenarios from each state, and $r(s)$ denotes the state of scenario s. With independent $E_{sj} \sim \mathcal{N}(0,1)$, the scenario components are

$$\xi^{(s)}_j = (A z^{(r(s))}_{\mathrm{req}})_j \exp\left\{0.035 E_{sj} - \tfrac{1}{2}(0.035)^2\right\}. \tag{EC.32}$$

The support size study in Section 6.3.3 uses 1,000 scenarios.

Contexts are independent draws from $\mathcal{N}(0, \mathrm{Id}_8)$. Let $V_{\mathrm{ctx}} \in \mathbb{R}^{8\times 2}$ contain the orthonormal columns of the QR factorization of an independent standard Gaussian matrix. The conditional probabilities are uniform within each state, with state probabilities given by an affine softmax

$$\pi_s(x) = \frac{4}{K_{\mathrm{ref}}} \frac{\exp\{x^\top V_{\mathrm{ctx}} v_{r(s)}\}}{\sum_{r=1}^4 \exp\{x^\top V_{\mathrm{ctx}} v_r\}}. \tag{EC.33}$$

CAP transforms positive demands by $[T(\xi)]_r = \log(\xi_r)$. Let $\underline{\xi}_r$ and $\overline{\xi}_r$ be the minimum and maximum values of component r among the reference scenarios. The decoder on $\mathbb{R}^{d_\xi}$ is

$$[g(y)]_r = \min\{\overline{\xi}_r, \max\{\underline{\xi}_r, \exp(y_r)\}\}. \tag{EC.34}$$

The decoder is piecewise differentiable, and automatic differentiation at clipping points follows the PyTorch derivative conventions for minimum and maximum operations.

Differentiating Eq. (44) gives the scenario gradient used to construct $\widehat{C}_b$. Here $\mathrm{logit}^{-1}(a) := (1+\exp(-a))^{-1}$ and $\odot$ denotes componentwise multiplication.

$$\phi_{\mathrm{CAP}}(z,\xi) = \frac{1}{d_\xi} A^\top \left[(c^u + c^o) \odot \mathrm{logit}^{-1}\left(\frac{Az-\xi}{0.35}\right) - c^u \right]. \tag{EC.35}$$

For CAP, differentiating the smoothing function gives $h'_\tau(a) = \{1+\exp(-a/\tau)\}^{-1}$ and $h''_\tau(a) = \exp(-a/\tau)/[\tau\{1+\exp(-a/\tau)\}^2] > 0$, with $h''_\tau(-a) = h''_\tau(a)$. Thus, the Hessian of the loss in Eq. (44) is $\nabla_z^2 \ell_{\mathrm{CAP}}(z,\xi) = d_\xi^{-1} A^\top D(z,\xi) A$, where $[D(z,\xi)]_{jj} = (c_j^o + c_j^u) h''_\tau((Az)_j - \xi_j)$. Each diagonal entry is continuous and strictly positive on the compact product of the feasible capacity set and the box defined by the demand bounds above, so the minimum over this product and $j = 1, \ldots, d_\xi$ is a positive number $\underline{h}$.

For any $v \in \mathbb{R}^{d_z}$,

$$\begin{aligned} v^\top \nabla_z^2 \ell_{\mathrm{CAP}}(z,\xi) v &= \frac{1}{d_\xi} \sum_{j=1}^{d_\xi} [D(z,\xi)]_{jj} (Av)_j^2 \\ &\geq \frac{\underline{h}}{d_\xi} \|Av\|_2^2 \geq \frac{\underline{h}\, \sigma_{\min}(A)^2}{d_\xi} \|v\|_2^2. \end{aligned}$$

The service matrix has full column rank, so $\sigma_{\min}(A) > 0$. For $z, z' \in \mathcal{Z}_{\mathrm{CAP}}(C)$, their joining segment is feasible. Taylor's formula with integral remainder therefore gives

$$\begin{aligned} &\ell_{\mathrm{CAP}}(z',\xi) - \ell_{\mathrm{CAP}}(z,\xi) - \langle \nabla_z \ell_{\mathrm{CAP}}(z,\xi), z'-z \rangle \\ &\quad = \int_0^1 (1-t)(z'-z)^\top \nabla_z^2 \ell_{\mathrm{CAP}}(z + t(z'-z), \xi)(z'-z)\, dt \\ &\quad \geq \frac{\underline{h}\, \sigma_{\min}(A)^2}{2 d_\xi} \|z'-z\|_2^2. \end{aligned}$$

Averaging this inequality with nonnegative scenario probabilities summing to one, or taking conditional expectation, preserves the same lower bound. Hence every complete CAP objective is strongly convex with $\eta = \underline{h}\, \sigma_{\min}(A)^2 / d_\xi > 0$ without an added quadratic term. Consequently, CAP uses $\eta_{\mathrm{reg,CAP}} = 0$. The prespecified center is $z_0 = 5\mathbf{1}_{d_z}$.

EC.2.2. Smoothed conditional value-at-risk portfolio optimization

We set $r_{\max} = 0.25$, $\alpha = 0.10$, $\lambda_\mu = 0.25$, $\lambda_w = 0.01$, and $\tau = 0.015$.

The master scenario set contains 1,000 scenarios for 128 assets. Let $H_s \sim \mathcal{N}(0, \mathrm{Id}_6)$ denote the factor vector and $C_s \sim \mathrm{Bernoulli}(0.12)$ the crash indicator for scenario s. For asset j, let a_j denote a normalized Gaussian loading, $r_j \sim \mathrm{Uniform}[0.5, 1.5]$ its crash exposure, and $\mu_j^0 \sim \mathcal{N}(0.004, 0.0015^2)$ its return intercept. Given independent $E_{sj} \sim \mathcal{N}(0,1)$, the return is

$$\xi_j^{(s)} = 0.245 \tanh\left(\frac{\mu_j^0 + 0.032 H_s^\top a_j + 0.008 E_{sj} - 0.075 C_s r_j}{0.25} \right). \tag{EC.36}$$

The settings for K_{ref} and d_ξ use nested prefixes. Contexts follow $\mathcal{N}(0,\mathrm{Id}_8)$. We independently draw $G_1^P, G_2^P, G_3^P \in \mathbb{R}^{8\times 7}$ with independent standard normal entries and $\omega^P \in [-\pi,\pi]^7$ with independent uniform entries. We construct the context map from the features

$$h^P_{\mathrm{lin}}(x) = \frac{x^\top G_1^P}{\sqrt{8}}, \qquad h^P_{\mathrm{per}}(x) = \sin\left(\frac{x^\top G_2^P}{\sqrt{8}} + \omega^P\right), \qquad h^P_{\mathrm{quad}}(x) = \tanh\left(\frac{(x\odot x - \mathbf{1}_8)^\top G_3^P}{\sqrt{8}}\right). \tag{EC.37}$$

Using these features, we set

$$\psi^0_P(x) = \tanh\left\{h^P_{\mathrm{lin}}(x) + 0.45[0.65h^P_{\mathrm{per}}(x) + 0.35h^P_{\mathrm{quad}}(x)]\right\}, \qquad \psi_P(x) = \left(\psi^0_{P,1:6}(x), \frac{1+\psi^0_{P,7}(x)}{2}\right). \tag{EC.38}$$

For $v_s = (H_s, C_s)$ and independent $\delta_s \sim \mathcal{N}(0, 0.10^2)$, the conditional probabilities are

$$\pi_s(x) = \frac{\exp\{-\|\psi_P(x) - v_s\|_2^2/(2\cdot 1.35^2) + \delta_s\}}{\sum_{r=1}^{K_{\mathrm{ref}}} \exp\{-\|\psi_P(x) - v_r\|_2^2/(2\cdot 1.35^2) + \delta_r\}}. \tag{EC.39}$$

PORT uses $[T(\xi)]_r = \operatorname{atanh}(\xi_r/0.25)$ and $[g(y)]_r = 0.25\tanh(y_r)$, with $\mathrm{dom}(g) = \mathbb{R}^{d_\xi}$.

In Eq. (45), $f((w,t)) = 0.01\|w\|_2^2/2$, and Q contains the remaining terms, including the linear threshold term. Writing $\omega_{\mathrm{tail}}(w,t,\xi) = \mathrm{logit}^{-1}\{(-\xi^\top w - t)/0.015\}$, we differentiate Q to obtain the scenario gradient used to construct $\widehat{C}_b$.

$$\phi_{\mathrm{PORT}}((w,t),\xi) = \left(-[0.25 + 10\omega_{\mathrm{tail}}(w,t,\xi)]\xi,\ 1 - 10\omega_{\mathrm{tail}}(w,t,\xi)\right). \tag{EC.40}$$

The leverage penalty is part of the operational loss and gives positive curvature in the directions of the portfolio weights, but no uniform positive bound has been established in the threshold direction. We therefore add a quadratic term with prespecified center $z_0 = (d_\xi^{-1}\mathbf{1}_{d_\xi}, 0)$ and coefficient $\eta_{\mathrm{reg,PORT}} = 10^{-6}$ only in the solution mapping. In both PORT and NET, the operational objective is convex, so the added quadratic term gives Eq. (5) with $\eta = \eta_{\mathrm{reg,app}} > 0$ for both population and finite-distribution objectives.

EC.2.3. Multicommodity network reservation

The network has node set $\{0,1,2,3\}^2$ and both orientations of every horizontal and vertical grid edge. In each replication, we lexicographically order all ordered origin–destination pairs whose row and column coordinates both differ, apply a fixed permutation for that replication, and retain the first 128 pairs as the master commodity list. For each commodity, one path moves horizontally and then vertically; the other moves vertically and then horizontally. We set $c = \mathbf{1}_{48}$, $\lambda = 10$, and $\rho = 0.05$. We generate 1,000 reference scenarios with 128 demand components and use nested prefixes for the settings of K_{ref} and d_ξ.

Define $b_j = 0.04 + (j-1)(0.06-0.04)/127$ and $g_j = 1 + \{(j-1) \bmod 4\}$. For each scenario, we draw $H_s \sim \mathcal{N}(0,\mathrm{Id}_6)$, $E_{sj} \sim \mathcal{N}(0,1)$, $R_s \sim \mathrm{Bernoulli}(0.12)$, and T_s uniformly from $\{1,2,3,4\}$. Define

$$M = \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}. \tag{EC.41}$$

Using the regime matrix in Eq. (EC.41), we generate the master scenarios as

$$\xi_j^{(s)} = b_j \exp\left\{0.35H_{s1} + 0.40H_{s,1+g_j} + 0.20E_{sj} + 0.90R_s M_{T_s,g_j} - \frac{0.35^2 + 0.40^2 + 0.20^2}{2}\right\}. \tag{EC.42}$$

We compute normalized total demand from all 128 components of each reference scenario, even when the setting retains fewer commodities, by

$$\bar{\xi}_s = \frac{1}{128}\sum_{j=1}^{128}\frac{\xi_j^{(s)}}{b_j}, \qquad s = 1,\dots,1{,}000. \tag{EC.43}$$

Within the current K_{ref}-scenario prefix, we order the scenarios by the normalized demand in Eq. (EC.43), breaking an exact tie by scenario index, and let $\mathcal{L}$ and $\mathcal{H}$ contain the lower and upper $K_{\mathrm{ref}}/2$ scenarios, respectively. Contexts follow $\mathcal{N}(0,\mathrm{Id}_8)$. Let $u = 8^{-1/2}\mathbf{1}_8$ and $\varpi(x) = \mathrm{logit}^{-1}(3u^\top x)$. The conditional distribution is

$$\pi_s(x) = \begin{cases} 2\{1-\varpi(x)\}/K_{\mathrm{ref}}, & s \in \mathcal{L}, \\ 2\varpi(x)/K_{\mathrm{ref}}, & s \in \mathcal{H}. \end{cases} \tag{EC.44}$$

The conditional law depends only on $u^\top x$, and the seven orthogonal context directions do not affect it.

NET transforms positive demands by $[T(\xi)]_r = \log(\xi_r)$ and reconstructs them by $[g(y)]_r = \exp(y_r)$, with $\mathrm{dom}(g) = \mathbb{R}^{d_\xi}$.

For fixed reservation z and demand ξ, the routing regularizer in Eq. (46) makes the recourse decision $q(z,\xi)$ unique. Danskin's theorem then gives the scenario gradient used to construct $\widehat{C}_b$.

$$\phi_{\mathrm{NET}}(z,\xi) = -2\lambda[L(q(z,\xi),\xi) - z]_+. \tag{EC.45}$$

The reservation cost gradient c belongs to the deterministic term f and is therefore not part of the scenario gradient $\phi = \nabla_z Q$. NET can be flat in reservation directions whose capacities are inactive. The added quadratic term has prespecified center $z_0 = 0.1 \cdot \mathbf{1}_{48}$ and coefficient $\eta_{\mathrm{reg,NET}} = 10^{-6}$, ensuring strong convexity.

EC.2.4. Quadratic problem with decision-irrelevant variation

We obtain V_{rel} and V_{noise} from the first four and remaining eight columns of the orthonormal factor in a reduced QR factorization of a 200×12 matrix with independent standard Gaussian entries. We fix each column sign so that its entry of largest magnitude is positive. These matrices specify the relevant and noise directions in the original realization space. The transformation and decoder are the identity, with $\mathrm{dom}(g) = \mathbb{R}^{200}$, and the prespecified center is $z_0 = 0$. For a probability law ν on $\mathcal{U}$, the operational loss, regularized solution, and scenario gradient are

$$\begin{aligned} \ell_{\mathrm{quad}}(z,\xi) &= \frac{1}{2}\left\|z - V_{\mathrm{rel}}^\top \xi\right\|_2^2, \\ z^{\mathrm{quad}}(\nu) &= \frac{\mathbb{E}_\nu[V_{\mathrm{rel}}^\top \Xi]}{1+\eta_{\mathrm{reg}}}, \qquad \phi_{\mathrm{quad}}(z,\xi) = -V_{\mathrm{rel}}^\top \xi. \end{aligned} \tag{EC.46}$$

In Eq. (EC.46), the deterministic quadratic term in z belongs to f, so the scenario gradient contains only the realization-dependent component. The quadratic basis objective uses $\eta_{\mathrm{reg}} = 0$ and is strongly convex with $\eta = 1$.

The five decision-relevant vertices form a regular simplex in $\mathbb{R}^4$ with squared vertex norms 4 and inner products -1 between distinct vertices before scaling by 1.5. For each scaled vertex, we draw ten independent $\mathcal{N}(0,\mathrm{Id}_4)$ perturbations, center them componentwise, and rescale the entire 10×4 array to have a root mean square of 0.1 over all entries. Adding these perturbations to the vertex gives u_{gh}^{rel}, for $g = 1,\dots,5$ and $h = 1,\dots,10$. We also draw ten independent $\mathcal{N}(0,\mathrm{Id}_8)$ vectors and center them componentwise. Writing the centered vectors as the rows of $E \in \mathbb{R}^{10\times 8}$, we whiten them

using the empirical covariance $E^\top E/10$ to obtain $e_r^\perp$, $r=1,\dots,10$, with empirical covariance Id_8. The scenarios are

$$\xi_{ghr}^{(\sigma)} = V_{\mathrm{rel}} u_{gh}^{\mathrm{rel}} + \sigma V_{\mathrm{noise}} e_r^\perp, \qquad \sigma \in \{1/4, 4\}. \tag{EC.47}$$

We generate each context by choosing uniformly among $\mu_{x,1}=(-5,-5)$, $\mu_{x,2}=(-5,5)$, $\mu_{x,3}=(5,-5)$, and $\mu_{x,4}=(5,5)$, then adding $\mathcal{N}(0,\mathrm{Id}_2)$ noise. Let $\tau_{\mathrm{gen}}(x)$ index the nearest mean and define $\gamma_g^{\mathrm{prob}} = 1.4(\cos(2\pi(g-1)/5), \sin(2\pi(g-1)/5))^\top$. The conditional probability of decision-relevant group g is

$$\pi_g(x) = \frac{\exp\{(\gamma_g^{\mathrm{prob}})^\top (x - \mu_{x,\tau_{\mathrm{gen}}(x)})\}}{\sum_{g'=1}^5 \exp\{(\gamma_{g'}^{\mathrm{prob}})^\top (x - \mu_{x,\tau_{\mathrm{gen}}(x)})\}}. \tag{EC.48}$$

Conditional on g, we choose the perturbation index h and noise index r uniformly from their respective sets of ten. The resulting reference scenario set has $K_{\mathrm{ref}} = 5\times 10\times 10 = 500$ elements.

For the reference decisions z_i^{ref} at the evaluation contexts, define $\bar{z}^{\mathrm{ref}} = n_{\mathrm{eval}}^{-1}\sum_i z_i^{\mathrm{ref}}$ and $\sigma_{\mathrm{ref}}^2 = n_{\mathrm{eval}}^{-1}\sum_i \left\| z_i^{\mathrm{ref}} - \bar{z}^{\mathrm{ref}} \right\|_2^2$. The normalized squared decision error of $\widehat{z}$, denoted by $\varepsilon^{\mathrm{norm}}(\widehat{z})$, is

$$\varepsilon^{\mathrm{norm}}(\widehat{z}) = \frac{n_{\mathrm{eval}}^{-1}\sum_i \left\| \widehat{z}(x_i) - z_i^{\mathrm{ref}} \right\|_2^2}{\sigma_{\mathrm{ref}}^2}. \tag{EC.49}$$

We report the ratio in Eq. (EC.49) without multiplying it by 100. For a chart basis $\widehat{V}_b^{\mathrm{orig}} \in \mathbb{R}^{200\times 4}$, obtained by restoring the component scales and orthonormalizing the columns, we define the minimum subspace alignment with the decision-relevant subspace as

$$\rho_b := \sigma_{\min}(V_{\mathrm{rel}}^\top \widehat{V}_b^{\mathrm{orig}}). \tag{EC.50}$$

EC.3. Computational Protocol

EC.3.1. Sampling and evaluation

For each setting within a replication, all methods use the same training and validation observations and are evaluated on the same contexts and realizations. We generate the reference scenario sets, context maps, and training, validation, and evaluation samples using separate random streams within each replication. PORT and NET also vary their physical problem parameters across replications. CAP keeps its service technology fixed and varies the resource demand states, contextual directions, and noise within each state.

For the simulations described in Section 6.2, let $x_1,\dots,x_{n_{\mathrm{eval}}}$ be the evaluation contexts and $\xi_i^{(1)},\dots,\xi_i^{(n_{\mathrm{draw}})}$ be independent draws from the known conditional law given x_i. These draws are independent of the training and validation data. For any decision rule $z(\cdot)$, we estimate its operational risk by

$$R_{\mathrm{MC}}(z) := \frac{1}{n_{\mathrm{eval}} n_{\mathrm{draw}}} \sum_{i=1}^{n_{\mathrm{eval}}} \sum_{\ell=1}^{n_{\mathrm{draw}}} \ell(z(x_i), \xi_i^{(\ell)}; x_i). \tag{EC.51}$$

For NET, we evaluate the operational loss by solving the recourse problem in Eq. (46) to convergence with the reservation decision fixed.

EC.3.2. Regularization and structure selection

We assess $\eta_{\mathrm{reg}} \in \{0, 10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ using the complete 500-scenario laws at 200 evaluation contexts for CAP and PORT and at three evaluation contexts for the larger NET joint quadratic programs. We use the coefficient specified for each application across all fitted methods, settings, and replications.

We use five internal folds for each candidate and replication in the assessment described in Section 4.5. In each chart, we set singular values of $\widehat{C}_b$ no larger than 10^{-12} times the leading value to zero. The number of remaining nonzero singular values is the effective rank. We apply Eq. (15) to these thresholded values with $\alpha_{\mathrm{ret}} = 0.9999$. A candidate is eligible only if every chart contains at least $n_{\min} = 50$ observations, has effective rank at least d_b, and achieves the retention target. In each chart, the minimum number of fitting observations across the five folds divided by the nominal number of parameters $K_b d_b + (K_b - 1)(d_x + 1)$ must be at least $c_{\min}^{\mathrm{par}} = 3$. We count distinct projected rows using Euclidean tolerance 10^{-10} and require K_b not to exceed the minimum number of distinct rows across fitting folds. After fixing the structure, we select one optimization iteration for all charts using validation operational loss.

EC.3.3. Training rules by method

For ICEO-A, we standardize the training contexts. The neural network has two hidden layers of width 32, each with a linear map followed by a ReLU activation, and a final linear layer with K_{ref} logits. Let a_θ denote the logit map, including this preprocessing, with network parameters θ, and define $p_\theta(x) = \mathrm{softmax}\{a_\theta(x)\}$. Using the numerical solution mapping $\widetilde{z}$, we fit ICEO-A by minimizing the empirical operational loss

$$\frac{1}{|I_{\mathrm{tr}}|} \sum_{i \in I_{\mathrm{tr}}} \ell\big(\widetilde{z}\{p_\theta(x_i), (\xi^{(1)}, \ldots, \xi^{(K_{\mathrm{ref}})}); x_i\}, \xi_i; x_i\big)\,. \tag{EC.52}$$

Atlas uses regularization coefficients of 10^{-5} for both scenario coordinates and probability slopes.

SOF uses the constrained approximate-risk criterion of Kallus and Mao (2023) with 100 bootstrap trees, no honesty split, a terminal node size of 10, and a minimum child fraction of 0.2. We aggregate duplicate realizations before the downstream solve.

For k-NN, the candidate set is $\{2^j : j = 0, \ldots, \lfloor \log_2 |I_{\mathrm{tr}}| \rfloor\} \cup \{|I_{\mathrm{tr}}|\}$. Validation selects k separately for each application, setting, and replication. Distances are Euclidean after standardizing each context component by its training mean and scale, and the selected neighbors receive uniform weight.

EC.3.4. Performance measures and timing

For method h, let $\widetilde{z}_h$ denote its numerical decision rule and z_i^{ref} the unregularized optimal decision under the known conditional law at evaluation context x_i. We report relative excess as

$$\mathrm{RER}_h = \frac{\sum_i \{R_0(\widetilde{z}_h(x_i); x_i) - R_0(z_i^{\mathrm{ref}}; x_i)\}}{|\sum_i R_0(z_i^{\mathrm{ref}}; x_i)|}. \tag{EC.53}$$

For Atlas, the reported structure includes $\widehat{B}$, each d_b and K_b, the query support $\max_b K_b$, and the stored support $\sum_b K_b$.

Construction time covers preparation, structure or hyperparameter selection, model fitting, and validation of fitted models. For Atlas and ICEO-A, it includes one complete fitting run of 1,600 updates. Worker initialization and shutdown are included; final quality evaluation is excluded. Wall time measures elapsed duration, and CPU time is the total processor time across the main process and its workers.

Deployment timing uses one representative seed and the batch sizes and core allocations in Table 5. It begins after the models and workers have been initialized. It includes prediction of the scenario distribution, communication between processes, downstream decision solves, and assembly

of the returned decisions. Each downstream optimization is initialized independently. We report median deployment times over five repetitions.

EC.3.5. Chart experiment guided by theory

For the chart problem in Eq. (47), we use the scenario and decision domain $[-4,4]^{10}$ and set $\eta_{\mathrm{reg}}=0$. The Hessian in z is the identity. The transformation and decoder are the identity on this box. We reject supports containing scenarios outside it, so the weighted mean is both the unconstrained optimum and a feasible optimum of the box-constrained problem.

For each replication, the QR factorization of a standard Gaussian 10×10 matrix provides mutually orthogonal two-column blocks $V_{\mathrm{c}}, V_1^{\mathrm{aux}}, \ldots, V_4^{\mathrm{aux}}$. The operating mode R is uniform on $\{1,2,3,4\}$. For $\omega_{\mathrm{dir}} \in \{0, \pi/4, \pi/2\}$, we define

$$V_r(\omega_{\mathrm{dir}}) = \cos(\omega_{\mathrm{dir}})V_{\mathrm{c}} + \sin(\omega_{\mathrm{dir}})V_r^{\mathrm{aux}}, \qquad \xi_{r,v} = V_r(\omega_{\mathrm{dir}})v, \quad v\in\{-1,1\}^2. \tag{EC.54}$$

The four corners from each mode form a reference set of 16 scenarios. At zero angle, Eq. (EC.54) yields only four distinct locations. We reuse the random matrix, mode draws, contexts, and uniforms for sampling scenario indices across angles, so only the mode-specific directions change.

Let h_r enumerate $(-1,-1),(-1,1),(1,-1),(1,1)$, and set $c_r = 2(h_{r1}, h_{r1}, h_{r1}, h_{r1}, h_{r2}, h_{r2})$. We independently draw e uniformly on $[-1/4,1/4]^6$ and s uniformly on $[-0.8,0.8]^2$. The observed context is $x=(c_R+e,t)$, where $t=\operatorname{atanh}(s)$ componentwise. The context domain $\mathcal{X}$ is the union of four products, each consisting of a mode box and the two-dimensional domain of t. We do not supply the true mode labels during fitting. Conditional on this context, only the four corners of mode R have positive mass, with

$$p(v\mid t) = \frac{\exp(v^\top t)}{\sum_{w\in\{-1,1\}^2}\exp(w^\top t)}, \qquad z^\star(x) = V_R(\omega_{\mathrm{dir}})\tanh(t). \tag{EC.55}$$

We evaluate relative excess in Eq. (EC.53) using the exact conditional mean in Eq. (EC.55) as the reference decision and conditional expected operational losses.

Each local corner probability is at least 0.01, and the two nonzero conditional covariance eigenvalues are at least 0.36. The feasible local simplex with coordinates $(-1,-1),(3,-1),(-1,3)$ contains all four corners; its vertices have norm at most $\sqrt{10}<4$ after mapping to the scenario space. Thus, a feasible $d+1$ local representation exists, while the representation using four corners also has an exact affine-softmax probability model. For a chart corresponding to mode r, the local set $\mathcal{Y}_b$ in Assumption 3 consists of standardized realization vectors whose decoded scenarios lie in the plane of that mode and in the box. On this set, the gradient is exactly affine in a two-dimensional projection. After restoring the component scales, the empirical basis also lies in that plane. These local statements concern the partition into four modes; merged candidates need not have the same dimension or signal margin.

For context regularity, let $C_r = c_r + [-1/4,1/4]^6$, $a_r(x) = [1-\operatorname{dist}(x_{1:6}, C_r)]_+$, and $w_r^{\mathrm{mode}}(x) = a_r(x) + \{1-\sum_j a_j(x)\}/4$. The unit neighborhoods of the four boxes are disjoint, so these nonnegative weights sum to one. Multiplying $p(v\mid t)$ by $w_r^{\mathrm{mode}}(x)$ defines a Lipschitz extension of the conditional law between the observed boxes and at the centroids of merged charts. It agrees with the sampled law everywhere on its context support.

Across three independent replications, we fit and report each $B\in\{1,2,4,8,16,32\}$ separately, using 1,600 parameter updates for each model. We select d_b from the training observations and set $K_b = d_b+1$.

EC.3.6. Quadratic basis experiment

The quadratic basis experiment uses three independent replications with training, validation, and evaluation sample sizes of 1600/200/200. Within each replication, the two noise levels share the contexts, group probabilities, decision-relevant values, base noise draws, and sampled scenario indices; only the scale of the eight-dimensional decision-irrelevant component changes. The two bases compared in Section 6.5 use the same chart assignments.